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OECD ECONOMIC SURVEYS

CANADA

1997

SPECIAL FEATURE

• THE TAX SYSTEM:

OPTIONS FOR CHANGES

OECD ECONOMIC SURVEYS

1996-1997

CANADA

ORGANISATION FOR ECONOMIC CO-OPERATION AND DEVELOPMENT

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BASIC STATISTICS OF CANADA

THE LAND

Area (thousand sq. km)	9 976
Agricultural area (1990, as per cent of total area)	6.4
Population of major cities, including metropolitan areas (thousands) (1995)	
Montréal	3 337
Toronto	4 344

THE PEOPLE

Population (1996)	29 925 429
Number of inhabitants per sq. km	3.0
Population, annual net natural increase (average 1992-1996)	178 800
Natural increase rate per 1 000 inhabitants (average 1992-1996)	6.2
Civilian labour force (1996)	15 145 400
Employment agriculture (1996)	452 900
Immigration (annual average 1992-1996)	232 629
Average annual increase in civilian labour force (1992-1996), per cent	1.0

PRODUCTION

GDP in 1996 (millions of Canadian dollars)	797 789
GDP per head (Canadian dollars)	26 700
Gross fixed investment (private and public) per head (Canadian dollars)	4 677
Gross fixed investment (private and public) as per cent of GDP	17.5
Origin of gross domestic product (1996, per cent of total, 1986 prices):	
Agricultural, forestry and fishing	2.8
Mining and quarrrying	4.4
Manufacturing	18.8
Construction	4.8
Non business sector	16.3
Other	55.6

THE GOVERNMENT

Government current expenditure on goods and services (1996, as per cent of GDP)	18.6
Government gross fixed capital formation (1996, as per cent of GDP)	2.1
Federal Government current revenue (1996, as per cent of GDP)	19.3
Federal direct and guaranted debt (1996, per cent of current expenditure)	276.6

Composition of Parliament (June 1997)	Number of seats House of Commons	Number of seats Senate
Progressive Conservative	20	48
Liberal	155	49
New Democratic	21	–
Bloc Québécois	44	–
Independent	1	3
Reform	60	–

FOREIGN TRADE

Exports (1996)	
Exports of goods and services, as per cent of GDP	38.4
Main exports (per cent of commodity exports):	
Wheat	2.0
Natural gas	2.4
Lumber and sawmill products	5.6
Pulp and paper	6.7
Other metals and minerals	10.0
Motor vehicles and parts	22.6
Other manufactured goods	28.4
Main customers (per cent of commodity exports):	
United States	79.0
EU	6.2
Japan	4.4
Imports (1996)	
Imports of goods and services, as per cent of GDP	35.3
Main imports (per cent of commodity exports):	
Industrial materials	18.5
Motor vehicles and parts	21.5
Producers' equipment	32.0
Consumer goods	10.8
Main suppliers (per cent of commodity imports):	
United States	75.9
EU	8.6
Japan	3.0

THE CURRENCY

Monetary unit: Canadian dollar

Currency unit per US dollar, average of daily figures:	
Year 1996	1.364

Note: An international comparison of certain basic statistics is given in an annex table.

This Survey is based on the Secretariat's study prepared for the annual review of Canada by the Economic and Development Review Committee on 8 September 1997.

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After revisions in the light of discussions during the review, final approval of the Survey for publication was given by the Committee on 24 September 1997.

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The previous Survey of Canada was issued in November 1996.

Assessment and recommendations

With economic activity picking up...

Following a short-lived rebound in 1994, the Canadian economy has expanded again at a robust pace since mid-1996, as both business and household demand surged in response to a substantial easing in monetary conditions which offset the damping effects of significant fiscal consolidation. However, with enterprises striving to improve productivity, the pick-up in activity has only been reflected in lower unemployment in recent months. Although real GDP growth in excess of 3½ per cent over the past twelve months to mid-1997 has meant that some economic slack has been taken up, a substantial margin of unused capacity still exists, exerting continued downward pressure on costs and prices. Despite buoyant import demand associated with the strengthening domestic economy, the current account of the balance of payments has remained in broad balance.

... prospects for sustained growth are favourable...

Both external and domestic circumstances point to continued solid growth at rates of just under 3½ per cent through 1998, as currently projected by the OECD Secretariat. Although the pace of expansion in the United States, Canada's major trading partner, may begin to slow, this should be partly offset by an improving outlook in other OECD countries. Thus, external demand can be expected to continue to underpin economic activity, with the help of a favourable competitive position. Prospects are also good for further increases in domestic demand. Given the usual lags in the effects of monetary policy, there should be

significant stimulus to growth and jobs still to come from past declines in interest rates to the current low levels. Business confidence has improved significantly, with the share of companies thinking it is now a good time to invest having reached a record high. Consumer confidence is also strengthening, albeit from low levels. At the same time, the amount of slack remaining in the economy should continue to limit price pressures, while Canada's external position is unlikely to become a policy constraint, given the substantial improvement in the current account in recent years.

... although there are risks to the outlook.

There are risks to this economic outlook, however. On the *downside,* these mainly reflect uncertainties about consumer behaviour, as the recent surge in household demand, which has occurred in the face of sluggish disposable income – with the saving ratio falling to a historically low level – may not prove sustainable. In particular, given the currently high personal debt, consumers could react negatively to a decline in the prices of their assets backing that debt. This would adversely affect demand and hence growth in the face of ongoing – albeit diminishing – fiscal restraint and still high unemployment. On the other hand, in view of the considerable cyclical momentum that the economy has shown of late, there are also important *upside* risks as household demand may gain considerable strength if labour market conditions improve significantly. In this case, growth well in excess of the rate projected by the Secretariat is not to be excluded.

A shift in monetary conditions will likely be required...

On balance, given the current economic outlook, a return to neutral monetary conditions will be required to prevent the emergence of inflationary pressure. Since the mid-1990s, with the turnaround in economic fundamentals (notably improved external and fiscal balance along with continued low inflation), the Bank of Canada has been successful in

bringing interest rates down below corresponding US rates across a broad portion of the yield curve. By the end of 1996, monetary conditions – taking into account both interest and exchange rate movements – had fallen to their level of greatest ease in recent decades. Since then, the Bank has sought to keep monetary conditions broadly unchanged, with a recent rise in short-term interest rates aiming to offset the effect of a weakening Canadian dollar. Such an orientation will have to change at some point as economic slack is taken up. The timing and speed of the resulting rise in short-term interest rates should depend on the speed at which the output gap closes and on movements in exchange rates.

... putting the inflation-targeting framework to the test.

The adoption of inflation targets (currently 1 to 3 per cent), highlighted the fact that monetary policy must necessarily be forward-looking because action by the Bank of Canada affects price developments with a long lag (of one to two years), implying that the authorities have to deal with evolving demand and cost pressures well ahead of time. However, such a framework does not lessen the obvious difficulties for monetary management in assessing and predicting the forces that affect inflation – in particular the output gap. Experience suggests that it is difficult to maintain inflation within a 2 per cent range and that trying to achieve a quick reversal of deviations from such a range may have adverse effects on activity. This is one of the issues to be considered when the authorities make a decision on a new inflation target beyond 1998. Another is the relative size of the costs and benefits of even modest further disinflation from the rate of around 2 per cent recorded in recent years. Consequently, the authorities might consider making another interim extension of the existing target to allow it to be tested over a full cycle before deciding on the ultimate objective consistent with price stability.

Expectations of a ''fiscal dividend'' are mounting...

Continued improvement in the fiscal positions of both the federal and provincial governments should help create the conditions for the maintenance of a low inflation and low-interest-rate environment. Over the past four years, with consolidation targets having been generally met or exceeded, the general government financial position has moved from a deficit of nearly 7½ per cent of GDP to broad balance. This implies that, abstracting from cyclical influences on spending and revenue, the ''structural'' deficit, which still exceeded 5 per cent in 1993, has been eliminated. Such fiscal consolidation has been largely achieved through reductions in spending on government programmes. The process of downsizing and restructuring has involved a great deal of hardship, contributing to job insecurity and low consumer confidence. It is, therefore, not surprising that, after all these sacrifices, pressures are mounting within the country for a ''fiscal dividend'', in the form of tax cuts and/or higher spending.

... but tackling the high public debt remains a priority.

With the federal and provincial governments' objective of balancing their budgets over the medium term already achieved or within reach, there is indeed a need to consider what should be done with any emerging budget surpluses. Given the relatively high level of public debt – at around 70 per cent, the net debt-to-GDP ratio remains the third highest amongst OECD countries – priority should be given to using these surpluses to put the ratio on a clear downward trend. This would help to further lower the risk premium on Canadian debt, reduce the country's vulnerability to economic shocks, and provide additional resources for the private sector. Pursuit of such an objective need not preclude, however, some additional spending in priority areas, such as human capital development and innovation, nor does it rule out targeted adjustments to the tax system.

Even so, changes to the tax system could be contemplated...

As discussed in the special chapter of the Survey, the Canadian tax system has evolved considerably over the past decade or so. The measures implemented reflect changes in the policy environment, including trade liberalisation and tax reforms in other OECD countries, the fiscal crisis brought on by high public debt levels, and the ongoing decentralisation of the Canadian federation. As a result, the system is now less distorting than in the past. Compared with value-added taxes used elsewhere, the Goods and Services Tax, introduced in 1991, has a very comprehensive base and ensures that traded products are treated neutrally on international markets. The personal income tax system is also broadly-based, with a limited number of brackets, reasonably generous treatment of retirement savings, and refundable tax credits at the bottom end. However, despite the efforts to flatten the rate structure, taxation on income has remained quite progressive, owing in part to the use of surtaxes at both the federal and provincial levels. In addition, partial indexation has resulted in ''bracket creep'' which, albeit contributing to the budget deficit reduction process, has tended to increase the tax burden and is unwelcome on grounds of transparency and accountability. With respect to corporate income taxation, there are some differential incentive effects across industries and activities, owing to preferential tax rates, imperfect capital deductions for different types of assets, and the inability of corporate groups to file consolidated returns. Hence, the need for further adjustments in the tax system to address these matters.

... to reflect the openness of the economy.

In this perspective, for a highly-open economy like Canada, a major concern must be to ensure that the private sector can respond to competitive pressures. While – despite their upward trend – personal income tax rates are close to the OECD average, they are quite high compared with the

United States, resulting in some locational advantage in particular for mobile persons at the upper end of the income scale. Some consideration should therefore be given to the adverse effects that high average income tax rates may have on the highly-skilled and entrepreneurial portion of the workforce. At the same time, increased use of refundable tax credits for personal incomes can offer a cost-effective means to take into account redistributive considerations in a way that mitigates adverse incentives to work. With respect to corporate income, since effective tax rates differ across sectors and investments, there would be scope for making taxation more neutral to provide a level playing field. This would tend to reduce existing rate differentials against the United States and ensure that resources are allocated in ways that best reflect the economy's comparative advantage.

This may imply changes to the tax mix...

Such an orientation would likely imply changes to the tax structure. Traditionally, Canada relies more on income taxes and less on payroll and consumption taxes than many other OECD countries. As the latter taxes are found to be economically less distorting, it might be possible to change the tax mix without running the risk of putting Canadian businesses at a competitive disadvantage. Moving in that direction would raise problems, however. First, the income tax serves equity objectives relatively well, contrary to payroll and consumption taxes. Second, while payroll taxes in Canada are relatively low, they have grown rapidly in recent years and there is some evidence that they may have adverse effects on employment in the short run. The large surplus in the Employment Insurance account rather suggests that the premium rates should be lowered, and pension reform will tend to increase the payroll tax burden anyway. Third, increased reliance on consumption taxes would have the desired effects only if it involved a shift

away from provincial sales taxes towards value-added taxes. More generally, any initiatives to change the tax mix risk being frustrated by developments at the provincial level.

... and would require close co-ordination with provinces.

Indeed, some of the difficulties with the existing tax system originate from provinces, which, together with the municipalities, account for about half of total tax revenues. Provincial retail sales taxes discriminate against goods relative to services, domestic relative to foreign products, and products using taxed business inputs (so far, only four provinces have adopted a VAT system). Thus, further sales tax harmonisation is particularly important. Furthermore, the additional burden on businesses through property and capital taxes is relatively high, and resource taxes vary widely across provinces. Federal and provincial income taxation is reasonably harmonised, but strains are apparent as provinces occupy more and more of the income tax room. As the two levels of government co-occupy the main tax fields, any unilateral federal initiative could be readily undone by the provinces. A co-ordinated approach is, therefore, required to ensure that adjustments to taxation have the intended benefits for the economy.

Past structural reforms should begin to bear fruit...

Besides taxation, keeping the emphasis on structural reforms more generally is necessary to further improve economic performance. Over the past decade or so, structural policy initiatives have opened up both domestic and international markets for Canadian producers and, at the same time, considerably reduced government involvement in economic affairs. Recent actions have maintained this course, tackling continuing structural impediments. Despite the reforms' focus on improving economic efficiency, the expected positive effects on productivity and potential output have yet to materialise. This reflects both the poor

macroeconomic situation over most of the 1990s and the substantial transitional costs associated with the adjustment to the reforms and to a more competitive environment. Nonetheless, it would seem reasonable to expect that, after an extended period of restructuring, such reforms will begin to bear fruit in the current more favourable macroeconomic circumstances. Even so, there is scope for further action in several areas.

... but more could be done to reduce high structural unemployment...

Still high structural unemployment reflects the fact that labour market reform has lagged initiatives in other fields. The recent overhaul of the unemployment insurance system and adjustments to social assistance programmes in some provinces go a long way towards bringing labour market policies in line with the reform agenda and will work to reduce structural unemployment in the period ahead. However, changes implemented thus far have modified, but not eliminated, work disincentives. In particular, as suggested in the recent OECD Job Study, further steps to tighten restrictions on the frequent use of the unemployment insurance system and to reduce the regional differentiation in the generosity of benefits could improve labour market flexibility. The hand-over of active labour market policies to some provinces is to be welcomed, as it will allow for greater flexibility in addressing local and regional conditions. This said, the federal government should not vacate this area completely since monitoring the progress of the new system, and ensuring that returning the unemployed to work is its primary focus, will become ever more important.

... and bolster efficiency through further product market reforms.

Gains in economic efficiency could also be achieved through more concerted efforts both to enhance labour and capital mobility and to free product markets through the Agreement on Internal Trade. The implementation of this Agreement appears lacking as does the will to move for-

ward with previous commitments. As a consequence, progress in the internal trading environment seems to be lagging compared with that made in reducing external barriers to trade through various bilateral, regional and multilateral arrangements. Regaining lost momentum and putting in place concrete measures to liberalise internal trade should, therefore, be a priority on the structural policy agenda. And while a lot has been achieved on the external front, more could be done to bring down high tariff rates in some areas (in particular for certain agricultural goods), which continue to impede efficient resource allocation.

Summary

In *sum*, after a ''growth pause'', the recovery of the Canadian economy has regained considerable momentum, owing to a substantial easing in monetary conditions that was facilitated by improved economic fundamentals, notably significant progress in fiscal consolidation. While prospects for sustained non-inflationary growth appear favourable, the inflation-targeting framework will be put to the test as the recovery matures and economic slack is taken up. The significant improvement in the fiscal position at the federal and provincial levels should provide scope both for reducing the high public debt-to-GDP ratio and for dealing with some deficiencies of the tax system. In combination with structural policies carried out over the past decade, further initiatives to reduce still high structural unemployment and continued reform in product markets should improve the economy's growth potential over the medium term.

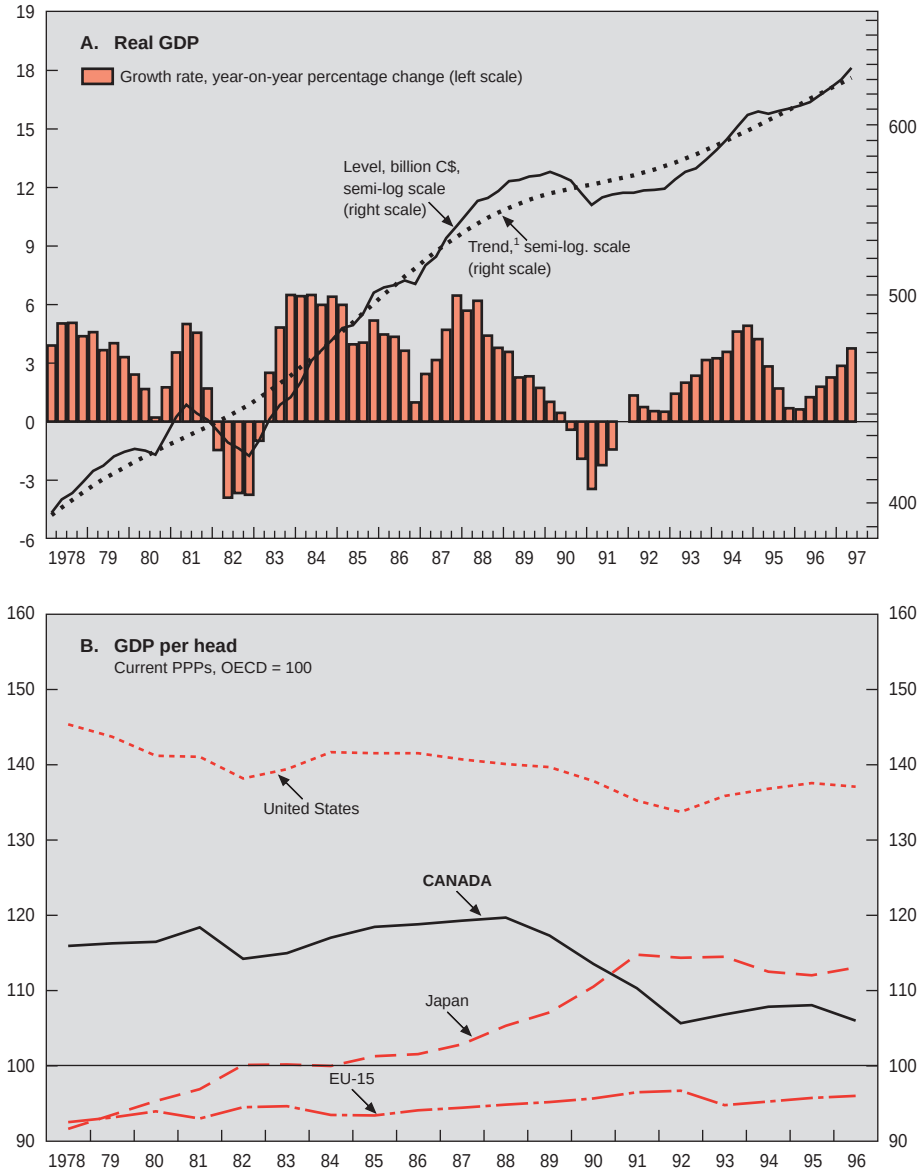
I. Recent trends and prospects

The current economic cycle in perspective

The Canadian economy is now in its sixth consecutive year of growth and activity has been quite strong in recent quarters. At the same time, inflation has remained low and the external current account has moved towards broad balance. This favourable performance needs to be seen in perspective, however. Recovery from the recession of the early 1990s has been relatively slow, with a spurt in activity in 1993-94 being short-lived (Figure 1). Trend growth has flattened markedly, and per capita real GDP has hardly risen in the 1990s. As a result, the latter has declined significantly relative to an OECD benchmark (from around 120 per cent in the late 1980s to about 107 per cent); relative to the United States, Canada's per capita income is now about 23 per cent lower, as compared with around 15 per cent ten years ago.

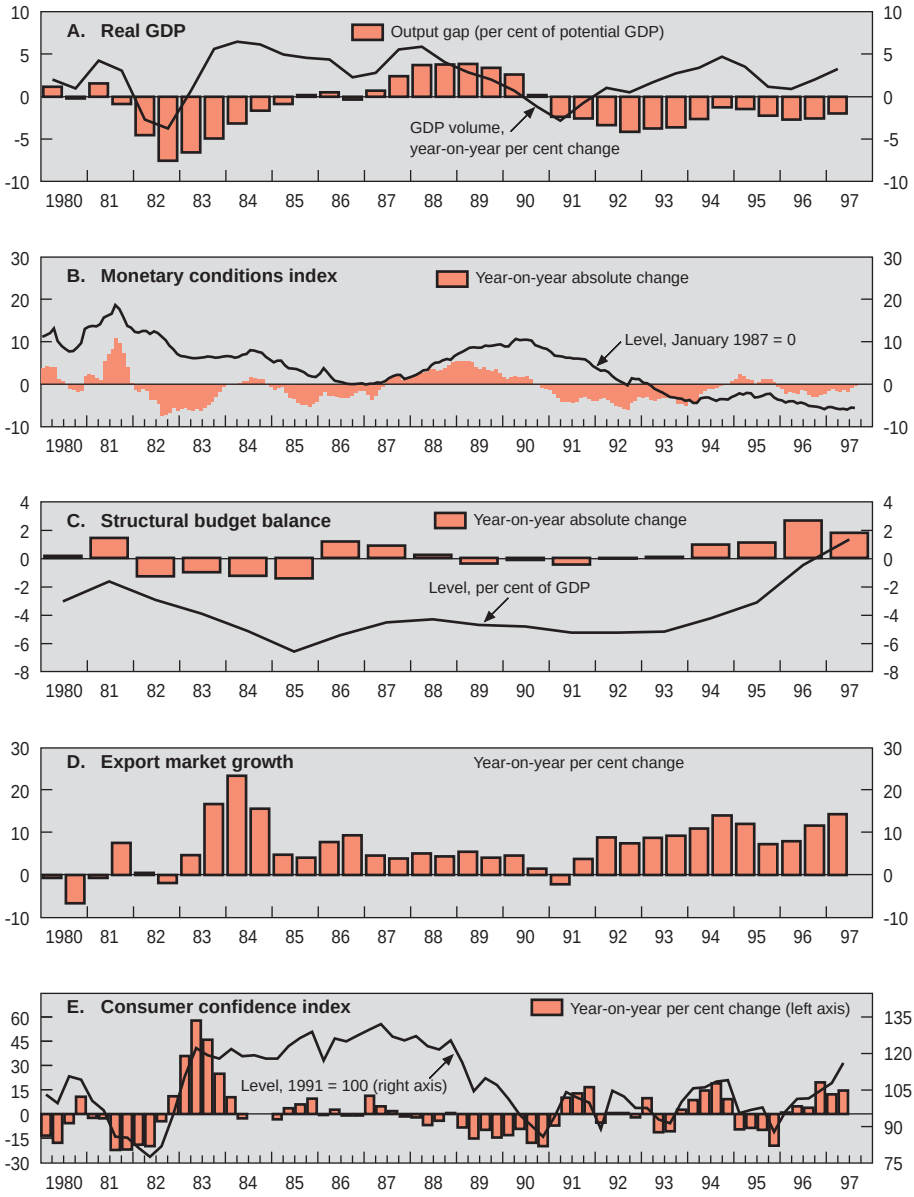
A number of factors have been put forward to explain why the Canadian economy has not fared better in recent years, despite an active programme of structural reforms over the past decade (see Chapter III). The most important appear to be extensive restructuring in response to both the reforms and increasingly competitive world markets, as well as the need to unwind the economic imbalances that had emerged since the 1970s, notably high public and external deficits and strongly rising costs and prices. Rapid disinflation in the early 1990s allowed a significant easing in monetary conditions. But this was partly reversed in 1994 owing to the turbulence in world financial markets and lingering concerns about the fiscal situation in Canada and the future of the federation. At the same time, fiscal consolidation started in earnest (Figure 2), withdrawing about 6 per cent of GDP from domestic demand since. In addition, growth in external demand slowed and consumer confidence collapsed (Figure 2), due in part to job insecurity associated with industrial and government restructuring.

Figure 1. **GROWTH PERFORMANCE OVER THE LONG TERM**



1. Hodrick-Prescott filter, $\lambda = 1\,600$.
Source: Statistics Canada; OECD, *National Accounts.*

Figure 2. **DETERMINANTS OF GROWTH**



Source: Statistics Canada, OECD Secretariat.

Although fiscal retrenchment still acts as a drag on growth, it has lessened and been outweighed by a renewed, significant easing in monetary conditions (Chapter II). Indeed, improved fundamentals (continued low inflation, narrowing public and external deficits) have allowed a marked fall in interest rates. As a result, as described in more detail in the subsequent paragraphs, the recovery has regained momentum and prospects for better economic performance over the medium term are favourable.

The recovery is regaining momentum

Following sluggish growth during 1995 and the first half of 1996, economic activity has accelerated, with real GDP expanding by more than 3½ per cent over the past twelve months. In stark contrast to developments in the first half of the 1990s, when the recovery relied heavily on exports, the recent upturn has been driven by domestic demand (Table 1) as both households and businesses responded to lower interest rates. Over the past year, the real foreign balance has actually been a drag on growth (Figure 3), given that the rise in imports exceeded that in exports. Despite above-potential growth, the output gap has remained large and unemployment high, continuing to exert downward pressure on costs and prices. With domestic demand being the main source of growth, the external current account has deteriorated somewhat.

Strengthening domestic demand

Private consumption has grown at an annual rate of around 5 per cent in recent quarters (Table 1) as lower interest rates are stimulating spending on durables. Surveys suggests that consumer confidence has improved significantly over the past year despite the fact that the unemployment rate did not start to decline significantly until mid-year and income gains have picked up quite gradually. With the economic outlook favourable and interest rates likely to have bottomed out, a record number of respondents think that it is now a good time to make a major purchase. As the pick-up in consumer spending occurred in the face of broadly stagnating real disposable income, the personal saving ratio has declined markedly, raising concerns that recent household demand levels may not be sustainable. However, as Figure 4 shows, while the saving ratio has come down sharply since the early 1980s, it is now comparable with that of the early

Table 1. **Demand and output**

Volume percentage change, annual rates

	1979/ 1973	1989/ 1979	1996/ 1989	1994	1995	1996	Q1 1997 Q4 1996	Q2 1997 Q1 1997
Private consumption	4.4	3.1	1.3	2.9	1.4	2.4	5.0	4.6
Government consumption	3.5	2.5	0.4	–1.7	–0.7	–1.8	–1.6	–0.2
Gross fixed investment	5.1	5.3	0.6	5.9	–0.1	6.4	15.2	9.3
Public	–0.3	3.8	2.6	6.4	2.7	–2.8	–0.2	5.5
Private	5.9	5.5	0.3	5.9	–0.6	7.8	17.4	9.8
Residential	3.4	4.5	–3.4	1.8	–15.1	11.5	10.1	10.6
Non-residential	7.8	6.0	1.9	7.6	5.2	6.6	20.0	9.5
Final domestic demand	**4.3**	**3.5**	**1.0**	**2.6**	**0.6**	**2.5**	**6.1**	**4.8**
Stockbuilding [1]	0.1	–0.1	–0.1	0.5	0.3	–0.8	–2.8	2.5
Total domestic demand	**4.4**	**3.4**	**0.9**	**3.1**	**1.0**	**1.6**	**3.1**	**7.4**
Exports of goods and services	4.6	5.2	7.7	14.7	12.0	4.5	23.6	2.8
Imports of goods and services	6.0	6.4	6.4	11.5	8.7	5.1	22.3	8.8
Foreign balance [1]	**–0.2**	**–0.4**	**0.4**	**0.9**	**1.2**	**–0.2**	**0.1**	**–2.7**
Error of estimate [1]	0.1	0.0	0.0	0.1	0.2	0.1	0.4	0.3
GDP at market prices	**4.2**	**3.1**	**1.2**	**4.1**	**2.3**	**1.5**	**3.7**	**4.9**
Agriculture	–0.2	2.3	0.9	3.2	1.4	2.1	–9.1	–5.6
Mining	–4.1	1.0	3.2	6.1	4.3	3.3	–3.1	4.3
Manufacturing	2.5	2.1	1.0	7.6	3.8	1.3	6.1	5.7
Construction	4.5	3.4	–2.8	2.9	–5.0	–2.0	8.4	12.0
Services	4.8	3.4	1.6	3.2	1.9	1.9	3.6	4.2
GDP at factor costs	**3.5**	**3.0**	**1.3**	**4.1**	**2.0**	**1.7**	**3.6**	**4.6**

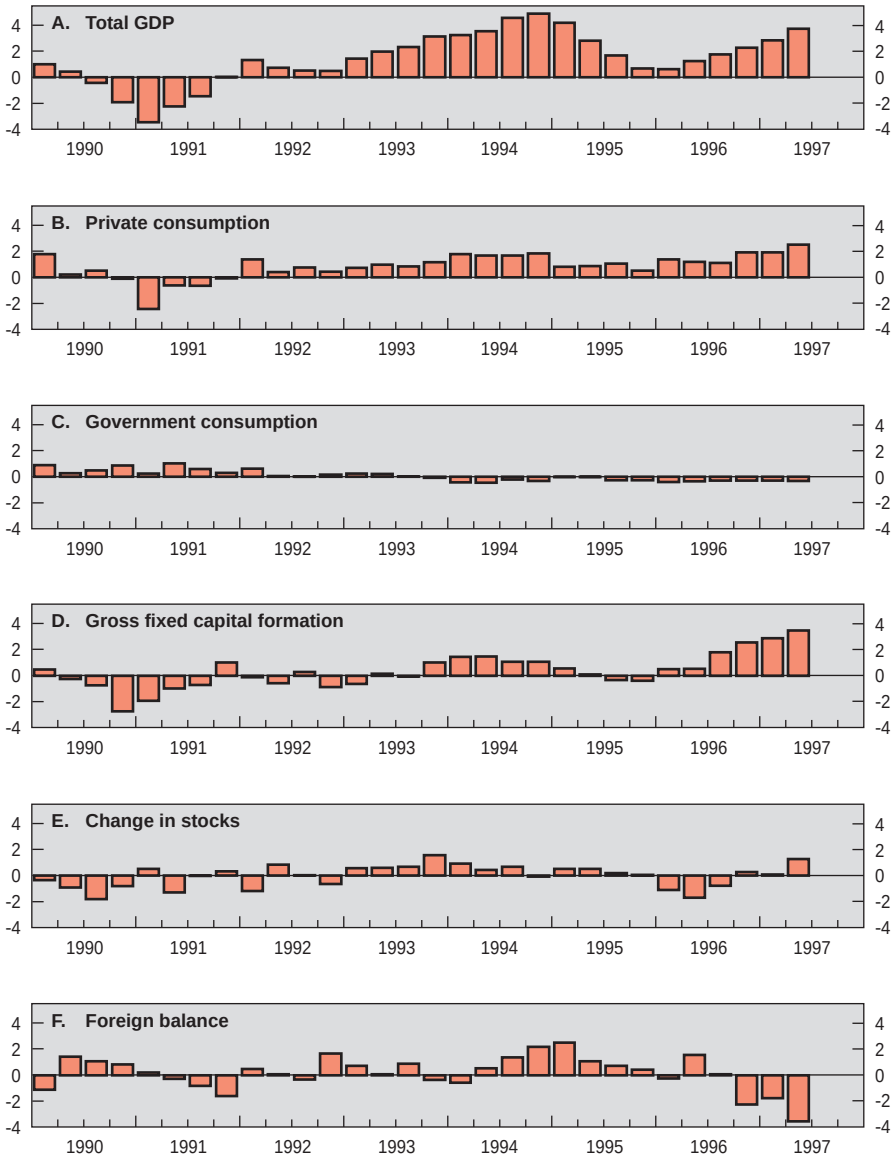
1. Contribution to GDP volume growth.

Source: Statistics Canada, *National Income and Expenditure Accounts*.

1960s when inflation and interest rates were similarly low. Moreover, the standard saving ratio does not take into account increases in asset values: once such additions to household net worth are included, the saving ratio appears much higher and has tended to rise in recent years. On the other hand, while the strong increase in households' indebtedness has been matched by the rise in the value of their assets, consumers now have less liquid assets relative to debt than in the past and may thus react more strongly to unexpected interest rate or income developments. There is also clearly the risk of an asset price reversal. Although the experience of the 1987 stock market crash suggests that consumer spending is

Figure 3. **CONTRIBUTIONS TO CHANGES IN GDP**

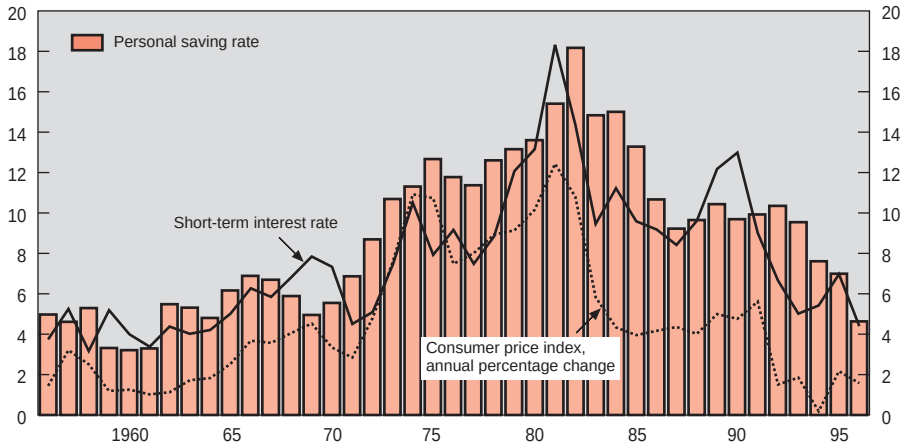
Per cent changes over 4 quarters



Source: CANSIM – Statistics Canada; OECD, *National Accounts.*

Figure 4. **HOUSEHOLD SAVING RATIO**

Per cent



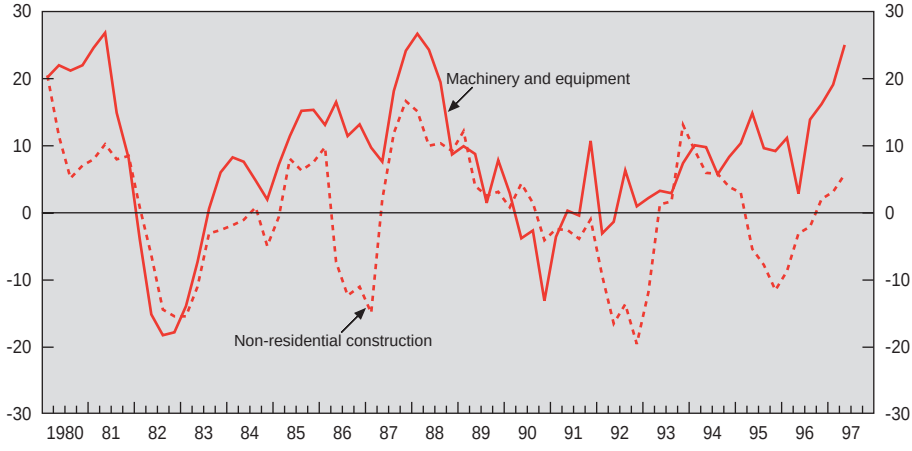
Source: CANSIM – Statistics Canada; OECD, *National Accounts.*

little affected by short-term movements in equity prices, the improved accessibility and use of mutual funds by Canadians may strengthen this relationship in the years to come.

Residential investment has recovered swiftly and vigorously in response to lower mortgage rates, growing at double-digit rates since the first quarter of 1996. However, this followed a dismal performance in 1995 and, indeed, over the current cycle (Table 1). Although mortgage rates are at their lowest levels in decades, they are not unusually low in real terms, at least at longer maturities. Together with sluggish employment and income growth, this probably explains why the level of housing activity has remained modest and generally below estimates of potential household formation.[1] In 1995-96, in particular, housing starts were considerably lower than estimated demographic requirements, suggesting the build-up of significant pent-up demand. While it is difficult to tell over what period that demand would tend to be eliminated, the fact that house prices are beginning to rise and the likelihood that mortgage rates will not fall further may now draw people into the housing market.

Business fixed investment has been similarly buoyant in recent quarters (Table 1) as profits and capacity utilisation improved again. Reflecting companies' restructuring efforts, capital spending has continued to be concentrated in machinery and equipment, particularly computers and other office equipment. Manufacturers and telecommunications companies both have boosted investment outlays considerably. Business construction, which tends to respond more strongly to fluctuations in aggregate demand (Figure 5), has recovered too, partially recouping the declines in 1995-96. Much of the gain has been in the oil and gas industry. Investment intentions surveys, which report substantial upward revisions to previous plans, suggest that this trend will continue, as firms increasingly focus on expanding capacity. Following substantial cuts in stocks which had restrained economic growth in 1995-96, firms have begun adding to inventories again in response to higher sales (Figure 3). Stock levels appear to be in good balance: after a temporary increase, the inventory-to-sales ratio has returned to its downward trend.

Figure 5. **BUSINESS FIXED INVESTMENT**
Year-on-year volume change



Source: CANSIM – Statistics Canada.

Narrowing output gap

The gradual nature of the current recovery has meant that economic slack has persisted over an unusually long period (Figure 2). Real GDP growth exceeded its potential rate in 1993-94, but thereafter the output gap widened again, reaching 2 per cent in the first half of 1996, according to OECD Secretariat estimates. Over the past year, output has once more grown faster than the economy's productive capacity, which is estimated by the OECD Secretariat to be expanding at a rate of 2½ per cent per annum. As a result, the output gap may have narrowed to under 2 per cent. Despite the recent shift in the composition of demand, capacity utilisation is still much higher in export- than domestic-oriented industries. Indeed, some exporters seem to have faced capacity constraints, although strong investment raised output capacity in the export sector by almost 8 per cent in 1995-96 (as compared with a rise of 2½ per cent in domestic-oriented industries).

Still high unemployment

Following some improvement in the labour market situation in 1993-94, little further inroads were made into unemployment until mid-1997. Since then, unemployment has resumed its downward trend (Table 2) but stayed well above its structural rate,[2] estimated by the OECD Secretariat to have fallen slightly in recent years to a range of 8 to 8½ per cent in response to reforms to the unemployment insurance system and social assistance programmes. One factor that has kept unemployment high is faster growth in the labour force, which is now expanding broadly in line with the working age population. As a result, the fall in the labour force participation rate observed during the first half of the 1990s has come to a halt. The growth of employment decelerated much less than that of output in 1995-96, possibly because firms were surprised by the duration of the ''growth pause'' and hence maintained their workforce, with a consequent marked deterioration in productivity performance and some pressure on profit margins. This explains why employment was relatively slow to respond to the pick-up in activity from mid-1996. Only in recent months has job creation started to regain momentum.

Overall employment has been strongly affected by public sector downsizing. After a sizeable run-up at the beginning of the decade, the trend in public sector employment began to decline significantly in late 1994, as various governments

Table 2. **Labour supply and demand**

Per cent change, annual rates

	1979/ 1973	1989/ 1979	1996/ 1989	1994	1995	1996	Aug.1997/ Aug.1996
Working age population	2.2	1.4	1.6	1.5	1.4	1.4	1.4
Labour force	3.2	2.0	1.0	1.1	0.7	1.5	1.6
Employment	2.9	2.0	0.6	2.1	1.6	1.3	2.1
Goods producing sector	1.6	0.6	–1.0	3.0	3.3	0.6	1.7
Service sector	3.6	2.6	1.2	1.8	1.0	1.5	2.3
	1974-79	1980-89	1990-96	1994	1995	1996	August 1997
Unemployment rate [1]	7.2	9.4	10.1	10.4	9.5	9.7	9.0
Participation rate [2]	61.7	65.8	65.8	65.3	64.8	64.9	64.9
Employment ratio [2]	57.3	59.7	59.1	58.5	58.7	58.6	59.1
	1979/ 1973	1989/ 1979	1996/ 1989	1994	1995	1996	
Memorandum items:							
Labour productivity	1.3	1.1	0.6	1.9	0.7	0.2	
Total factor productivity [3]	0.8	0.5	–0.1	2.2	0.4	0.4	

1. Per cent of labour force.
2. Per cent of population 15 years and over.
3. Business sector.

Source: Statistics Canada.

and government-owned enterprises started the process of restructuring. Since then, employment in this sector has fallen by 6 per cent. While most of the planned federal cuts have already occurred, adjustments at the provincial level have yet to be fully implemented. Certain services have moved to the private sector, together with some of the employees involved. In addition, it is probable that many of those who left the public sector have become self-employed, helping to boost the growth of this category, in particular in business and personal services. Self-employment also accounts for most of the job creation in trade, transportation and communications.

Another notable feature of labour market developments is the marked deterioration in job opportunities for youths. Notwithstanding some pick-up in recent months, the employment ratio for 15 to 24 year olds has fallen by 11 percentage points over the 1990s. In addition, almost half of those who do find jobs are now working part-time, compared with one-fifth seven years ago, although many of them would prefer a full-time job.[3] However, about one-third of the sharp fall in

youth labour force participation reflects a strong rise in school attendance (see Chapter III of the 1996 Survey). This is not surprising, given the marked preference of employers for the more educated and skilled worker: in 1996, for instance, employment among post-secondary graduates rose by almost 3 per cent (more than double the average increase); at the same time, demand for workers with less than a high school diploma dropped by 3½ per cent.[4]

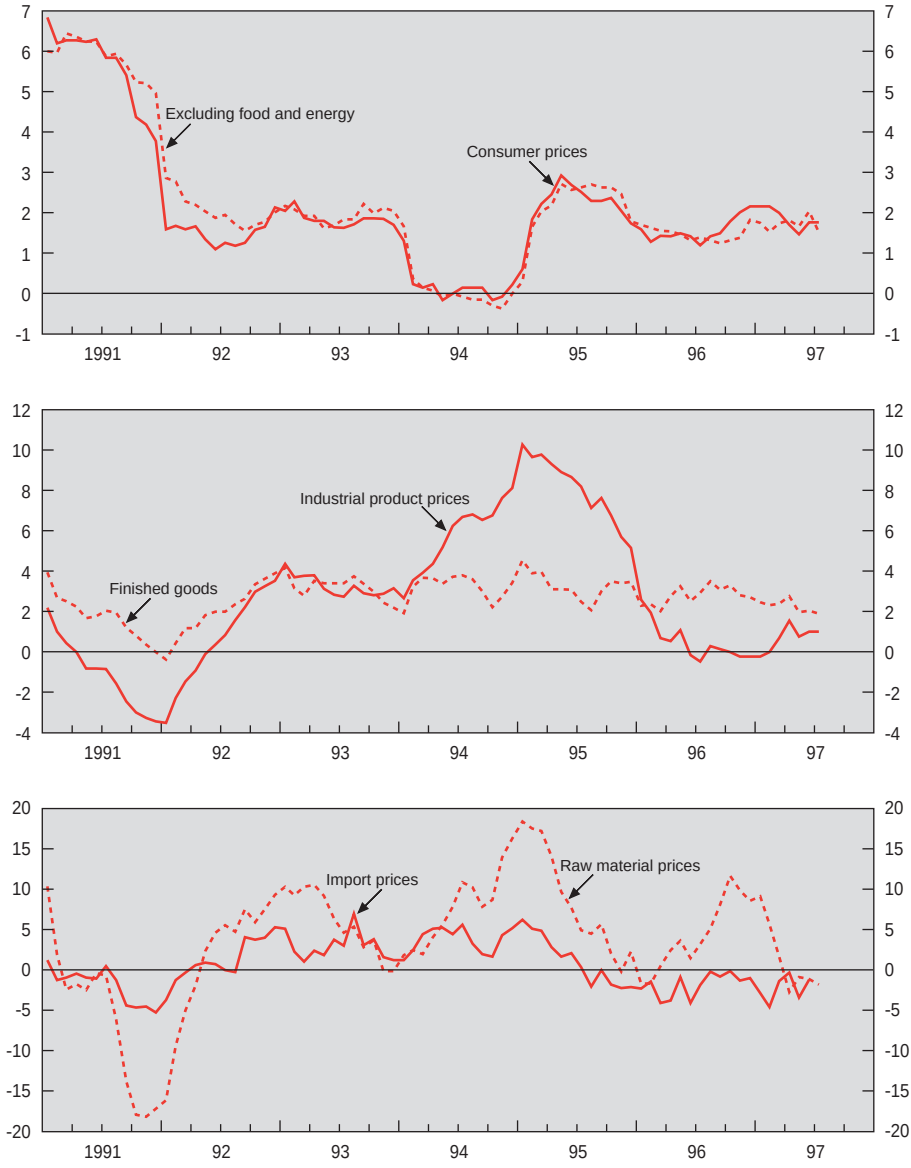
Low inflation

With a persistent large output gap, inflation has remained low. The twelve-month increase in the consumer price index (CPI) picked up temporarily in late 1996 to reach around 2 per cent before falling back more recently. To a large extent, this resulted from fluctuations in food and energy prices (Figure 6). The Bank of Canada's measure of core inflation, which in addition to these two items also excludes the effect of indirect taxes, has remained within the lower half of the official 1 to 3 per cent target range, measuring 1½ per cent in August 1997. Some of the limited rise in core inflation since mid-1996, when it approached the lower bound of the target band, reflects efforts by firms to raise depressed profit margins in the face of the strong recovery in household demand. Producer prices of finished goods have shown diminishing increases over the past year.

Given continued significant labour market slack (that is, unemployment in excess of its structural rate), wage increases have remained moderate. Still, historical relationships would have suggested even weaker wage growth in response to this cyclical unemployment, prompting a debate as to whether or not nominal wage rates are downwardly rigid at low rates of inflation (see Chapter II). Over the past two years or so, nominal wage gains have tended to rise somewhat (Table 3). However, after accounting for inflation, the increase in compensation per employee in 1996 just offset the decline in the preceding year. Moreover, to a considerable extent, recent income gains have reflected rising overtime work and shifts in the composition of the workforce. Excluding such factors, underlying wage increases are estimated to have been in a range of 1½ to 2 per cent, broadly in line with collective settlements, implying little change in real terms. The upward trend in settlements has been largely due to developments in manufacturing, in particular the automotive industry.

Figure 6. **INFLATION INDICATORS**

Percentage change over 12 months



Source: CANSIM – Statistics Canada.

Table 3. **Wages, prices and profits**

Percentage change over corresponding period

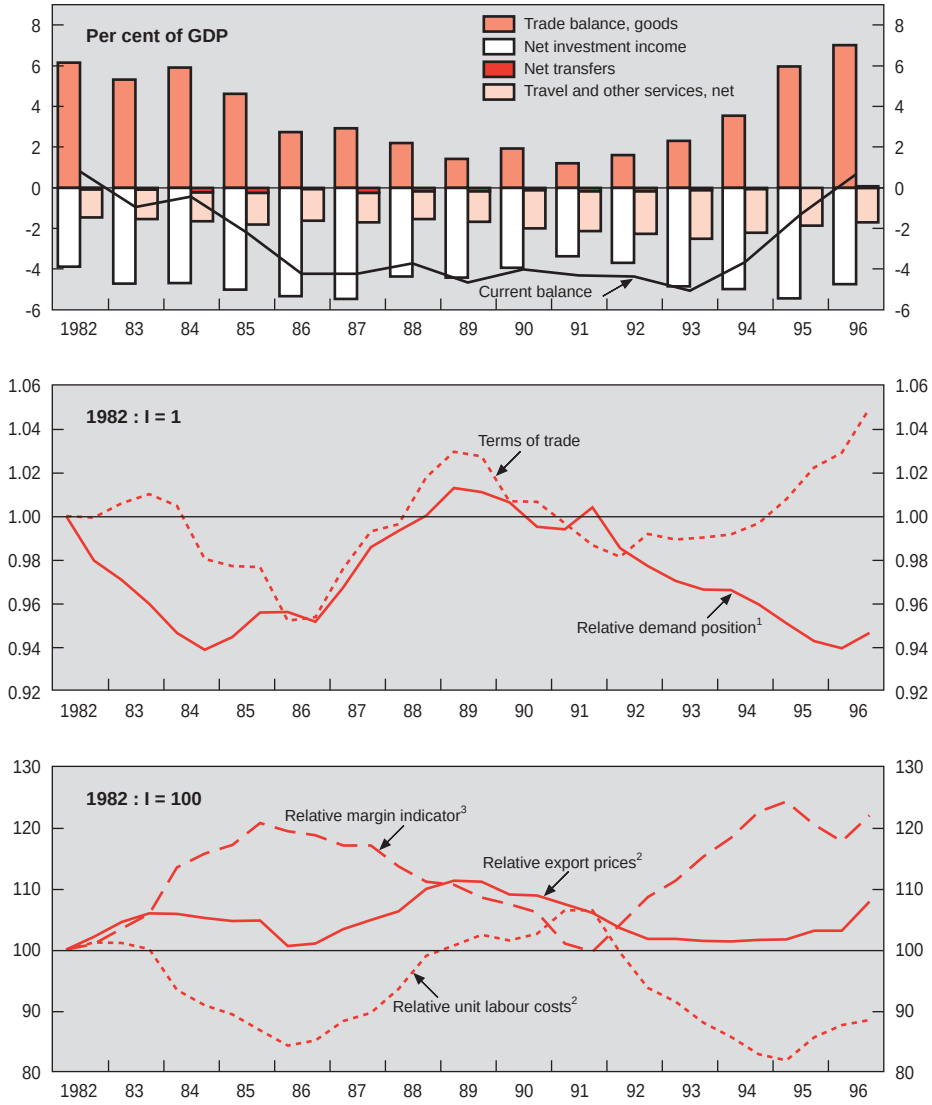
	1979/ 1973	1989/ 1979	1996/ 1989	1994	1995	1996	Q1 1997/ Q1 1996	Q2 1997/ Q2 1996
Wages								
Compensation per employee	10.8	6.8	2.8	0.7	1.4	2.5	4.5	4.6
Hourly earnings in manufacturing	11.6	6.3	3.0	1.6	1.4	3.2	3.3	1.8
Major collective settlements		6.4	2.1	0.4	0.9	1.0	1.5	1.9
Unit labour costs								
Total economy	9.3	5.6	1.7	–1.3	0.8	1.3	0.9	0.5
Manufacturing	9.4	4.9	0.6	–2.5	0.3	3.7	1.0	1.0
Profits								
Pre-tax	14.8	5.3	0.9	34.4	13.1	0.2	23.2	25.1
After-tax	15.2	4.0	0.9	46.7	16.9	–1.9	26.3	28.9
Per unit of output	10.6	1.9	–0.5	27.7	10.4	–2.2	18.4	18.9
Prices								
GDP deflator (current weights)	9.2	5.7	1.7	0.7	1.5	1.3	1.7	0.9
GDP deflator (fixed weights)	9.8	6.0	2.0	0.8	2.5	1.5	1.4	0.8
Consumer price index	9.2	6.5	2.5	0.2	2.2	1.6	2.1	1.6
Private consumption deflator	8.6	6.1	2.2	0.7	1.6	1.2	1.7	1.5
Import price deflator	10.7	2.0	1.4	4.0	1.3	–1.3	–2.3	–1.0
New housing price index			–1.1	0.1	–1.2	–2.0	–0.1	0.9

Source: Statistics Canada; OECD, *National Accounts*.

Broad external balance

The external current account has improved markedly in recent years, moving into a small surplus in 1996 for the first time since the early 1980s (Figure 7). This reflected favourable developments in Canada's terms of trade as well as the country's relative cyclical position and restored international competitiveness. The improvement in the terms of trade has been the result of stronger commodity export prices and a compositional shift among imports towards (lower-priced) office machinery. Weaker economic growth in Canada than in the United States – its major trading partner – has tended to strengthen the real trade balance, favouring exports relative to imports. Owing to both exchange rate depreciation and low domestic inflation, Canada's competitive position – as measured by unit labour costs on a common-currency basis – has greatly improved and rarely been better in the post-war period. However, these factors have become less supportive of trade performance more recently: the rise in the terms of trade has petered out;

Figure 7. **THE CURRENT BALANCE AND ITS MAJOR COMPONENTS**



1. Index of Canadian final domestic demand divided by an export share weighted index of final domestic demand of major trading partners.
2. In manufacturing. Common currency.
3. Ratio of relative export prices to relative unit labour costs.

Source: Statistics Canada, OECD Secretariat.

the pick-up in activity in Canada has led to a less favourable relative demand position; and cost and price competitiveness has weakened a little since 1995, partly due to a stronger Canadian dollar. These developments have caused the current account to swing back to a deficit position.

The recent deterioration in the current account, like the previous improvement, has largely reflected movements in the merchandise trade balance (Table 4). Apart from less favourable terms of trade, the decline in the trade surplus in recent quarters has resulted from a rebound in import volumes associated with the strengthening of domestic demand, in particular for machinery and

Table 4. **Balance of payments**

C$ billion, annualised

	1993	1994	1995	1996	1997 Q1	1997 Q2
	Seasonally adjusted					
Goods exports	190.4	227.9	264.9	280.6	295.5	293.6
Goods imports	177.6	208.6	231.2	239.6	263.2	271.8
Trade balance	12.8	19.3	33.7	41.0	32.3	21.8
Travel and other services, net	–13.4	–11.8	–10.4	–9.9	–8.8	–8.3
Investment income, net	–26.9	–27.2	–30.7	–27.7	–26.4	–25.9
Transfers, net	–0.6	–0.3	–0.0	0.4	–0.5	–0.3
Current balance	–28.0	–20.1	–7.4	3.8	–3.4	–12.7
	Not seasonally adjusted					
Current balance	–28.0	–20.1	–7.4	3.8	–13.7	–17.6
Capital account	10.7	10.2	6.8	8.1	6.3	6.3
Canadian assets (net flows)	–27.3	–48.9	–37.5	–68.3	–92.2	–29.3
Canadian liabilities (net flows)	51.3	57.3	34.1	53.3	115.9	46.0
Financial account [1]	24.0	8.4	–3.5	–15.0	23.7	16.7
Statistical discrepancy	–6.7	1.5	4.1	3.1	–16.3	–5.4
	Seasonally adjusted					
Memorandum items:						
Current balance						
US dollars billion	–21.7	–14.7	–5.4	2.8	–2.5	–9.1
Per cent of GDP	–3.9	–2.7	–1.0	0.5	–0.4	–1.5

1. A minus sign denotes an outflow of capital resulting from an increase in claims on non-residents or a decrease in liabilities to non-residents.

Source: Statistics Canada.

equipment which has high import content. However, it also reflects surprisingly weak export performance, given the robust growth experienced in the United States. The apparent loss in volume market share over the past year or so can be partly explained by the composition of US demand, major labour conflicts in the automobile sector, and possible capacity constraints in some export industries. Moreover, in contrast to previous years, export growth is no longer benefiting from exchange rate depreciation. Developments in the invisibles balance have not provided an offset to the recent deterioration in the trade account. While there has been a continued decline in interest payments owing to both reduced borrowing from abroad and lower interest rates, strongly rising profits by foreign multinationals on their Canadian operations have tended to increase the deficit on investment income.

The short-term outlook

Continued robust growth

For the remainder of 1997 and during 1998, the Canadian economy is expected to expand at a healthy pace (Table 5), at rates in excess of the estimated growth in potential output. Given the lags involved, the monetary easing that has taken place over 1995-96 (see Chapter II) should continue to underpin domestic demand. At the same time, exports are expected to play again a major role in keeping growth sustained, although the projected slowdown in the United States implies some deceleration in the expansion of Canada's markets. Low interest rates, improving consumer confidence as employment growth picks up, and pent-up demand should support household spending on housing and consumer durables, while business investment is expected to become more broadly based, spreading from machinery and equipment to construction. With an increase in labour force participation, unemployment is projected to decline only gradually, remaining above its estimated structural rate, and, notwithstanding above-potential output growth, a continuing margin of spare capacity is expected to keep inflation well within its current 1 to 3 per cent target range. The external current account may remain in a small deficit position but should cease to deteriorate as the recovery broadens to expenditure components with lower import propensities.

Table 5. **Short-term projections**

Percentage changes, seasonally-adjusted annual rates, volume (1986 prices)

	1993 C$ billion[1]	1997	1998	1997		1998	
				I	II	I	II
Private consumption	436.5	4.0	3.1	5.0	3.5	3.0	2.9
Government consumption	152.2	–1.2	0.3	–1.2	–0.2	0.4	0.5
Gross fixed investment	128.9	13.2	7.3	15.0	8.7	7.1	6.6
Public[2]	15.8	0.2	3.4	2.4	3.2	3.6	3.1
Private residential	43.1	14.4	6.7	13.2	8.2	6.3	5.9
Private non-residential	70.0	15.3	8.2	18.0	9.8	7.9	7.4
Final domestic demand	**717.6**	**5.1**	**3.6**	**6.1**	**4.1**	**3.5**	**3.4**
Change in stockbuilding[3]	1.1	0.4	0.1	–0.1	0.3	0.0	0.0
Total domestic demand	**718.7**	**5.5**	**3.7**	**5.9**	**4.3**	**3.5**	**3.4**
Exports of goods and services	209.4	7.6	7.6	9.0	8.8	7.2	7.2
Imports of goods and services	212.5	12.4	8.0	15.0	9.5	7.5	7.4
Change in foreign balance[3]	**–3.2**	**–2.1**	**–0.3**	**–2.5**	**–0.5**	**–0.3**	**–0.3**
Statistical discrepancy[3]	–2.7	0.2	0.0	0.4	0.0	0.0	0.0
GDP at market prices	**712.9**	**3.6**	**3.4**	**3.8**	**3.9**	**3.3**	**3.2**
Inflation							
GDP implicit price deflator		1.1	1.6	0.4	1.4	1.6	1.7
Private consumption deflator		1.5	1.4	1.5	1.3	1.4	1.4
Industrial production		**3.9**	**3.9**	**3.5**	**3.8**	**4.0**	**3.9**
Unemployment rate[4]		**9.2**	**8.8**	**9.5**	**9.0**	**8.9**	**8.8**
Current balance (US$ billion)		**–6.1**	**–6.1**	**–5.8**	**–6.4**	**–6.4**	**–5.9**

1. Current prices.
2. Excluding nationalised industries and public corporations.
3. Contribution to GDP volume growth.
4. As a percentage of the labour force.

Source: OECD Secretariat.

The above projections, which are based on information available in September 1997, embody the following assumptions:

– World trade expands at an annual rate of 8 per cent over the eighteen months to end-1998 while the growth of Canada's export markets drops below that rate, given a projected slowdown in US growth (to 2½ per cent in 1998).

- The average OECD crude oil import price remains broadly stable at just over $18 per barrel.
- With constant nominal exchange rates from September 1997, Canada's import prices rise at an annual rate of 1 per cent in the period ahead.
- The cyclically-adjusted general government financial balance improves by 2 per cent of GDP over the two years to 1998.
- Canadian short-term interest rates increase gradually over the projection period while long-term interest rates decline somewhat, to below corresponding US rates.

Risks and uncertainties

Given that the Canadian economy is so closely linked to that of the United States, unexpected changes in the US economic and policy environment would clearly have a substantial effect on the domestic outlook. However, recent developments do not point to significant shifts in growth prospects and/or monetary policy settings in Canada's southern neighbour. On the domestic side, risks appear to be evenly balanced. On the one hand, it might be premature to assume that a self-sustaining expansion is assured, given continued fiscal consolidation, relatively high personal debt levels, a historically low saving ratio, and still high unemployment. Although fiscal restraint is diminishing, so is the stimulus from past monetary easing, and, as discussed earlier, while high personal debt is backed by assets, liquidity constraints or asset price reversals could still lead to household retrenchment, in particular in the absence of a sustained improvement in employment and income trends. On the other hand, stronger employment growth could fuel consumer confidence and unleash pent-up demand, which could further spur activity and job creation. With such a ''virtuous circle'', the recovery could gather considerable cyclical momentum and the output gap be closed much faster than projected. This could entail inflationary pressure, calling for a stronger rise in interest rates than assumed to avoid an overshoot of the inflation target.

A medium-term scenario

Although the outlook is not without risks, Canada's medium-term economic prospects appear favourable. With much improved macroeconomic fundamen-

tals, structural reforms should begin to have the expected positive effects on economic efficiency and potential output growth. Investment in machinery and equipment, which generally embodies the most advanced technological knowledge, has been strong in recent years (Figure 5). This bodes well for future productivity trends, since such investments allow workers to take advantage of more efficient methods and techniques.[5] Moreover, there is evidence that Canadian firms are quickly catching up to the United States in the use of advanced technologies, which should also enhance productivity performance.[6] This should allow potential output growth to rise gradually from its estimated current rate of

Table 6. **A medium-term scenario**

Average annual percentage change, volumes

	1988-1996	1996-2002
Private consumption	1.5	3.1
Government consumption	0.9	0.5
Fixed capital formation	1.3	7.0
Stockbuilding [1]	0.0	0.0
Total domestic demand	1.3	3.6
Exports	6.8	7.0
Imports	6.4	7.3
Real foreign balance [1]	0.1	–0.2
Real GDP	1.4	3.4
Output gap [2]	–0.8	–0.5
Potential output	2.2	2.9
Labour productivity	0.6	1.4
Employment	0.8	2.0
Participation rate (per cent)	66.1	65.6
Unemployment rate (per cent)	9.6	8.5
Private consumption deflator	2.5	1.5
GDP deflator	2.1	1.6
Private compensation per employee	3.1	2.9
Current balance [1]	–3.1	0.1
Real exchange rate	–2.4	–0.2
Short-term interest rate (per cent)	8.0	4.5
Long-term interest rate (per cent)	9.1	6.3

1. Per cent of GDP.
2. Per cent of potential output.

Source: OECD Secretariat.

2½ per cent to perhaps 3 per cent. At the same time, the structural rate of unemployment is assumed to fall to 8 per cent in response to labour market reforms (see Chapter III).

On the basis of these assumptions, an OECD Secretariat scenario depicts real GDP growth more than doubling in the 1997-2002 period compared with the rate of 1½ per cent achieved since the late 1980s (Table 6). This would imply growth in excess of the 2 per cent projected for the OECD area (up from 2¼ per cent). Underlying this profile is a stronger expansion of domestic demand, which is expected to fall into line with longer-term trends following its weakness in recent years. With robust growth of foreign demand, the external current account would be in modest surplus. At the same time, inflation would stay at the current low levels, as slack in product and labour markets is absorbed only gradually. Needless to say there is considerable uncertainty about the exact size of the output gap and hence the pace and duration of growth required to close the gap. Moreover, the future path of potential output crucially depends not only on the assumed improvement in the productivity trend but also on the development in the labour force participation rate in coming years. Finally, improved growth performance, as depicted in the scenario, is based on the assumption that interest rates will remain low, which requires the achievement of continued low inflation and an ongoing move by the federal and provincial governments towards less vulnerable public debt positions, as discussed in the following chapter.

II. Macroeconomic management

The recent performance of the Canadian economy has greatly benefited from the policy framework put in place since the early 1990s. Determined efforts to regain control of public finances have significantly reduced fiscal deficits and brought budget balance within reach. Along with the adoption of inflation-control targets, this has contributed to a marked decline in interest rates to below the levels prevailing in the United States, thereby facilitating the resurgence of economic growth. The following paragraphs examine the progress in budget consolidation, focusing on the issue of what to do with possible fiscal surpluses once they emerge. This is followed by an overview of monetary management, with particular emphasis on the appropriate policy stance in the period ahead and the inflation target to be adopted for the period beyond 1998.

Fiscal policy

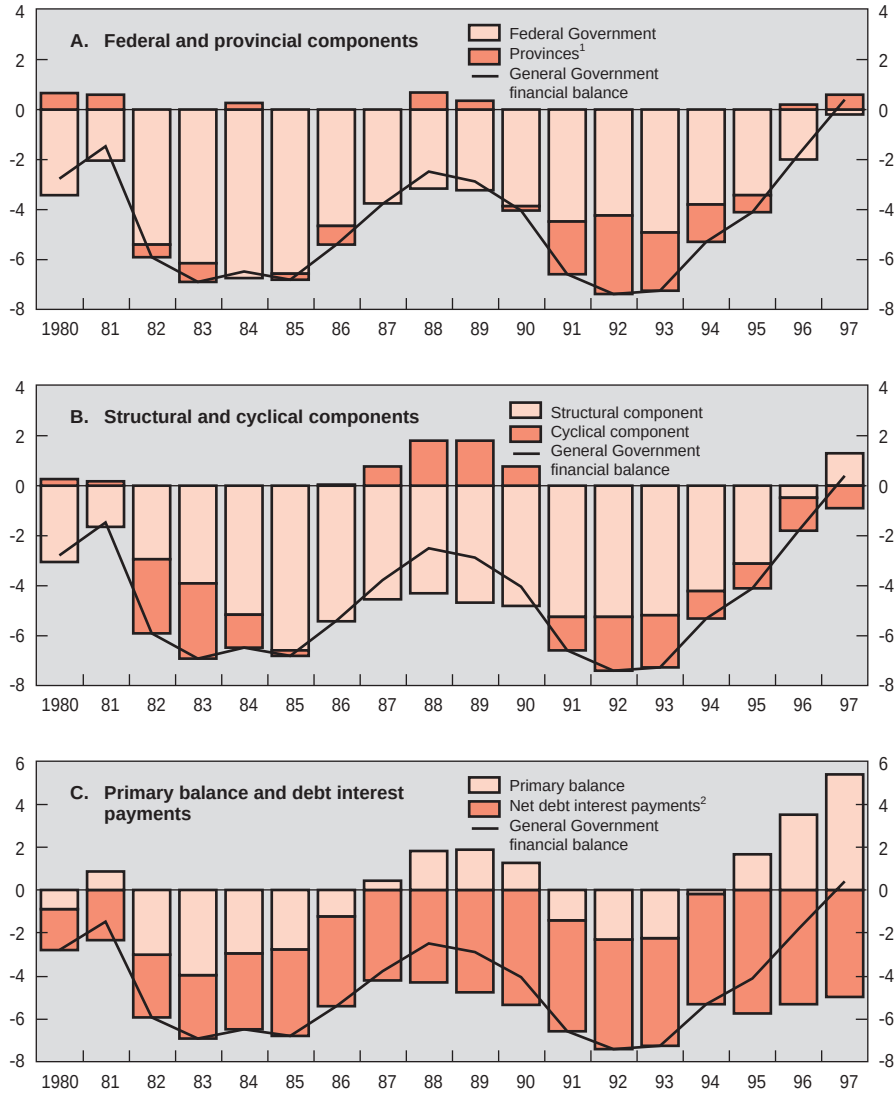
Significant progress in budget consolidation

Within the OECD area, Canada stands out as one of the countries where substantial progress in fiscal consolidation has been achieved. On a national accounts basis, general government finances have moved from a deficit of almost 7½ per cent of GDP in 1992-93 to broad balance (Figure 8). As a result of this turnaround, which largely reflects a decline in programme spending, Canada's budget position is likely to be the most favourable among the seven major economies in the current year, following more than a decade when its deficit was consistently the second highest (behind Italy).

Developments at both the federal and provincial level have contributed to this impressive improvement in the fiscal position (Figure 8, first panel). In contrast to the federal government, which was already running substantial deficits

Figure 8. **DECOMPOSITION OF THE GENERAL GOVERNMENT FINANCIAL BALANCE**

Per cent of GDP (National Accounts Basis)



Note : 1997 data are OECD Secretariat projections.

1. Including local government, hospitals, Canada and Quebec Pension Plans.
2. Interest payments are shown with a negative sign.

Source: CANSIM – Statistics Canada; OECD, Secretariat estimates.

in the 1980s, provincial finances deteriorated significantly only in the early 1990s, necessitating a sharp fiscal correction. More recently, budget consolidation at the provincial level has been made more difficult by cuts in federal transfers, although provinces in turn have reduced their transfers to local governments. Nonetheless, the latter have managed to maintain broadly balanced budgets.

According to OECD Secretariat calculations, two-thirds of the decline in the general government deficit since 1993 have been due to discretionary measures, the remainder reflecting some narrowing in the output gap. Adjusted for cyclical influences, the budget balance has moved from a deficit of over 5 per cent of potential GDP into a surplus (Figure 8, second panel). Although such a sharp tightening in the fiscal stance, which culminated at over 1½ percentage points of GDP in 1996, represented a major drag on economic activity in the short term, it had become inevitable given the explosive growth in public debt (see below). Indeed, as earlier efforts to rein in the structural deficit proved to be short-lived, the actual deficit had reached new record levels following the cyclical downturn in the early 1990s. Given the resulting rise in debt service charges, a primary surplus (excluding debt interest payments) of about 5 per cent of GDP has been required to achieve broad budget balance (Figure 8, third panel).

Federal government

With the 1994 and 1995 Budgets, the stage was set for the elimination of fiscal deficits at the federal level. They provided for direct government savings equivalent to 3 per cent of GDP by 1997/98, largely coming from spending cuts; in addition, they included ‘‘prudence factors’’ in the form of contingency reserves and projections based on conservative assumptions for the key macroeconomic variables. As a result, after repeated slippage relative to fiscal plans in the early 1990s, targets have been over-achieved since 1994 (Table 7). This tendency has become more pronounced over time. Although the final outcome for the 1996/97 fiscal year (ending in March) is not yet available, latest estimates suggest that the budget deficit was at least 1 percentage point smaller than the target of 3 per cent of GDP (public accounts definition). This significant undershoot reflects higher-than-estimated tax collections, lower-than-anticipated programme spending, lower-than-assumed interest rates leading to declining debt

Table 7. **Federal government deficit: projections and outcomes**

Public accounts basis; fiscal year (starting 1 April); per cent of GDP

	1990/91	1991/92	1992/93	1993/94	1994/95	1995/96	1996/97	1997/98	1998/99
1990 Budget	4.5	4.4	3.2	2.1	1.2	0.7			
1991 Budget	4.2	3.7	2.7	1.7	1.1				
1992 Budget		4.6	3.8	2.9	1.7	1.0	0.6		
1993 Budget			5.2	4.5	3.8	2.5	1.6	0.9	
1994 Budget				6.4	5.4	4.2	3.0		
1995 Budget					5.1	4.2	3.0		
1996 Budget						4.2	3.0	2.0	
1997 Budget							2.4	2.0	1.0
Outcome	4.8	5.1	6.0	5.9	5.0	3.7	(2.0)[1]	..	..

1. Latest estimate.

Source: Department of Finance; OECD Secretariat.

service payments, and the fact that (as in previous years) the substantial contingency reserve was not needed.

With the actions taken in 1994-95 being judged sufficient to keep the deficit on a downward track, the two most recent federal Budgets contained few new discretionary measures. Indeed, the February 1997 Budget provided for some targeted tax relief and strategic investments to encourage growth and job creation over the medium term, although the overall fiscal impact of these measures is small. Initiatives included: *i)* increased tax assistance to help students and their families cope with rising post-secondary education costs and to assist workers in enhancing their skills; *ii)* the creation of a Canada Foundation for Innovation to modernise and enhance research infrastructure at post-secondary institutions and hospitals (see Chapter III); and *iii)* substantial enrichment of the Child Tax Benefit to reduce disincentives to work and to provide greater aid to children in low-income families (see Chapter IV).

Recent developments suggest that once again the ‘‘prudence factors’’ in the current Budget will not be required. Both short- and long-term interest rates have been considerably lower than assumed and economic growth seems set to exceed Budget projections. Hence, the generous contingency reserve included in the expenditure estimates is unlikely to be needed. Combined with the better-than-expected outcome for 1996/97, this points to a significant downward revision of

deficit targets for this year and next (2 and 1 per cent of GDP, respectively, on a public accounts definition) in the Fall Fiscal Update. Financial requirements have fallen sharply, to the point that the government no longer has to raise new money on financial markets.

Actions taken by the government since late 1993 will have reduced the federal deficit in the fiscal year 1998/99 by around C$ 28 billion from what it otherwise would have been. Of this, about 90 per cent is due to expenditure reductions, with spending on programmes declining by almost 14 per cent over the five years to 1998/99 (from almost 17 per cent to less than 12 per cent of GDP). Measures include a 10 per cent reduction in transfers to other levels of government and a substantial cut in federal government employment (55 000 positions, that is 15 per cent, are eliminated). Through Programme Review exercises (see 1995 Survey), all aspects of departmental spending have been reconsidered, with a view to ensuring that resources are directed to the highest priorities.

Provincial governments

Given the negative reaction of financial markets to rising debt levels, as evidenced by worsening borrowing conditions, most provinces reacted promptly to the deterioration of their fiscal position in the early 1990s. By 1995/96, six of the ten provinces had balanced their budgets, and the others had begun to implement plans to eliminate deficits by the end of the decade. In 1996/97, four provinces (Alberta, Manitoba, Saskatchewan and New Brunswick) moved into significant fiscal surplus positions (Table 8). Nonetheless, according to latest estimates, the combined provincial deficit still amounted to 1 per cent of GDP, given that the two largest provinces (Ontario and Quebec) have been laggards in fiscal consolidation. Latest provincial budgets suggest little change in the overall fiscal position, as some provinces are planning to reduce their surpluses, broadly offsetting continued deficit reductions in others. However, most provincial budgets include prudence factors, similar to those applied by the federal government. In any case, financial markets have responded positively to recent fiscal developments: the spread between federal and provincial bond rates has narrowed, while rating agencies have upgraded their assessments of a number of provinces.

Although all the provinces are committed to budget discipline, the strategy adopted to this end differs widely. The Western provinces – albeit not highly

indebted (Table 8) – have put most emphasis on responsible fiscal management, delaying tax cuts despite the achievement of substantial surpluses (British Columbia being an exception). The Atlantic provinces have been in an intermediate position: some of them have recorded budget surpluses even with tax reductions, others have remained in deficit because of their difficult economic situation (due to the dependence on specific primary resources, notably fishing, see Box on Newfoundland in Chapter III). Among the two largest provinces still in a fiscal deficit position, Ontario – which accounts for almost half of the provincial debt – has decided to cut simultaneously expenditure and income taxes, implying an only gradual reduction in deficits which are to be eliminated by 2000/01. By contrast, although it has been the last province to adopt a fiscal consolidation course, Quebec expects to balance its budget a year earlier, ruling out overall tax cuts (a reform of the personal income system will be financed largely by an increase in indirect taxes).

Table 8. **Provincial deficits and debt**

Public accounts basis; fiscal year (starting 1 April); percentages

	Proportion of total provincial expenditure	Ratio of deficit/surplus (+/–) to provincial GDP		Ratio of net debt to provincial GDP	Proportion of total provincial net debt
	1995/96	1996/97	1997/98 [3]	1995/96	1995/96
Ontario	35.6	2.3	1.9	32.3	46.0
Quebec	25.9	1.9	1.2	35.2	28.0
British Columbia	13.7	0.4	0.2	11.9	5.5
Alberta	8.8	–2.8	–0.2	13.6	5.3
Manitoba	3.5	–1.3	–0.1	26.0	3.1
Saskatchewan	3.1	–1.4	–0.1	31.0	3.5
Nova Scotia	2.8	0.0	0.0	46.4	4.0
New Brunswick	2.7	–0.8	–0.2	34.1	2.5
Newfoundland	2.1	0.3	0.2	48.6	2.2
Prince Edward Island	0.5	0.3	0.6	39.3	0.4
Total [1, 2]	**100.0**	**1.0**	**1.0**	**28.4**	**100.0**
Memorandum item:					
Federal government		2.0	2.0	74.0	

1. Including territories.
2. Total provincial deficit/debt as a proportion of national GDP.
3. Budget projections.

Source: Department of Finance; OECD Secretariat.

Outlook and longer-term policy options

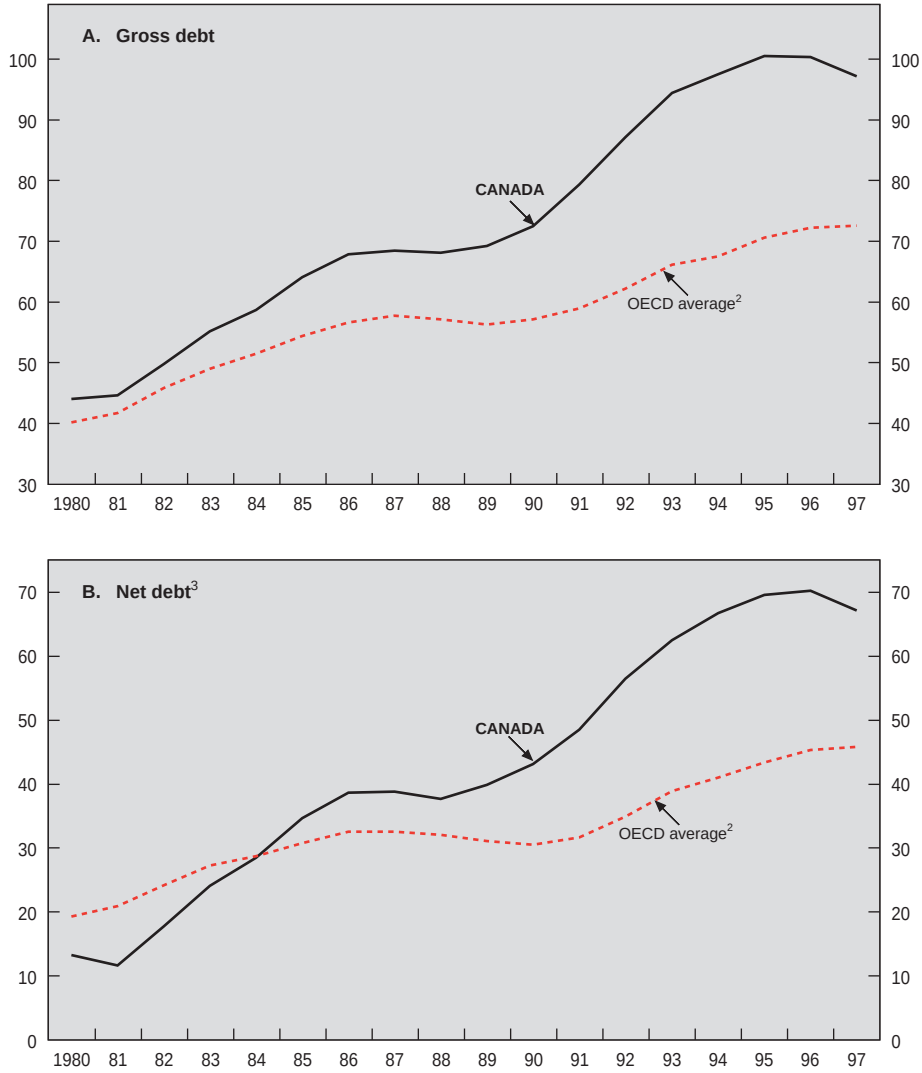
As the objective of balancing the budget over the medium term is likely to be attained by the federal government over the next year or so and has already been achieved by some provinces, designing a new longer-term fiscal strategy is necessary. Indeed, the public debate in Canada has already centred on the question of what to do with any emerging budget surpluses. Should they be maintained to achieve declines in the public debt, or should such a ''fiscal dividend'' be used to finance spending increases and/or tax reductions?

Being in the lead on budget consolidation, Canadian provinces have contributed to this debate through recent legislated restraints on deficits, debt accumulation, and tax increases. A majority of provinces and territories now have a balanced budget legislation in place, ranging in coverage from New Brunswick's four-year cumulative balanced budget requirement to Manitoba's annual balanced budget requirement that also includes non-compliance penalties through ministerial salary cuts (of up to 40 per cent). In the Yukon territory there is an election trigger if any debt is accumulated. Several jurisdictions have legislated debt repayment plans. In general, these require the reduction or elimination of debt over an extended period of time. In Alberta, however, debt retirement, originally scheduled for a 25-year period, has been accelerated to a nine-year period, with annual minimum debt repayments. British Columbia's Financial Management Plan does not schedule any specific paydowns of debt but incorporates targets for a reduction in the debt-to-GDP ratio. Finally, in a number of provinces a referendum is required to decide on major tax increases.

It remains to be seen how these rules will perform over the medium term with business cycles and changes in government. Moreover, while they may help reduce borrowing costs for Canadian provinces, international experience suggests that they are not necessarily optimal for national governments because they may introduce too much fiscal rigidity and create incentives to reduce transparency. For the federal government, a framework along the lines of New Zealand's Fiscal Responsibility Act, for example, might be more appropriate. The latter calls for generating budget surpluses until public debt has been reduced to ''prudent levels'', and budget balance over the cycle thereafter. This allows some flexibility for economic fluctuations while giving priority to the debt issue. Indeed, while recent declines in both federal and provincial deficits are impressive, Canada's net public debt-to-GDP ratio is very high both by historical and international

Figure 9. **GOVERNMENT DEBT**[1]

Per cent of GDP



1. General government; National Accounts definitions. 1997 data are OECD projections.
2. Weighted average using 1991 GDP weights and exchange rates.
3. Financial liabilities less financial assets.

Source: OECD, *National Accounts;* Secretariat estimates.

comparison (Figure 9). In 1996, it was the third highest among OECD countries, exceeding the area average by 25 percentage points (see last year's Survey for a more detailed discussion). Hence, priority should be given to putting the debt ratio on a clear downward path, with a view to further lowering risk premia on Canadian debt, reducing the country's vulnerability to economic shocks, and providing additional resources for the private sector. Such an approach need not exclude some additional spending in priority areas, such as human capital development and innovation, nor does it rule out targeted adjustments to the tax system.

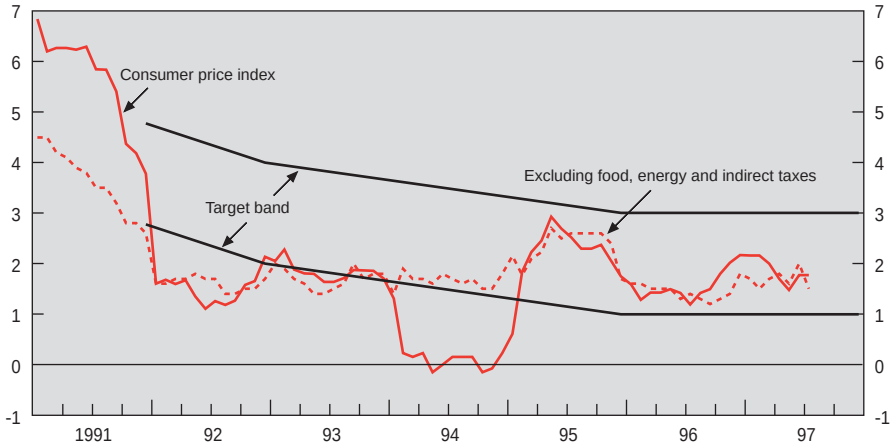
Monetary policy

The framework

Canada was the second country (after New Zealand) to introduce explicit targets for inflation control. The motivation for using this approach was to make the goal of monetary policy more concrete by committing the authorities to a well-defined medium-term strategy and to anchor inflation expectations. The first set of targets, which was jointly announced by the Government and the Bank of Canada in early 1991, aimed at reducing inflation to a range of 1 to 3 per cent by the end of 1995. Subsequently, this target band has been extended until the end of 1998 (Figure 10). While the inflation-control strategy is couched in terms of the twelve-month increase in the overall Consumer Price Index (CPI), in the short run the Bank focuses on a measure of core inflation which excludes from the headline CPI certain volatile components (food and energy) and temporary effects of indirect taxes.

So far, the authorities have been successful in keeping core inflation within the target ranges, apart from a temporary undershoot following the inception of the new framework (Figure 10). Since inflation-control targets have been in place, inflation has averaged about 2 per cent, as compared with 6½ per cent in the 1980s. While the persistence of economic slack has played a key role in bringing inflation down, the existence of the targets themselves has probably also contributed to better price performance by providing a great deal of discipline to, and transparency in, the conduct of monetary policy and thereby lowering inflation expectations. These expectations, which exercise an important influence on

Figure 10. **INFLATION TARGETS**
Percentage change over 12 months



Source: CANSIM – Statistics Canada; Bank of Canada.

the future path of inflation, are affected both by actual price developments and by the credibility of the Bank's inflation-control targets. As credibility grows, inflation expectations should increasingly reflect the mid-point of the target range. Based on the Consensus Forecast, the expected rate of inflation has indeed gradually declined from 4 per cent at the beginning of the 1990s to around 2 per cent more recently. Another measure often used to proxy inflation expectations is the difference between nominal and indexed bond yields. This differential has decreased as well but is still at around 3 per cent.[7] Although long-term interest rates reflect a number of factors (public debt, political uncertainty), the steepness of the yield curve (see below) might also suggest that the credibility of low inflation over the long term has not yet been fully established.

To provide a firmer grounding for low inflation expectations in financial markets, over the last few years the Bank has taken a number of measures aimed at enhancing the transparency of monetary policy, including:

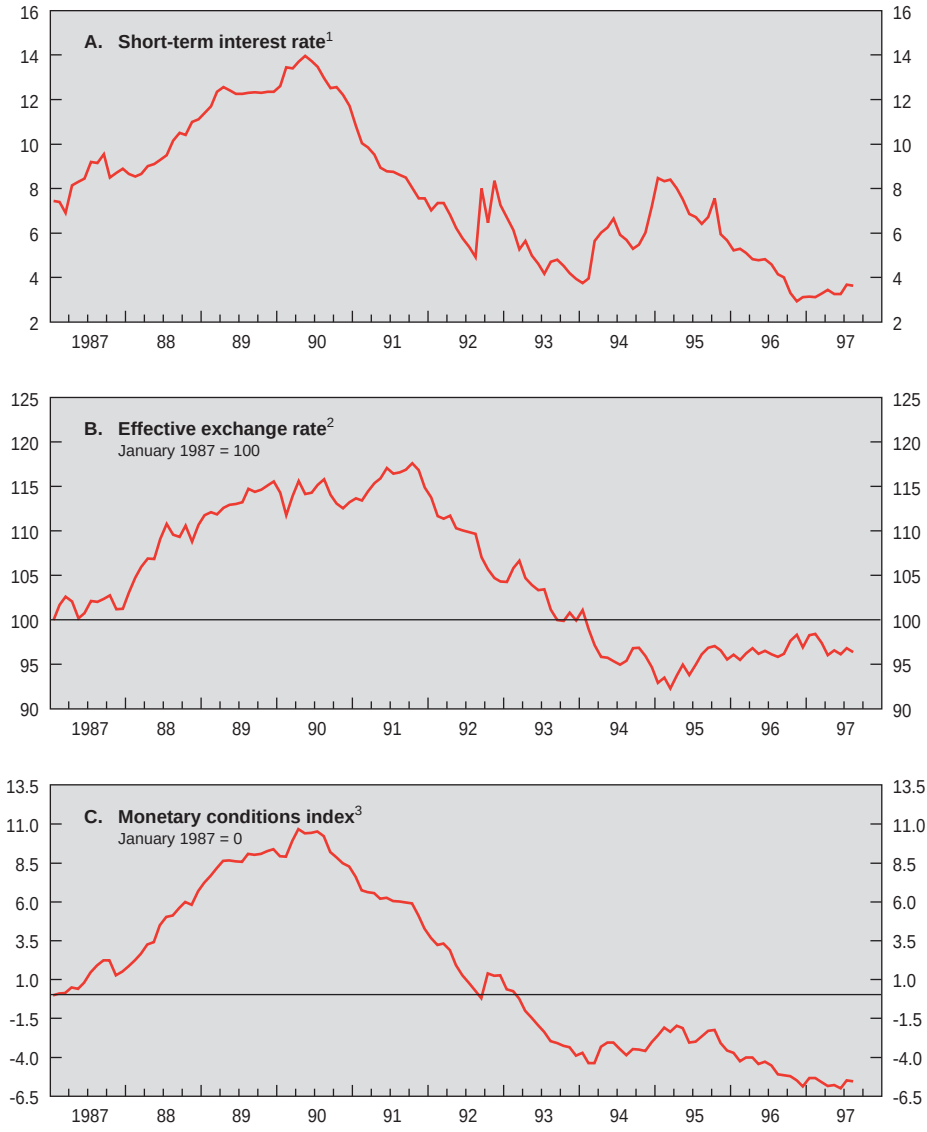
- Initiating the semi-annual Monetary Policy Report.
- Publishing its Monetary Conditions Index (MCI), a short-hand measure of the stance of policy which combines movements in the effective exchange rate and short-term interest rates, and whose desired path is constantly reassessed in the light of new information to ensure that it remains consistent with the achievement of the inflation objective.
- Publishing estimates of the output gap.
- Regularising the appearance of the Governor in front of the House Finance Committee.
- Issuing press releases outlining the reasons for changes in the target bands.
- Initiating a regional outreach programme for the Governing Council.
- Enhancing the Bank's regional presence.
- Introducing a 50-basis point wide target range for the overnight money market financing rate, with the Bank Rate set at the upper limit of the operating band.
- Using ''asymmetric'' foreign exchange intervention as a ''signalling tool''.

These measures should help market participants understand the stance of monetary policy better, making policy actions more predictable, which should add to the credibility of monetary policy.

Recent changes in monetary conditions

From mid-1995, the Bank of Canada moved decisively to loosen monetary conditions (Figure 11) given the strong fiscal adjustment and sluggish aggregate demand. Indeed, with the output gap widening again, there was a risk that inflation might fall below the target range had the Bank not provided more monetary stimulus to the economy. The task of the authorities was facilitated by increasing credibility of the inflation-control targets and the improvement in the government's fiscal position, which allowed a decline in interest rates to below US levels without downward pressure on the exchange rate. In contrast, in previous years, credibility problems and associated financial-market volatility had often put a constraint on the implementation of monetary policy. By late 1996, monetary conditions, as defined above, had fallen to their level of greatest ease in recent decades.

Figure 11. **MONETARY CONDITIONS**



1. 3-month corporate paper rate.
2. Canadian dollar against G10 currencies.
3. Weighted average of changes in the interest rate and in the effective exchange rate.

Source: Bank of Canada.

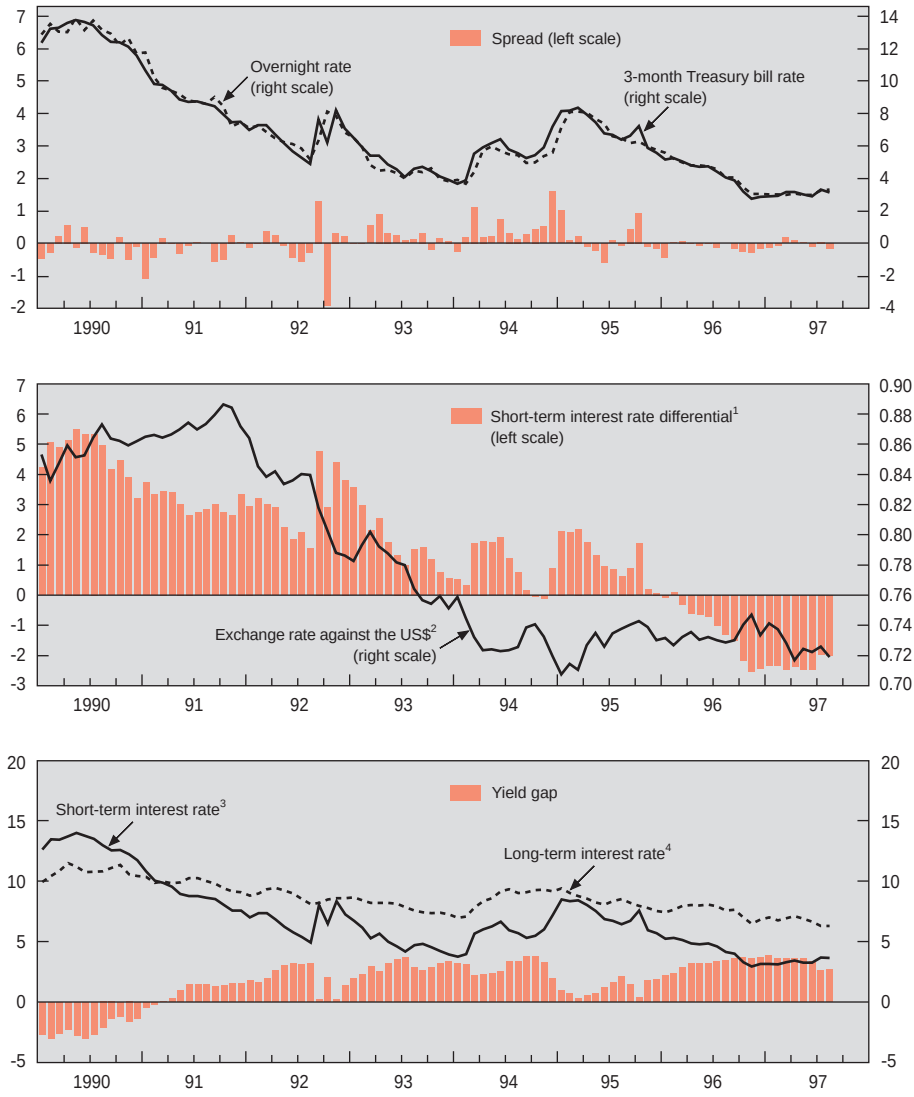
The easing of overall monetary conditions was largely achieved through a decline in interest rates, as the effective exchange rate tended to appreciate in the face of a strengthening in the external position (Figure 11). From May 1995 to November 1996, the Bank lowered its target for the overnight financing rate 20 times by a quarter point to a range of 2¾ to 3¼ per cent, a cumulative reduction of 500 basis points. Except during the run-up to the Quebec referendum on sovereignty in October 1995, markets tended to anticipate the Bank's actions, as evidenced by the negative spread between the three-month treasury bill rate and the overnight rate (Figure 12). This reflected increased policy credibility, with financial markets recognising that lower interest rates were consistent with Canada's fundamentals. By late 1996, Canadian short-term interest rates had fallen 250 basis points below corresponding US rates (Figure 12). The only previous occasion in recent history when this occurred for an extended period was from 1969 to 1973, when Canadian inflation was low and the current account position strong.

Since the autumn of 1996, the Bank has sought to keep monetary conditions broadly unchanged, as there have been increasing signs that the economy is responding to lower interest rates. Indeed, the last few reductions in the overnight rate target in October and November were intended to prevent monetary conditions from tightening as the Canadian dollar strengthened. In early 1997, the Canadian dollar weakened *vis-à-vis* the surging US dollar but showed considerable strength against the other major currencies, remaining broadly stable in effective terms. The tightening of monetary policy in the United States in late March resulted in higher Canadian short-term interest rates and in a weaker Canadian dollar. As the MCI stayed in its recent range, the authorities at first left the overnight rate target unchanged. In late June 1997, however, when increasing downward pressure on the Canadian dollar pushed the MCI below its previous low recorded at the end of 1996, the Bank raised its operating band for the overnight interest rate to 3 to 3½ per cent, considering that the economic situation did not warrant a further easing in monetary conditions.

Although interest rates have declined across the maturity spectrum, the yield curve has steepened considerably (Figure 12). It is true that improved fundamentals (lower inflation, declining public sector borrowing, broad external balance) have led to a significant decrease in the risk premia that investors require on Canadian assets. However, while Canada/US interest differentials from short-

Figure 12. **INTEREST RATE AND EXCHANGE RATE DEVELOPMENTS**

Per cent



1. *Vis-à-vis* the United States.
2. US$ per Can. $.
3. 3-month corporate paper rate.
4. Over-10-year Government bond yield.

Source: CANSIM – Statistics Canada; OECD Secretariat.

term to ten-year maturities have turned strongly negative, longer-term Canadian interest rates have fallen below corresponding US rates only recently.

The challenges ahead

While the new policy framework has proved successful, it has not yet been tested over a full economic cycle. With the economy gathering momentum, the challenge for policy makers in the period ahead will be to encourage monetary conditions that both support a durable expansion and preserve Canada's low trend rate of inflation. Moreover, a decision on the inflation target beyond 1998 will have to be made.

Should the monetary stance be tightened?

With the introduction of inflation targets, monetary policy is forward-looking, relying heavily on forecasts and other indicators of future inflation. This is because policy actions by the Bank of Canada affect inflation with a long lag (of one to two years), so that emerging pressures must be recognised at an early stage if a breach of the target is to be avoided. In the Bank's currently used model, inflation is determined by expectations and the deviation of output from its potential level (with excess demand assumed to be more inflationary than excess supply is deflationary). According to the OECD Secretariat's projections, the output gap will narrow markedly but not be closed by the end of 1998. However, estimates of potential output are difficult to make and have a wide margin of error around them.[8] Moreover, the momentum of the recovery might well prove stronger than currently projected by the OECD Secretariat, leading to a closing of the output gap before the end of 1998. On balance, given the current economic outlook and the fact that monetary conditions are presently very stimulative, a return to neutral monetary conditions will be required. Although there does not seem to be a pressing need for a rapid tightening at this time, a move in this direction might prove necessary to prevent inflation from overshooting the target, all the more so because in the presence of ''asymmetric'' inflation effects corrective action may be very costly in terms of output losses once excess demand emerges (see Annex II of the 1995 OECD Survey of Canada).[9]

Given the uncertainties surrounding estimates of the output gap, the Bank also relies on a broad range of other economic and financial indicators to arrive at

a judgement as to whether the policy stance needs to be changed, in particular monetary and credit aggregates (Table 9). Broad definitions of money have been historically correlated with near-term inflation. Nonetheless, they have become more difficult to interpret over the past five years or so because, with the strong expansion of mutual funds, holders of deposits have been switching on a large scale into equity and bonds. Even so, the generally moderate growth in broad monetary aggregates would rather suggest that inflation will continue to be subdued. The most preoccupying development has been the rapid increase in narrow money in recent quarters. In large part, this reflects the adjustment of money balances to a low-interest environment made possible by low inflation. Econometric estimates at the Bank suggest that the stock adjustment process may not yet be complete, and that unusually fast M1 growth might continue a little longer. However, over time the rate of narrow money growth clearly has to come

Table 9. **Money and credit**

Percentage change

	1993	1994	1995	1996	Seasonally adjusted[1]	
					1997 Q1	1997 Q2
Monetary growth						
Gross M1[2]	9.1	12.1	5.4	10.1	24.1	12.9
M1[3]	10.8	11.6	5.0	10.7	22.5	5.4
M2	3.1	2.2	4.0	3.1	–0.9	–2.3
M3	4.8	3.5	4.9	5.3	7.1	2.7
M2+	3.6	2.3	4.1	3.9	2.0	–1.4
Credit expansion						
To business sector	0.5	4.2	4.5	4.4	8.0	7.1
of which:						
Short term	–6.9	0.2	4.4	1.2	7.1	11.5
To household sector	6.5	6.8	4.4	4.9	8.1	6.0
Consumer credit	2.4	7.4	6.6	7.1	12.6	12.3
Residential mortgages	7.9	6.6	3.7	4.1	6.5	3.9
Memorandum items:						
Nominal GDP growth	3.3	4.8	3.9	2.8	4.1	3.5
Income velocity (GDP/M1)	–6.7	–6.1	–1.0	–7.1	–15.1	–1.9
Income velocity (GDP/M2)	0.2	2.6	–0.1	–0.3	5.0	5.8
Income velocity (GDP/M3)	–1.4	1.3	–0.9	–2.4	–2.8	0.7

1. Annual rate of change over preceding period.
2. Currency outside banks plus demand deposits.
3. Gross M1 minus private sector float, *i.e.* items in transit through the payment clearing system.

Source: Statistics Canada; OECD.

down: historical experience suggests that in the long run a rate of 3 to 5 per cent would be consistent with the inflation target range. Close monitoring of the aggregate in the period ahead is therefore required.

In summary, while there does not seem to be a need for a rapid return of monetary policy to a neutral stance, this assessment might change in the light of new information regarding the strength of the economy. In this event, a rise in short-term interest rates would be required, the size of which would depend on the movement of the exchange rate. This would not necessarily imply an increase in long-term rates, as improved fundamentals can be expected to maintain low risk premia for Canadian borrowers.

What inflation target beyond 1998?

The current inflation target expires at the end of 1998. When it was announced in December 1993, the authorities indicated that post-1998 targets ''would be consistent with price stability''. Both the Bank of Canada and the Department of Finance have done research on this issue. Relevant questions include: the measure of inflation that should be used; the costs and benefits of changing the mid-point of the target; the appropriate width of the target band; and the horizon over which the target should be achieved.

There is no ''ideal'' price measure to be used as a target variable. Most countries are using the CPI, with the focus of policy on a measure of ''core'' inflation that excludes volatile components (Table 10). The CPI's advantages are that it is best known to the public, frequently and promptly published, and rarely revised. Further, a recent study.[10] suggests that if the monetary authority stabilises CPI inflation, it should also do a reasonable job of controlling inflation as measured by other indicators. The CPI, as other indices, has a number well-known biases resulting from base weighting, imperfect adjustment for quality changes, new goods, outlet substitution, and the formula used. For Canada, the same study estimates that the CPI overstates the ''true'' rate of inflation by 0.5 per cent (with an upper bound of 0.7 per cent). This bias needs to be taken into account when considering what inflation objective might be consistent with price stability.

Apart from the bias in the price index, the choice of the mid-point of the inflation target range depends on the long-run costs and benefits of lowering inflation, and the transitional costs associated with such a move. Regarding the

Table 10. **Inflation targeting regimes in OECD countries**

	Date	Variable[1]	Target range[2]	Target horizon	Caveats[5]	Operational target	Policy instrument	Published projections	Response to a breech of target
Australia	1993	CPI (underlying)	2-3 per cent on average	Over the business cycle	None	Not explicit	Overnight cash rate	No	Not explicit
Canada	February 1991	CPI (underlying)	1-3 per cent	End-1998	Explicit	Monetary conditions index	Overnight rate operating band	No	Not explicit
Finland	February 1993	CPI (underlying)	2 per cent	1995 onwards	None	Not explicit	Tender rate	2 years ahead	Not explicit
Italy		Private consumption deflator	2.5 per cent	Annual[3]	None	Not explicit	Not explicit	3 years ahead	Not explicit
New Zealand	March 1990	CPI (underlyng)	0-3 per cent	August 1998[4]	Explicit	Monetary conditions index	Settlement cash balances	Up to 2-3 years ahead	Reserve Bank Governor can be dismissed
Spain	January 1995	CPI	Close to 2.5 per cent	End-1997[3]	None	Not explicit	Certificates; 10-day auction rate	Up to 3 years ahead	Not explicit
Sweden	January 1993	CPI	2 ± 1 per cent	1995 onwards	None	Not explicit	Repo rate	Up to 1-2 years ahead	Not explicit

Table 10. **Inflation targeting regimes in OECD countries** *(cont.)*

	Date	Variable[1]	Target range[2]	Target horizon	Caveats[5]	Operational target	Policy instrument	Published projections	Response to a breech of target
United Kingdom	October 1992	RPIX	2.5[6]	Up to 2 years ahead	None	Commercial bank's base rate	Money market dealings rate	Up to 2 years ahead	Letter from the Bank of England indicating appropriate response

1. The price indices that these variables refer to are as follows: Australia: consumer price index excluding mortgage interest and credit debt charges, goods and services subject to government regulation or influence and goods and services whose prices are seasonal; Canada: consumer price index excluding food, energy and temporary effects of indirect taxes; Finland: CPI excluding indirect taxes, subsidies and housing capital costs; New Zealand: CPI excluding significant changes resulting from changes to indirect taxes, government charges, terms of trade, direct effects from interest rates and natural disasters; United Kingdom: retail price index excluding mortgage interest payments.
2. The range refers to the current target. Previous intermediate targets have been as follows: Canada: 3 per cent, plus or minus 1 per cent between February 1991 and the end of 1992; 2.5 per cent, plus or minus 1 per cent from 1993 to mid-1994; and 2 per cent, plus or minus 1 per cent until end-1995; New Zealand: 3-5 per cent between March 1990 and end-1992; and 0-2 per cent between 1993 and end-1996; and United Kingdom: 2.5 per cent or less between October 1992 and June 1997.
3. Italy's target horizon is determined year-by-year. In the current context, 2.5 per cent is the target for 1997; and 2 per cent for both 1998 and 1999. For Spain, the target horizon is close to 2.5 per cent by end-1997 and close to 2 per cent during 1998.
4. The Policy Target Agreement which sets out the inflation targeting framework is an agreement between the Minister of Finance and the Governor of the Reserve Bank and expires at the end of the Governor's term of office, after which it is renegotiated.
5. Caveats for Canada include large supply shocks and the effects of indirect taxes; for New Zealand: significant changes in the terms of trade; changes in indirect taxes or the GST, natural disasters, a significant price level impact arising from changes to government levies, and movements in interest rates that cause a significant divergence between the change in the CPI and the change in CPI excluding the interest cost component.
6. If inflation deviates more than 1 percentage point on either side of the target, a letter from the Bank of England to the Chancellor of the Exchequer must be forthcoming.

Source: OECD Secretariat.

potential benefits of further disinflation, a number of factors are thought to play a role, such as reduced transaction costs, greater certainty concerning future nominal and real variables, and the removal of distortions created by the interaction of inflation with the tax system.[11] In contrast, some recent studies.[12] have explored the possibility that downward nominal wage rigidities may lead to costs associated with reducing inflation to very low levels, although there is no conclusive evidence on this point.[13] The traditional approach to measuring such costs is to examine the lost output that is required per unit of disinflation. Estimates of this ''sacrifice ratio'' differ widely, however, and suggest that the ratio likely varies over time. Indeed, there is some evidence that the Phillips curve gets flatter at low rates of inflation, in which case the cost of further disinflation could become potentially very large.[14]

The typical band width used by central banks in bringing down the rate of inflation has been 2 percentage points. This is generally well below what would be needed to include a high percentage of inflation outcomes based on empirical work with historical data.[15] Stochastic model simulations for Canada[16] suggest that inflation can be expected to be outside such a band between 30 and 40 per cent of the time; a widening of the target range to 3 percentage points would reduce this likelihood to 15 per cent. The reason for choosing, nevertheless, such a narrow band was to focus attention on a path that clearly showed a downward trend during the disinflation period. Once credibility has been attained, it might be argued that a band more in line with historical experience would be both justified and feasible.

The appropriate width of the target band also depends, however, on how fast the monetary authority would like to see inflation return to the band. The New Zealand experience shows that trying to reverse any overshoot quickly requires large interest and exchange rate movements that can produce substantial fluctuations in output. In this case, a wide target band is appropriate because it minimises the amount of time that inflation is outside the band. By contrast, planning for a gradual return of inflation to the target band allows for smaller interest and exchange rate movements that tend to minimise fluctuations in output. Although this can be at the expense of more persistent deviations of inflation from its target bands, the latter do not need to be so wide. In any event, how the monetary authorities will respond to breaches of the target bands should be made as transparent as possible to keep market participants from overreacting.

The background note provided by the Bank of Canada at the time of the announcement of the current target bands made this point clear and was helpful in this regard.

In summary, there are several outstanding issues yet to be resolved with respect to what should be the ultimate inflation target. In this regard, a further interim extension of the existing target in order to experience a full economic cycle with it might provide valuable insights that could be used in deciding on a longer-term inflation objective consistent with price stability.

III. Structural policies

An extensive array of structural reform measures have been undertaken in Canada over the past decade or so, involving: freeing trade and investment through various channels, including bilateral, regional and multilateral negotiations; breaking down traditional ownership barriers in financial markets; reforming the tax system; and reformulating policies in several areas (such as Competition, Transportation and Telecommunications) to reflect the needs of a modern economy (Table 11). Three salient features of the microeconomic policy environment stand out. First, the forces for change have emanated mainly from external sources and the necessity to enhance international competitiveness (to wit, the push for free trade, tax reform, and financial deregulation). Second, the thrust of the initiatives has been increasingly to roll back the extensive involvement of government in economic activity, to lighten heavy-handed regulation, and to let markets functions more freely. Third, actions toward the labour market have significantly lagged moves in other areas, and this has probably contributed to the slow adjustment and weak economic performance of the past decade.

Indeed, over the last ten years, total factor productivity growth in Canada has continued to decline relative to historical achievements, and compared with other OECD countries (Table 12). There is no consensus on why productivity has performed so dismally, although resistance to change and the difficult past macroeconomic environment (see Chapter II) probably offer some of the possible explanations.[17] In such a situation, the beneficial effects of structural reform are likely to be slower to come through.[18] Thus, it could simply be that more time is needed for the Canadian economy to adapt to the vigorous programme of microeconomic reform that has been implemented as well as to the knowledge-based economy more generally. Or, as some observers suggest, slow growth might be typical of a period of deep structural adjustment in which the knowledge-based economy cannot perform at its full potential until its productive,

Table 11. **Summary of major structural reform measures in Canada: 1985-96**

Policy area	Measures taken
International trade	Participated actively in the Uruguay Round of multilateral trade negotiations under the auspices of the GATT; concluded a free trade agreement with the United States in 1989 which was largely extended to include Mexico in 1994; made a broad commitment in 1994 to pursue future free trade and investment liberalisation within the context of APEC.
Interprovincial trade	In 1995, the Agreement on Internal Trade was implemented to reduce interprovincial trade barriers, with specific obligations in ten economic sectors, including government procurement, labour markets and investment.
Business regulation	Implemented new Competition Act in 1986 and a new National Transportation Act in 1988; in 1985 replaced the Foreign Investment Review Agency with Investment Canada and dismantled the National Energy Program; in 1993 established a new Telecommunications Act.
Financial markets	Since 1987, the elimination of the historical separation of ownership of commercial banks and securities dealers; since 1988 permitting foreign-owned financial institutions to own Canadian securities dealers; and from 1992, allowed federally-regulated financial institutions to offer most financial services either directly or indirectly through subsidiaries or networks; the elimination of primary reserve requirements for chartered banks.
Taxation	In 1988 a lowering of personal income tax rates and the reduction in the number of tax brackets; a cut in the basic corporate tax rate; and in 1991, the replacement of the federal sales tax with the Goods and Services Tax.
Labour markets	Minor changes to the Unemployment Insurance (UI) system in the early 1990s; active labour market policies organised under the Employability Improvement Program; in 1996, introduction of the Employment Insurance Act (EI) which replaced the UI system; altered the funding of the new EI system; and moves to hand over ALMPs to provincial responsibility.
Regional development	Introduced Atlantic Opportunities Agency and Western Diversification Program in 1987.
Science and technology	Introduced refundable R&D tax credit for small firms; broadened the definition of R&D for tax purposes; created Networks of Centers of Excellence in 1988.
Industrial support	Replacement of subsidies to enterprises, through making financial support repayable and increased use of various tax incentives and credits.
Privatisation and public sector reform	Since 1992, more than 50 agencies and other government enterprises have been eliminated or consolidated and the number of government departments has been reduced from 32 to 23; 40 crown corporations and government holdings privatised since the mid-1980s. In 1994, announced a Programme Review with the intention of reducing the overall cost of government and to focus expenditures on high priority activities.

Source: Various OECD Economic Surveys for Canada; and Department of Finance, Canada.

Table 12. **Productivity in the business sector**

Percentage changes at annual rates

	Total factor productivity[1]			Labour productivity[2]			Capital productivity		
	1960[3]-73	1973-86	1986-96[4]	1960[3]-73	1973-86	1986-96[4]	1960[3]-73	1973-86	1986-96[4]
United States	2.5	0.4	0.4	2.6	0.8	0.6	2.3	–0.4	0.0
Japan	5.7	1.1	1.1	8.4	2.5	2.1	–2.3	–3.0	–1.8
Germany[5]	2.6	1.1	0.7	4.5	2.1	0.9	–1.4	–1.2	0.3
France	3.7	1.4	1.4	5.3	2.6	2.2	0.6	–1.0	–0.3
Italy	4.5	1.4	1.4	6.4	2.1	2.5	0.5	–0.1	–0.8
United Kingdom	2.6	1.3	1.2	3.9	2.2	1.2	–0.3	–0.7	1.1
Canada	2.0	0.2	–0.2	2.9	1.3	0.8	0.2	–2.0	–2.1
Total of above countries[6]	3.4	0.8	0.7	4.5	1.6	1.2	0.5	–1.1	–0.4

1. TFP growth is equal to a weighted average of the growth in labour and capital productivity. The sample-period averages for capital and labour shares are used as weights.
2. Output per employed person.
3. Or earliest year available, *i.e.* 1962 for Japan and the United Kingdom; 1965 for France; and 1966 for Canada.
4. Or latest available year, *i.e.* 1994 for Germany; 1995 for Japan, France, and the United Kingdom.
5. The two first averages concern Western Germany. The percentage changes for the period 1986-95 are calculated as the weighted average of Western Germany productivity growth between 1986 and 1991 and total Germany productivity growth between 1991 and the latest year available.
6. Aggregates were calculated on the basis of 1992 GDP for the business sector expressed in 1992 PPPs.

Source: OECD.

organisational and policy structures are altered and much of the old physical capital stock is replaced.[19] It would be reasonable to expect, however, that the benefits of structural initiatives should soon be felt, especially as economic growth picks up and capacity utilisation increases as projected over the coming period.

The following paragraphs report on reform progress that has been made more recently in a number of areas: labour markets; pensions; international trade; interprovincial trade; financial markets; privatisation and public sector reform; and science and technology. A concluding section details unfinished business in the structural policy field.

Recent progress

Labour market

In July 1996, the government introduced an Employment Insurance (EI) scheme which replaces the previous Unemployment Insurance (UI) system. The

new EI scheme represents the most fundamental reform of the UI programme since 1971. The objective of the new policy is threefold: to reduce the cost of providing income benefits[20] (annual savings once the programme is fully in place are expected to amount to C$ 2 billion per year, with C$ 800 million reinvested in active re-employment benefits); to lessen dependency and work disincentives prevalent within the old framework; and to address prior inequities, such as those affecting part-time employees. It incorporates a significant number of changes – regarding benefits; eligibility requirements; duration; experience rating; and clawbacks – which were described in detail in last year's Survey.

It is far too early at this stage to derive lessons from modifications to the unemployment insurance system since EI is being implemented in stages (Table 13). Nonetheless, two problems have already emerged and are being addressed.[21] *First*, in areas of high unemployment and weak employment prospects, the switch from a weeks-based to an hours-based method of calculating

Table 13. **Phasing-in of Employment Insurance: key dates**

Elements taking effect from 1 January 1996
- Reduction in maximum insurable earnings for premium purposes.
- Reduction of premium rates for workers, from C$ 3.00 to C$ 2.95 per C$ 100 of earnings and for employers from, C$ 4.20 to C$ 4.13.
- Lowered income threshold for clawback of benefits of high income claimants.

Elements taking effect from 1 July 1996
- Tightened entry requirements for new and re-entrants (from 20 to 26 weeks).
- Reduction in the maximum entitlement period (from 50 to 45 weeks) for benefit purposes.

Elements taking effect from 1 January 1997
- Move to hours-based system (between 420 and 700 hours to qualify for benefits, 910 hours for new and re-entrants).
- Reduction of premium rates for workers to C$ 2.90 per C$ 100 of earnings and to C$ 4.06 for employers.
- Collection of premiums on first dollar earned; elimination of weekly minimums and maximums.
- Refund of premiums to workers earning less than C$ 2,000.
- Small businesses facing large premium increases due to first dollar coverage or new hires receive temporary refund in 1997 and 1998.

Elements with a progressive phase-in
- Reduction on benefit rate related to previous benefit history.
- Increased clawback for high income earners with more than 20 weeks of past benefits (beginning with 1997 tax year; by 2001 claims will be based on a full 5 years, or 60 months of benefit history).

Elements with a timed phase-in
- Family Income Supplement top-up for claimants in low-income families (beginning in January 1997, top-up allows benefits to reach 65 per cent of insurable earnings, rising to 80 per cent in 2000).

Source: Human Resources Development Canada.

benefits could create work disincentives. This is because including ''small weeks'' (*i.e.* with earnings of less than C$ 150) in the calculation of earnings while at the same time allowing individuals to discount some weeks (up to twelve in high unemployment areas) with no earnings, would lower benefits below levels that would be attained if an employee had instead chosen not to work.[22] The issue is being dealt with through small weeks adjustment projects in 29 high unemployment regions, where small weeks will either be bundled into large weeks or be excluded altogether in the benefit rate calculation. *Second*, for employees currently in the labour force, the eligibility for insurance is based on the number of hours worked (420-700 depending on the regional unemployment rate). Nevertheless, for people entering the labour force for the first time, or re-entering after an absence of two years, a minimum of 910 hours of insurable employment is required. This could pose difficulties for some people in accessing the programme, particularly for those displaced from the labour force for exogenous reasons such as in Newfoundland as a result of the closure of the northern cod fishery (see Box 1). [In this case, a temporary regulatory change has ensured that individuals who have collected Atlantic Groundfish Strategy (TAGS) benefits are not considered re-entrants to the labour market.]

As mandated by the EI Act, the federal government has offered the provinces options to participate more directly in all aspects of planning, designing and delivering active labour market policies (ALMPs). Up to now, eight provinces have signed such agreements (excluding Ontario, Saskatchewan and the territories). It is projected by the government that aggregate funding of up to C$ 1.95 billion annually will be available to support the labour market agreements at maturity in 2001/02.[23] The arrangements between the federal government and the provinces are of two types: co-management, which involves shared responsibilities and harmonisation of programmes; and full devolution of ALMP responsibility to the province.[24] In both instances, it is agreed that national standards be upheld, access to the National Employment Service be preserved, and accountability be assured. The latter is to be judged against three indicators: the number of current EI claimants served; the number of returns to employment; and the savings to the EI account.

The accords are for an indefinite period, with a provision for a three-year review to determine if the desired results are being achieved. In the past, labour market policy was designed at the national level, which did not always allow for

Box 1. **Labour market support: the case of Newfoundland**

Newfoundland continues to experience the highest unemployment rate among Canada's provinces at around 20 per cent. The province has a history of relatively high unemployment, but several factors have acted in recent years to accentuate long-standing labour market problems: the moratorium on the northern cod fishery in 1992; the winding down of the Hibernia offshore oil construction project; lower federal transfers to the province; and changes to the unemployment insurance system. In response to the closure of the cod fishery, the federal government introduced a long-term support programme – the Atlantic Groundfish Strategy (TAGS) – to offset income lost by affected fish harvesters and plantworkers in Quebec and the Atlantic provinces, particularly Newfoundland. TAGS was implemented in 1994 as a five-year programme designed to provide labour market adjustment, training options and income replacement. It had a C$ 1.9 billion budget and was due to expire in May 1999. However, TAGS will end when funding is exhausted, which may be as early as May 1998. A total of around 28 000 Newfoundlanders are eligible to receive TAGS benefits and the overall average eligible benefit rate is C$ 285 per week. A number of participants exit the programme each year but nearly 55 per cent are expected to remain until support ends.

The winding down of the TAGS programme and changes to the EI system pose a particular challenge to both federal and provincial policy makers as the majority of TAGS participants* have low education levels, and have acquired almost no training during the support period. Tightened restrictions for unemployment insurance eligibility for re-entrants will prevent many from accessing those benefits, as well as ALMPs which are available only to those employees who have made UI claims over the past three years. A large question remains concerning what will happen to these people when TAGs ends. In this regard, the federal government announced that it would begin a review of long-term post-TAGs issues in the autumn of 1997. The review is to involve the relevant provinces and industry stakeholders.

A partial reopening of the (southern) cod fishery this year should help somewhat, as should the ongoing development of the Voisey's Bay nickel mining project, and efforts at the provincial level to diversify the economy. The Newfoundland Department of Development and Rural Renewal is trying a local approach to development out of necessity, focusing on comparative advantage, for example, in the areas of aquaculture, marine technology, dimension stones, and adventure tourism. This ''zoning approach'' to development – which establishes local control and decision making with central resources and co-ordination – could strengthen the economic base, without hampering mobility and adjustment. Indeed, though income support is considered a legitimate role for the EI system within the region, labour mobility should be encouraged through the continued reduction in the population's dependence on unemployment insurance. Other forms of direct income supplementation might be more appropriate, however unpopular they may be, to avoid further deleterious effects on the fragile labour market.

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* Approximately 2 per cent of TAGS clients are currently less than 25 years of age, 63 per cent are between the ages of 25 to 44, and 35 per cent are age 45 or more. Nearly 42 per cent have less than a Grade 9 education, while almost 72 per cent have not completed high school, and only 9 per cent have formal training beyond high school.

the flexibility needed to address local or regional conditions. The handing over of ALMPs to provinces should serve to rectify this bias, while the remaining provinces and territories not already part of the arrangement should be brought on board as soon as possible. Nonetheless, such a move does create some concerns which must be guarded against. Chief among them is the need to adopt more rigorous evaluation systems than currently being used by the provinces, to ensure that policies which have demonstrated past success are given high priority; and to monitor the types of active measures being pursued to avoid potential abuse through, for example, the widespread use of wage subsidies which could simply lead to the ''cycling'' of workers between federal (EI) and provincial programmes of income support.

Persistently high unemployment and a decline in labour force participation rates among youths beginning in the late-1980s are worrying developments in Canada. Since 1989, the unemployment rate for those aged 15 to 24 has averaged around 15½ per cent while their participation rate has fallen by 9 percentage points to 61.5 per cent in 1996. In addition, as noted in Chapter II, the employment ratio for this age group has declined by 11 percentage points over the 1990s. Hence creating employment opportunities for young Canadians is a key challenge facing the country. Earlier this year, the government announced the Youth Employment Strategy, the major goal of which was to bridge the gap between school and work. It introduced new funding of C$ 315 billion over three years, building on the existing government's investment in youth of over C$ 2 billion. There are two key components to this strategy: a new internship programme in areas deemed to have long-term employment potential, in particular in science and technology, environment, and international trade and development; and improved access to information and services relating to job postings, careers, educational courses and government projects. Another aspect to the Strategy involves the existing Youth Service Canada and Youth Internship Canada programmes, which are being refocused toward ''at risk'' youths, who face barriers to full labour market participation, because of limited education, problems with the law, disabilities and single teenage parentage.

Removing disincentive effects in the labour market involves an important provincial element, as provinces provide social assistance (SA) which does not depend on previous employment or past earnings, and which can be received for an unlimited duration (subject to family wealth and income means testing).

Provincial governments are, however, moving quickly to correct this perverse incentive effect. Since 1995, the government of Ontario, for example, has cut benefit rates for SA by 21 per cent in order to: *i)* reduce the generosity of the system which had increased significantly in the mid-1980s; *ii)* bring it back into line with those in other provinces and; *iii)* reverse the increase in caseloads and the likelihood that some labour would fall into an ''unemployment trap''.[25] To date, SA caseloads in Ontario have fallen by around 14½ per cent and recent evidence suggests that a major reason for leaving the scheme has been employment related. A survey of people who left SA found that 62 per cent of respondents did so because they had gained some form of employment, with 46 per cent of those attaining full-time jobs.[26] In a different study examining the effects of SA cuts on lone mothers, the reduction of benefit rates was found to have led to an increase in the employment rate by nearly 6 percentage points by 1996.[27] Thus reducing social assistance can apparently have positive effects on labour market behaviour as intended.

More recently, the Ontario government has put into place a ''Workfare'' programme which obliges social assistance recipients deemed to be ''employable'',[28] to engage in specific activities in order to receive these benefits. As of May 1997, 21 municipalities are operating these schemes on a pilot basis with 18 000 recipients participating, but this is considerably lower than the anticipated enrolment of over 50 000. Indeed, all Ontario municipalities are required to have Workfare fully in place by January 1998. Municipalities are currently negotiating terms and conditions with the provincial government and a ''softer'' form of workfare appears to be emerging, which involves choice between taking part in employment support, community involvement or employment placement programmes. Municipalities which already have measures in place are finding it difficult to screen recipients, find useful placements and to monitor their work. The Ontario experience in this regard is important as most provinces and territories are reforming their social assistance systems and placing increasing emphasis on some form of work as a condition of entitlement to benefits.

Pension reform

Following the overhaul of Old Age Security announced in the 1996 budget, the government tabled legislation regarding a major policy shift concerning the

second pillar of the country's publicly-provided pension system, the Canada Pension Plan (CPP). The proposed changes to the CPP, which are expected to become effective next year,[29] reflect the outcome of a statutory review of the system which the federal and provincial governments undertook over 1996-97 with a view to strengthening its financing and sustainability. The thrust of the new measures – which are in line with assessments made in last year's OECD Survey – is fourfold: to improve inter-generational fairness; to move away from the current pay-as-you-go financing toward fuller funding, building a much larger reserve fund (from around two years of benefits to around five); to increase contribution rates rapidly over the next six years to 9.9 per cent of contributory

Tableau 14. **Comparison of existing Canada Pension Plan and proposed changes**

	Status quo	Proposal
Reserve fund	Equal to 2 years of benefits and declining.	Growing to 5 years of benefits.
Contribution rates	Rising to 10.1 per cent by 2016.	Rising to 9.9 per cent by 2003, then held steady.
Year's basic exemption	Currently C$ 3 500 indexed to wages.	Frozen at C$ 3 500.
Year's maximum pensionable earnings	Indexed to wages.	No change.
Eligibility for disability benefits	Must work and contribute in 2 of last 3 or 5 of last 20 years.	Must work and contribute in 4 of last 6 years.
Retirement pensions for disability beneficiaries	Based on year's maximum of pensionable earnings (YMPE) when recipient turns 65, then indexed to prices.	Based on YMPE at disablement with subsequent price indexing.
Combined survivor-disability benefit	Ceiling equal to maximum retirement pension plan plus larger of two flat-rate components.	Ceiling is one maximum disability pension.
Death benefit	6 months retirement benefits, maximum of C$ 3 580; grows with wages.	6 months retirement benefits to maximum of C$ 2 500 and frozen.
Investment policy	Invested in non-negotiable provincial bonds.	New funds invested in a diversified portfolio of securities.
Normal retirement	Age 65.	No change.

Source: Department of Finance, Canada.

earnings and to cap them from then on; and to adopt a diversified portfolio approach to the management of the fund. (Table 14 provides a brief comparison of the existing CPP system with the new initiatives.)

There are some concerns that the steep rise in CPP contributions will have an adverse effect on the labour market. However, because of the close linkage between benefits and contributions, this levy may be closer to a user charge than a tax (see also Chapter IV). If the rise in rates is perceived as putting the Plan on a sounder financial footing and thus ensuring expected future benefits, then the net effect on employment may not be negative. Moreover, labour market reforms and the improving economic situation should allow for an acceleration in the decline of EI premiums (especially as the EI surplus mounts), somewhat offsetting the rise in CPP contributions.

Trade liberalisation

A key plank in Canada's structural reform efforts has been trade liberalisation. Canada is implementing the Free Trade Agreement (FTA) with the United States, which was widened to include Mexico in 1994 (NAFTA). It participated actively in the Uruguay Round and follow-on multilateral negotiations on Information Technology, Basic Telecommunications and Financial Services; and, together with its APEC partners, it is committed to freeing trade and investment in the Pacific region by the year 2010.[30] The trade-exposed sectors of the economy have adjusted well to the liberal trading environment, and trade volumes have expanded by an annual average of over 9 per cent during the 1990s, compared with below 6 per cent a decade earlier. Indeed, a recent study concluded that bilateral Canada-US trade grew more quickly in sectors liberalised by the FTA than in others still protected, and that trade with all other countries in such liberalised sectors grew faster than trade in general.[31] However, the same study also suggested that the FTA has not yet resulted in widely expected productivity and real wage growth. In manufacturing, for example, the sectors that have experienced above average earnings have been largely those already exposed to free trade before the FTA was put into place, while those newly exposed by the agreement have fared less well (although still better than the average for the economy as a whole).

The average applied tariff for imports from all sources fell to below 1½ per cent in 1996 from 4½ per cent in 1981, while for trade with the United States

tariffs affected by the FTA will be completely eliminated by next year. Nonetheless, some tariff rates remain high – particularly for agricultural and food products affected by the tariffication of prior quantitative restrictions under the WTO agreement in this sector – although these are being brought down gradually. Excluding farm products subject to tariffication, the average most-favoured-nation tariff stood at 6.4 per cent in 1996, but it can exceed 250 per cent for a number of agricultural and food products.[32] The government has announced that a simplified Customs Tariff will be introduced at the beginning of 1998, with the objective of improving the competitive position of Canadian industry and making the tariff system more transparent and simple, so as to lessen the regulatory burden and associated costs for both business and government. Among the more substantial changes are: a consolidation of the seven existing tariff schedules into a single one; the reduction of tariffs on a wide range of manufacturing inputs; an acceleration by one year of certain Uruguay Round tariff commitments; and the elimination of most ''nuisance'' rates (those of less than 2 per cent). Net duty savings to business and consumers as a result of the modifications are estimated to be C$ 90 million in 1998. The government has also invited Canadian industry to submit requests for early elimination in the second round of accelerated tariff cuts under NAFTA. (Tariffs within NAFTA will be substantially eliminated by 2004, except for those on several sensitive goods for which the phase-out period will extend to 2009.)

Canada concluded with Israel another free trade agreement which took effect from January 1997. The primary motivation for such an accord from the Canadian side is the desire to restore the country's competitive position in the Israeli market which had been eroded by US and European preferential arrangements with that country.[33] The Canada-Israel pact liberalises virtually all trade in industrial goods; covers 80 per cent of bilateral agri-food exchange (but excludes supply-managed products); contains a binding dispute settlement mechanism; and leaves most other areas governed by the rights and obligations of the World Trade Organisation. In November 1996, Canada also initialled an ''interim'' free trade agreement with Chile, which is to provide both a bridge for that country's eventual accession to NAFTA and a beachhead for broader trade liberalisation in the Americas.[34] The Canada-Chile arrangement entered into force in July 1997. It covers trade in goods and services; investment; dispute settlement; and has a number of other features which are modelled after NAFTA. Moreover, as in the

case of NAFTA (and the Israeli agreement), it excludes cultural industries, supply-managed agricultural products and the Auto Pact. Two-way trade between Canada and Israel and Canada-Chile totalled C$ 504 and C$ 672 million, respectively in 1996, as compared with Canada-Mexico trade of more than C$ 7.2 billion in the same year.

Canada is, in addition, actively participating in the negotiations for a Multilateral Agreement on Investment (MAI) being pursued in the OECD context. The goal of the MAI is to create multilateral investment rules which, at their basis, have national treatment obligations, to allow the free transfer of funds and to open access for, and ensure protection of, both foreign and domestic investment. Canada's goals in the negotiations are to obtain NAFTA-like foreign investment rules, while protecting sensitive areas such as cultural industries[35] and the ability to undertake both research and regional development initiatives. Canada also seeks to extend the MAI to non-OECD countries and encourages early accession by those countries, once an agreement is reached between OECD Members.

Interprovincial trade

The Agreement on Internal Trade, which came into effect on 1 July 1995, aims to reduce inter-provincial barriers to trade, investment and labour mobility within Canada, which have arguably become more burdensome than those between Canada and its external commercial partners. It is similar in spirit, intent and coverage – including, for example, a dispute settlement procedure – to international trade agreements to which Canada is a party. The federal government and seven provinces (Newfoundland, Prince Edward Island, Nova Scotia, New Brunswick, Quebec, Manitoba and Alberta) have passed legislation to implement their obligations under the Agreement, while the remaining signatories have considered regulatory measures adequate to assure their compliance with it. Markets have been opened for pan-Canadian suppliers – most extensively for government procurement and investment. With regard to procurement, significant progress has been achieved; around C$ 59 billion worth of goods and services is now covered out of a total market valued at around C$ 137 billion. Negotiations are under way to include municipalities, academic institutions, social services and hospitals (the MASH sector) and many more Crown corporations. Successful completion of these talks could more than double this coverage. In other areas, such as labour mobility, where an ambitious work programme is

Table 15. **Agreement on Internal Trade: outstanding issues**

Procurement (Chapter 5)
– Negotiation of special provisions to cover MASH (municipalities, publicly funded academic institutions, social services agencies and hospitals) procurement have not yet reached a consensus; discussions also continuing regarding broader coverage of Crown corporations.

Investment (Chapter 6)
– Work programme to reconcile corporate registration and reporting requirements which should ease the compliance burden for firms operating in more than one jurisdiction not yet completed; target date for implementation is 1 December 1998.

Labour mobility (Chapter 7)
– Implementation being coordinated by the Forum of Labour Market Ministers; at the end of 1996 more than 400 occupational regulatory bodies had been invited to initiate work toward compliance with the obligations of the Agreement, that work is still underway.

Consumer-related measures and standards (Chapter 8)
– Co-operative enforcement of consumer protection across boundaries to be studied.

Agricultural and food goods (Chapter 9)
– Work is underway to remove impediments to trade in agricultural and food goods through the development of a sanitary/phytosanitary protocol and the eventual elimination of technical barriers in a number of areas;
– An agreement in principle has been reached to revise the scope and coverage of this Chapter through the development of general principles that would apply to measures affecting internal trade in agricultural and food goods. A tentative deadline of December 1998 has been set for a revised Chapter.

Alcoholic beverages (Chapter 10)
– An *ad hoc* Committee has been formed to review exceptions, reservations and grand-fathered measures with a view to their eventual reduction or elimination.

Energy (Chapter 12)
– No agreement reached.

Source: Industry Canada.

under way to recognise the qualifications of workers from other provinces and to reconcile differences in occupational standards, the Agreement is a work in progress. Overall, it represents an important first step and negotiations are continuing to expand its scope and coverage (Table 15).

Financial markets

Major financial market reform has taken place in Canada in two phases: the first being in 1987 with the breakdown of ownership barriers between commercial banks and securities dealers; the second occurring in 1992 with the freeing up of regulations that prohibited financial institutions of any type from offering a broad range of services, either directly or indirectly through subsidiaries or

networks. Continuing with this deliberate strategy, a number of amendments to the financial institutions legislation were put into place in June 1997, in order to deal with consumer protection issues, ease the regulatory burden on market participants and fine-tune certain provisions in the relevant statutes. In addition, the government staked a commitment to implement before year's end a new entry regime for foreign banks, which will permit them to operate branches directly in Canada. At this point, the scope of this commitment is unclear, although the government has proposed that foreign bank branches operating in Canada would not be able to accept retail deposits. The government's view is that, for prudential reasons, acceptance of retail deposits should be confined to an insured subsidiary. This would put Canada at odds with other OECD countries, the majority of which permit such activities. The government has yet to announce whether a foreign bank would be allowed to operate both a retail subsidiary and a wholesale branch. Permitting that arrangement would help to encourage competition in the banking sector.

In a move to provide better protection for consumers of financial services, the government has implemented a number of changes. These relate to *i)* obliging financial institutions to establish procedures governing the collection, retention, use and disclosure of customer information; *ii)* improve disclosure of the cost-of-credit; and *iii)* ensure that these institutions adopt a policy on tied-selling (the practice of requiring the purchase of a package of goods or services) and create guidelines for dealing with complaints in this area. A number of steps are being taken to reduce the regulatory burden facing the financial sector, and should thus help lower banks' costs and lead to gains in competitiveness. Banks which do not accept retail deposits will be permitted to opt out of Canada Deposit Insurance Corporation (CDIC) coverage, and subsequently to avoid reporting requirements associated with CDIC membership. Several initiatives are being introduced to streamline the self-dealing regime (those involving financial transactions between a financial institution and persons related to it) which will allow subsidiaries of a federally-regulated financial institution to transact with each other. Foreign banks will be treated more fairly by allowing regulated ones to hold other financial institutions directly, instead of through the Schedule II bank by authorising near banks to operate freely – provided they do not accept retail deposits – and by allowing them to own non-bank financial institutions. Finally, the legislation will permit greater latitude for joint ventures within the financial sector and for

enhancing access to capital markets for insurance companies, including permitting them to issue participating shares.

Privatisation and public sector reform

Privatisation and commercialisation has been an important aspect of the federal government's Programme Review which it undertook in 1994 as a means to providing more efficient service delivery and in helping to lower its financial obligations. As called for in the 1995 budget, Canadian National Railways has now been turned over to private hands; a substantial portion of the government's 70 per cent stake in Petro-Canada has been sold; and Transport Canada's air navigation assets have been transferred to NavCan, a not-for-profit private sector corporation. These transactions have generated net proceeds of C$ 4.4 billion since 1996. In addition, the Canada Communication group (the government's printing operations) was sold to a private sector purchaser. In the future, the government intends to move ahead with new privatisation and commercialisation schemes in a pragmatic fashion, pursuing such avenues where it is both appropriate and feasible. Prospective candidates will probably include its remaining stake (18 per cent) in Petro-Canada, its 11 per cent interest in National Sea Products and Theratronics International Limited, a manufacturer, developer and marketer of cancer treatment and advanced computerised treatment planning systems.

The government's Programme Review exercise has led to savings from rationalisation and efficiency improvements achieved through changing the way it delivers its goods and services. However, the magnitude of these savings is difficult to identify since they are bound together with programme cuts and other fiscal consolidation measures. In contrast, the provincial government of Ontario in its 1997 budget highlights fiscal operational savings of C$ 2.3 billion in 1997-98, or 4 per cent of total expenditures. This is a result of ongoing efforts to restructure administrative services and to control costs, by, among other things: adopting standardised procedures; streamlining processes; integrating and co-ordinating functions between ministries; and out-sourcing services to the private sector. The Ontario government is also striving to raise the accountability of the public sector, and a Public Sector Accountability Act is expected to be introduced shortly. This Act will oblige public sector organisations in Ontario to function more like their private sector counterparts in a number of ways, particularly: in applying similar financial reporting requirements; disclosing to the

public a business plan; and benchmarking their activities against both the public and private sector to ensure that best practices are being followed.

Science and technology

The strength of both federal and provincial fiscal consolidation efforts in recent years has led to mounting resource constraints on educational and research institutions and growing concerns that an out-migration of scientific personnel (particularly to the United States) is occurring because of a lack of adequate research infrastructure. Indeed, Canada's ''innovation gap'' compared to other OECD countries – that is, a relatively limited amount of activity in the field of science and technology (see 1995 Survey) – appears to reflect, in part, more limited efforts in basic research and development.

To help close this gap, the federal government introduced in the 1997 budget a proposal to create the Canada Foundation for Innovation. The main objectives of the Foundation will be to increase the country's ability to carry out world-class scientific research and technology development; to expand research and job opportunities for young Canadians; and to promote productive networks and collaboration among post-secondary educational institutions, research hospitals and the private sector. The instrument for achieving these goals is an up-front investment by the federal government of C$ 800 million, which will enable the Foundation to contribute about C$ 180 million annually over five years to promote partnerships with the private and voluntary sectors and provincial governments to support research infrastructure projects. These projects are to be directed toward key needs in the areas of health, environment, science and engineering and will mainly cover capital costs in modernising infrastructure; establishing computer networks and communications linkages; creating research data bases and information processing capabilities, as well as upgrading laboratories and installations.[36] Contributions by the Foundation are expected to average 40 per cent of eligible project costs, but in no case will exceed 50 per cent.[37] The Foundation will be created as an independent, not-for-profit corporation with a board of directors who will be responsible for its activities.

The federal government has taken and is taking additional steps to promote science and technology. The recent Budget announced the renewal of the Networks of Centres of Excellence (NCE) programme with stable funding of C$ 47 million per year as well as additional funding for the Industrial Research

Assistance Program (IRAP) to stabilise its funding at C$ 97 million per year. The NCE programme supports university-industry research partnerships while the IRAP programme encourages research and innovation by small and medium-sized enterprises.

Unfinished business

Microeconomic policies in Canada have opened up both domestic and international markets and significantly lessened government involvement in economic affairs. Although improving macroeconomic conditions should now allow these policy shifts to bear fruit, there is no room for complacency as unemployment levels remain high and productivity growth remains low. Notwithstanding some backsliding compared with initial proposals, recent labour market reforms are a step in the right direction, being broadly consistent with the recommendations made in last year's Survey regarding the implementation of the *OECD Jobs Strategy* (Table 16). But they could go much further in reducing the regional generosity of the unemployment benefit system and in strengthening experience rating. The handover of active labour market policies to some provinces is encouraging and the remaining jurisdictions should be brought on board. Nevertheless, the federal government should not view the occasion as an invitation to vacate the ALMP area, since monitoring the progress of the system and ensuring that incentive structures are not abused will become increasingly important, as will helping young Canadians to find their way back into the labour force and employment. In the area of public pensions, it is to be hoped that the proposed legislation will be passed without significant material changes which could undermine the future financial soundness of the system.

Substantial economic benefits could also be achieved by freeing inter-provincial trade and investment through carrying out the existing commitments in the Agreement on Internal Trade and by broadening and deepening this Agreement. Even now, the external and internal trading environments remain out-of-line; much has been and is being done to liberalise the former, while progress appears to have stalled with regard to the latter. It goes without saying that the longer internal barriers persist, the longer Canadians are denied the demonstrated benefits of freer trade, and the more difficult it will be to sustain internal momentum for it. That is not to suggest, however, that more could not be

Table 16. **Implementing the OECD Jobs Strategy – an overview of progress**

Reforms adopted in 1996-1997 comprise several important elements to improve labour market flexibility. However, it is too early to gauge their impact on labour market performance.

Jobs Strategy proposal	Actions taken	OECD assessment/recommendations
I. Reform unemployment and related benefit systems – increase extent of experience rating; – reduce regional dependence on unemployment insurance; – monitor eligibility to disability benefits; – increasing threshold of allowable earnings to social assistance benefits; – make passive income support conditional; – increasing the working income supplement to low-income working families.	– in 1996, Unemployment Insurance (UI) system revamped under the Employment Insurance (EI) Act; EI system increases experience rating; tightens eligibiilty requirements; and reduces both the extent and duration of benefits; – social assistance levels cut in Ontario; and introduction of a mandatory workfare programme for recipients of assistance deemed employable; – a two-step doubling of the working Income Supplement for low-income working families.	– the EI system should help lessen work disincentives; however, the variation in regional generosity still largely remains and experience rating could be further strengthened; – make sure that workfare will add to human capital accumulation and does not undermine ALMPs; – ensure that the announced overhaul of the child benefit system reduces welfare traps.
II The wage formation system – grade minimum wages by age.	– no action.	
III. Improve labour force skills and competence – address adult literacy problems; – reduce barriers to mobility for workers; – develop the Pan-Canadian Indicators Program.	– work under way to recognise occupational qualifications and competencies across provinces.	– Agreement on Internal Trade which includes labour mobility has been in place since mid-1995 and negotiations seemed to be bogged down. A deadline for the completion of this work would help impart a sense of urgency.
IV. Active labour market policies – target ALMPS; – allow provinces greater responsibility for ALMPs; – investigate other forms of employment assistance; – maintain the operation of a ‘‘one-stop’’ shop for ALMPs.	– since 1996, 8 provinces have opted to accept more direct responsibility of ALMPs; – in 1997, the government announced the Youth Employment Strategy; – in 1997 it also announced measures to help persons with disabilities make the transition to employment.	– potential for substantial gains as ALMPs should now be better tailored to regional needs and as duplication of efforts is eliminated; however, provinces must now adopt a more rigorous evaluation of the system than heretofore.
V. Enhance product market competition – Continue implementation of the labour mobility chapter of the Agreement on Internal Trade.	– implementation of the labour mobility chapter of the Agreement on Internal Trade is ongoing.	– see point III above; and extend the Agreement to include the procurement of Crown corporations and the MASH sector; and energy.

Source: OECD.

achieved on the international front. On the contrary, while the tariff system is being revamped, action should also be forthcoming to bring high tariffs down, especially, but not exclusively, on supply-managed agricultural and food products.

With regard to the financial sector, recent reforms are to be welcomed, but the government needs to fully specify the conditions under which foreign banks will be allowed to operate branches in Canada. Concerning science and technology support and public sector reform efforts, the government faces the dual challenge of accountability and transparency. Substantial public funds are directed toward closing Canada's ''innovation gap'' and even more resources will be devoted in future, but outcomes remain inconclusive. As a result of programme reviews, public monies are probably being used more efficiently and savings are being realised, but this could be made more transparent. On both fronts, the accountability of government programmes and their efficient delivery remains an issue. Finally, there is scope for further adjustments to the tax system, as discussed in the following chapter.

IV. The tax system: problems and options for changes

The Canadian tax system has undergone many changes in the past two decades. Several of these have been in response to developments in the external environment, which have also been faced by other OECD countries. Some of these changes have been a result of broad reforms, such as those concerning the income tax in 1988 and the sales tax in 1991; others have been more gradual to address fiscal realities or constraints that have arisen during that period. Now that the federal government has made considerable progress in reducing its budget deficit, attention has turned again to tax issues, given the expectations of a ''fiscal dividend'' associated with the possible emergence of surpluses in the near future. While the latter would not initially be large, they would provide an opportunity for dealing with deficiencies in the tax system that have built up over time.

In this context, three particular aspects of the policy-making environment stand out. *First,* increasing international competition resulting from trade liberalisation, especially with the United States, and the rapid growth of low-wage countries has put pressure on Canadian policy makers to streamline the tax system so that the private sector is better prepared to respond to these challenges. Income tax rates in the United States remain significantly lower and proposed reductions to capital gains taxation will put renewed pressure on Canada to follow suit. *Second,* the enormous growth of public sector debt as a result of unrelenting deficits has put considerable strain on fiscal budgets as well as on tax levels, creating not only tax fatigue amongst the Canadian population but also an impetus for rationalising the tax system. *Third,* the responsibility for revenue raising in Canada, which, to some extent, has always been shared with the provinces, has become increasingly decentralised in recent years. This has put pressure on the framework of tax harmonisation in Canada and raised the need for provincial participation in any reform.

It is with these features in mind that the Canadian tax system as it has currently evolved is first briefly reviewed in this chapter. This will provide the basis for considering some of the major issues that have faced the system, and how tax policy has responded to them in the past and should tackle them in the future.

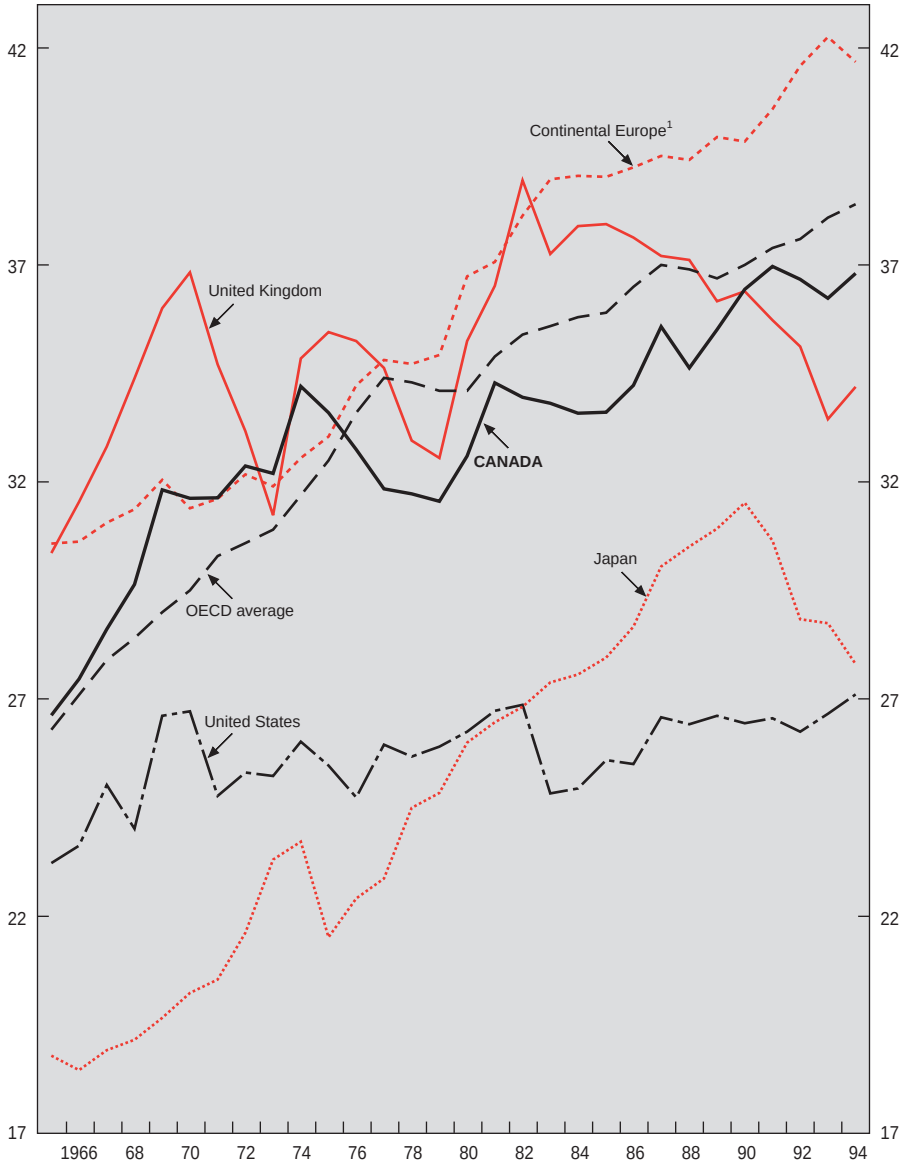
Main features of the system

As in other industrialised countries, the Canadian tax system consists of several types of taxes, which are described in Annex I. Although the structure of each of the main types of tax bears much similarity to those found elsewhere, there are some unique features. In particular, the tax mix reflects the fact that revenue raising responsibilities are divided between the three main tiers of government – federal, provincial and municipal – with each level having significant discretion over tax policy. The following discussion will focus largely on taxes levied at the federal and provincial levels, whose tax systems are inextricably linked.[38]

Relative to other OECD countries, the following aspects of the Canadian tax system are worth noting. *First,* at around 36 per cent of GDP, tax revenues are close to the OECD average, being lower than in continental European countries, but significantly higher than in the United States, Canada's major trading partner (Figure 13). The tax-to-GDP ratio has grown gradually for the past thirty years, both in Canada and in the rest of the OECD countries, with Canada's relative position changing little. *Second,* Canada relies more on personal income taxes than any other G7 country (Figure 14). Notwithstanding their recent rise, payroll taxes tend to be less important in Canada, which has, together with the United Kingdom, the lowest ratio of social security contributions to GDP. Taxes on goods and services are lower than in Europe, but considerably above those found in the United States. Among other levies, property taxes are relatively important (see below), while unlike most other OECD countries, inheritance or bequest taxation is not a feature of the Canadian tax system.

Third, the Canadian tax system is highly decentralised and is gradually becoming more so. Figure 15 depicts how the share of total tax revenues for the

Figure 13. **TAX REVENUE AS A PERCENTAGE OF GDP**



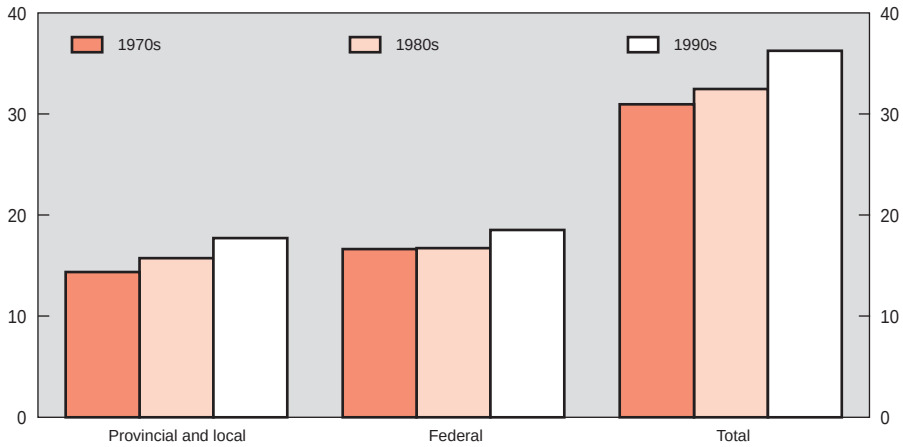
1. France, Germany, Italy.
Source: OECD, *Revenue Statistics, National Accounts.*

Figure 14. **TAXES BY TYPE AS A PERCENTAGE OF GDP**

1. France, Germany, Italy.
Source: OECD, *Revenue Statistics, National Accounts.*

Figure 15. **TAX REVENUES BY LEVEL OF GOVERNMENT**

Per cent of GDP



Source: Statistics Canada, *National Accounts.*

federal and provincial/local governments has changed over the past three decades. As can be seen, although revenues as a proportion of GDP have been rising for each level, the provincial/local share has increased at a slightly higher rate. As a result, the 36 per cent of GDP collected as tax revenues is now roughly equally divided between the federal level and provincial/municipal governments.

The federal government relies relatively more heavily on income taxation and on payroll taxes than the provinces (table 17). It currently obtains more than half of its revenues from the personal and corporate income tax. Payroll taxes (earmarked for social insurance programmes) account for one-seventh and consumption taxation, including both the Goods and Services Tax (GST) and various excise taxes and duties, for over one-fifth. The provinces rely on indirect taxes to a similar extent, but significantly less on income taxes and payroll taxation, the latter mainly used to finance health and social insurance programmes. Instead, a substantial part of their revenue comes from other sources, in particular property taxes and various natural resource levies. These tax mixes have changed little since the mid-1980s. Federal reliance on income, consumption and payroll taxes

Table 17. **Federal and provincial tax mixes**

Share of total revenues, per cent

Tax Type	Federal government		Provincial governments	
	1985/86	1995/96	1985/86	1995/96
Income	52.0	55.0	33.8	37.0
Consumption	20.1	22.3	22.7	23.0
Payroll	10.6	13.4	8.9	8.1
Property	–	–	1.4	4.5
Resources	4.2	–	10.0	4.5
Other	13.1	9.3	23.2	22.9

Source: Statistics Canada; OECD.

have all risen slightly at the expense of other revenues (Table 17). At the provincial level, income and property taxes have become more important, while the use of natural resource levies has declined.

The steepness of the personal income tax structure has been reduced markedly as a result of the reforms in the late 1980s, which limited the number of brackets while broadening the tax base (see 1990 Survey). At the same time, considerable progressivity has been maintained, especially at the lower end of the income scale. Indeed, for the lowest-income persons with children, average tax rates have turned negative, owing largely to the institution of refundable tax credits, as discussed below.[39] In addition, high-income surtaxes have contributed to the progressive nature of the tax system[40] if account is taken of income growth since the 1980s, average tax rates have generally tended to rise, while marginal tax rates have increased significantly especially for middle-income earners (Table 18). As a result, Canadian marginal personal income tax rates for that category have moved from below to above the OECD average (Table 19).

With the income tax reform implemented in the late 1980s, the corporate tax rate has dropped too (Table 20). But virtually all of this decline has been at the federal level; provincial tax rates have fallen little, except for small business rates. Indeed, some provincial corporate tax rates have actually increased after the reform. There are preferential rates for small business at both the federal and provincial level. Moreover, tax credits are available in high unemployment regions and for research and development.

Table 18. **Marginal and average personal income tax rates for a typical single individual**

Per cent

Earnings (1997 C$)	Marginal tax rates				Average tax rates			
	1984	1988	1992	1997	1984	1988	1992	1997
0	2.3	2.3	3.0	2.9	0.0	0.0	0.0	0.0
5 000	4.1	4.3	5.4	5.9	0.0	0.0	0.0	0.0
10 000	11.3	30.0	29.1	29.0	1.2	4.4	6.2	8.2
20 000	30.9	30.0	31.1	31.0	12.6	15.1	15.9	16.6
25 000	30.9	30.0	31.1	31.0	15.4	17.2	17.9	18.3
30 000	32.3	30.0	36.1	50.1	17.5	18.6	19.2	19.7
40 000	33.8	41.1	43.7	40.8	20.9	21.7	23.2	24.7
50 000	36.8	41.1	41.5	40.8	23.9	25.5	26.9	27.9
60 000	39.3	41.1	41.5	45.5	26.3	28.1	29.3	30.1
70 000	44.1	41.1	46.3	47.0	28.9	30.0	31.3	32.4
80 000	44.1	45.8	47.7	47.0	30.8	31.8	33.3	34.3
90 000	44.1	45.8	47.7	47.0	32.3	33.3	34.9	35.7
100 000	47.7	45.8	47.7	47.0	33.4	34.6	36.2	36.8

Notes: Single individual with no dependant and income from employment only.
Earnings were converted to 1997 dollars using the growth in average production workers' wage.
A representative level of provincial income tax is included (*i.e.*, 47 per cent of basic federal tax in 1984, 55 per cent in 1988, 55 per cent in 1992, and 54 per cent in 1997).
Marginal tax rates reflect the Goods and Services Tax Credit (GSTC), Canada pension Plan (CPP) and the Employment Insurance (EI) contributions to best reflect the impact at the margin on take-home pay.
Average tax rates measure the progressivity of the income tax system only and do not reflect the GSTC and CPP/EI contributions. This approach was chosen because the purpose of the GSTC is to compensate lower income families for the impact of the 1991 sales reform, while CPP and EI are earmarked to finance CPP and EI benefits.

Source: Department of Finance.

Although the provinces together raise about the same amount in tax revenues as the federal government and have virtually independent authority over both the type and level of taxes they choose to levy, their policies are affected by decisions at the federal level and by the amount of revenue that is transferred to them from the federal government.[41] Finally, it is worth mentioning that fiscal decentralisation does not stop with the provinces. The relative importance of property taxes (including business occupancy taxes) is higher in Canada than in any other OECD country (until the late 1980s, Canada ranked second to the United Kingdom, Figure 16). These taxes apply both to residential property and to business property. While these are largely municipal taxes, provinces have significant indirect control over them given that municipal responsibilities are determined by provincial legislation.

An important aspect of Canadian taxation concerns the interaction between the federal and provincial tax systems. The Canadian federation is fiscally more

Table 19. **Marginal tax rates by income level**

Single person

	66 per cent of APW[1]		100 per cent of APW[1]		200[2] per cent of APW[1]	
	1978	1995	1978	1995	1978	1995
United States	31.6	29.9	37.6	29.9	46.5	42.9
Japan	10.1	15.0	13.7	19.4	24.4	28.1
Germany	38.2	50.6	53.2	52.0	48.6	48.8
France	23.0	28.0[3]	29.5	35.6[3]	30.1	37.0[3]
Italy	19.8	34.3	25.3	34.3	32.7	41.1
United Kingdom	39.5	35.0	39.5	35.0	33.0	40.0
Canada	29.8	31.4	33.1	45.9	46.1	48.1
Australia	33.5	39.5	33.5	35.5	47.5	48.5
Belgium	35.5	54.8	46.1	54.8	48.8	61.8
Denmark	41.5	51.7	55.9	54.5	66.7	66.3
Finland	37.5	46.7	49.3	53.1	57.9	58.7
Netherlands	44.3	48.4	50.8	55.9	50.0	50.0
Norway	42.6	35.8	47.6	45.3	69.6	49.5
Spain	20.7	30.3	21.7	32.5	23.6	30.4
Sweden	41.7	37.2	59.7	37.2	81.7	56.5
Average	32.6	37.9	39.7	41.4	47.1	47.2

1. Average production wage.
2. Above this income level, capital income is likely to be signifiant.
3. 1994 figure.

Source: OECD, Directorate for Financial, Fiscal and Enterprise Affairs.

Table 20. **Federal and provincial top corporate tax rates**

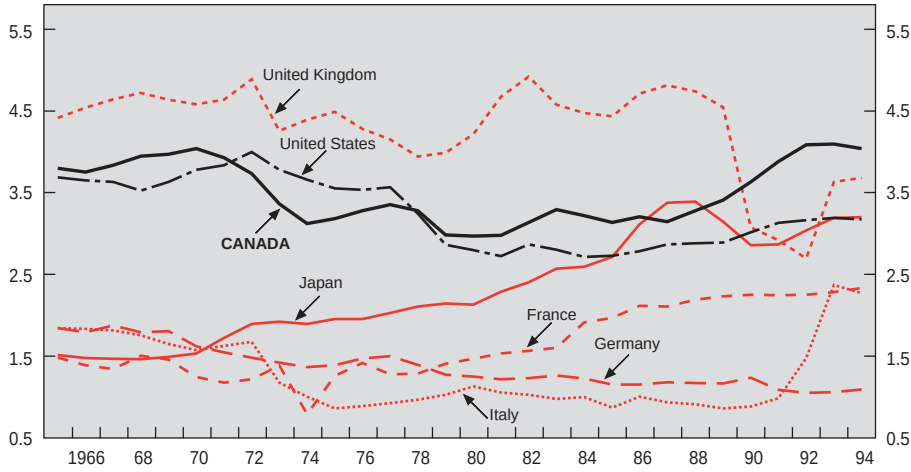
Per cent

	Federal		Provincial[1]		Combined[2]	
	General	Small business	General	Small business	General	Small business
1972	45.5	23.3	12.0	11.0	47.5	24.3
1987	46.6	24.9	15.0	10.0	51.6	24.9
1996	39.1	23.1	15.5	8.0	44.6	21.1

1. Averaged over provinces.
2. Taking into account 10 per cent credit allowed.

Source: Treff and Perry (1997).

Figure 16. **TAXES ON PROPERTY AS A PERCENTAGE OF GDP**



Source: OECD, *Revenue Statistics; National Accounts.*

decentralised than other OECD countries, judging from the proportion of public expenditures and revenues accounted for by lower-level governments and the fact that federal and provincial governments co-occupy most of the major tax bases. this is an important dimension of the tax structure as the extent of harmonisation of the federal and provincial tax systems can affect administration, collection and compliance costs as well as the efficiency of the tax system. Given the extent of decentralisation of revenue-raising responsibility in Canada, provincial tax systems are nevertheless quite similar. Indeed, one of the major components of the federal-provincial transfer systems, Equalization, whose main features are described in Annex I, relies on provincial tax bases being harmonised to some degree.[42]

There are two formal agreements for administering federal and provincial taxes: the Tax Collection Agreements (TCAs) and the Comprehensive Integrated Tax Coordination Agreements (CITCAs), both outlined in Annex I. TCAs are agreements signed by the federal government with individual provinces according to which the federal government assumes responsibility for collecting their income taxes, provided they abide by the base and rate structure set by the federal government.[43] CITCAs are the means of harmonising the Goods and Services

Tax (GST) with provincial sales taxes, and have been so far negotiated between the federal government and three of the Atlantic Provinces (New Brunswick, Newfoundland and Nova Scotia). For these provinces, the federal government administers a common value-added tax system, referred to as a Harmonized Sales Tax (HST).[44]

Concerns with the system

In Canada, like in many OECD countries, increased international competition, the globalisation of international capital markets and fiscal stringency resulting from high levels of public debt pose a challenge for tax policy, which is to ensure that the tax system conforms to the realities of the modern international economy while at the same time preserving the special features of the Canadian fiscal setting and social objectives. In this section the implications of international competitiveness for the taxation of business and labour income are touched on first. Then the issue of the mix of taxes among the broad tax types is taken up, given the differences that exist in this respect between Canada and other OECD countries. Finally, more specific issues of tax reform are dealt with, including the taxation of capital income at the personal level, the progressivity of the tax and transfer system, and the indexation of the income tax system. Problems for tax policy that arise in the increasingly decentralised Canadian federation are discussed separately in the next section.

Taxes and international competitiveness

The high degree of openness of the Canadian economy has essentially three broad implications for tax policy. *First,* in order to ensure that resources are allocated in ways that best reflect the economy's comparative advantage, taxes and other instruments of policy should be as neutral as possible to provide a level playing field. *Second,* it is increasingly recognised that higher growth rates can bring dynamic efficiency gains. Thus, tax policy should avoid discouraging investment as well as saving, both of which can contribute to growth. *Third,* important sources of growth and productivity include the process of invention and innovation, the willingness of people to become entrepreneurs, and the propensity of firms to engage in risky ventures. Here, as well, the tax system can be an important influence.

There are a number of characteristics of tax systems that might be conducive to these aspects of competitiveness:

- Taxes on businesses should avoid introducing differential marginal incentive effects on the various types of decisions firms take. Deductions for capital expenditures should reflect the proper cost of capital usage.[45]
- Taxes on labour payments, on sales and on property values should not vary systematically across different activities.
- Average tax rates on business and on personal incomes should not be significantly higher than those in competing jurisdictions to ensure that internationally mobile businesses and skilled persons are not discouraged from locating for tax reasons.
- Taxation should not discriminate against innovation, entrepreneurship and human capital formation and, given the externalities involved, might even encourage them.

All in all, these objectives of competitiveness suggest that taxes on business should apply to bases that are relatively broad and with rates that are as low and uniform as possible. As well, business income tax systems should aim to make marginal effective tax rates for various types of investments broadly uniform by allowing deductions that reflect imputed capital costs.

Business taxation in Canada fares reasonably well by these criteria, especially since the reforms of the late 1980s and early 1990s, when, as noted, income tax bases were broadened and rates reduced. The GST has ensured that domestic industries are treated evenly with foreign goods and services at the federal level. Statutory tax rates for manufacturing are comparable to those in the United States, and for certain taxes, like payroll taxes, rates are lower in Canada than in other OECD countries. Savings for retirement are sheltered relatively generously while substantial incentives exist for investing in R&D. Some potential sources of concern remain, however:

- Relative to the United States, personal income tax rates are quite high, especially for upper income persons who are subject to federal and provincial surtaxes. For example, according to OECD Secretariat calculations (Table 19), the marginal tax rate for a person earning twice the average production wage is 48 per cent in Canada compared with 43 per cent in the United States (and 47 per cent in all OECD countries).[46] Similar differentials exist for average tax rates. This could make it more

difficult to attract and retain highly skilled persons who are likely the most mobile internationally.

- The disincentive imposed on investment by the business income tax system tends to be relatively high for large firms. A recent study by Chen and McKenzie (1996) found that the marginal effective tax rate (METR) on investment in manufacturing was higher in Canada than in all countries studied except for Germany and Japan (Table 21).[47] Similar patterns prevailed for services, with the differential between Canada and the United States, in particular, being even larger.
- Considerable unevenness of tax treatment of various sectors remains. Mining as well as oil and gas have especially low METRs (Figure 17) due to generous write-offs for locating, acquiring and exploiting resource properties.[48] Manufacturing industries have the next lowest ones. By contrast, relatively high METRs are found in construction and trade industries. This intersectoral variation in METRs, which is compounded by differing tax rates across the provinces, suggests potential efficiency losses that could impede economic growth.[49]
- Although payroll tax rates are low compared with other OECD countries, the same is not the case for total taxes payable on labour, including both income tax and social security contributions (Table 22). The average effective tax rate on labour income in Canada is 29 per cent, which is higher than all except the continental European OECD Members. Moreover, Canada's relative position in this respect has worsened over the past two decades.
- The provincial sales tax systems that are not yet harmonised with the federal GST discriminate against goods relative to services and adversely affect domestically produced relative to foreign goods.
- Despite the favourable income tax treatment of savings for retirement,[50] some types of capital income that are unsheltered are taxed at relatively high rates. Dividend and capital gain receipts are more favourably treated than interest income.[51] This could provide a disincentive to save through interest-bearing securities, thereby reducing funds available to capital markets and for supplementing retirement savings. It also could represent an incentive to avoid taxes by, for example, moving assets offshore.[52]

Table 21. **Marginal effective tax rates for large firms with domestic investors**

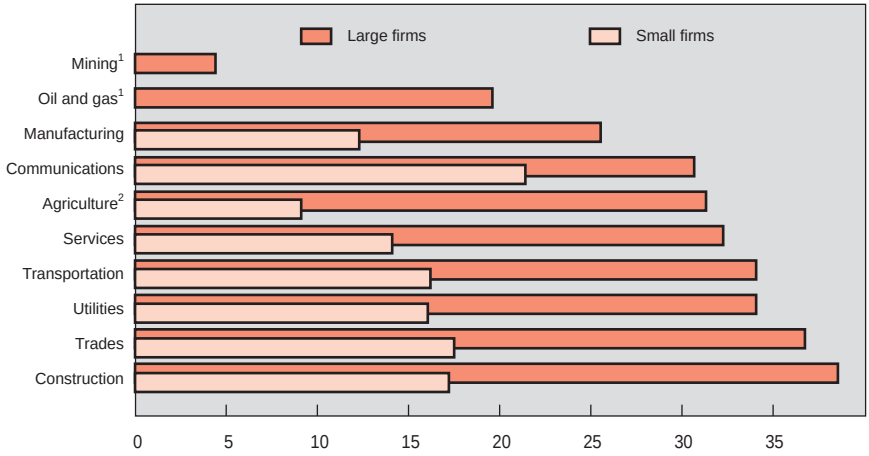
Per cent

	Manufacturing	Services
Canada	25.5	32.2
United States	21.5	19.9
United Kingdom	20.2	19.0
Germany	27.5	33.1
France	21.9	25.5
Italy	22.1	34.1
Japan	32.0	33.9
Mexico	16.5	17.7
Hong Kong	11.9	3.7

Source: Chen and McKenzie (1996), Table 2.

Figure 17. **INTERSECTORAL MARGINAL EFFECTIVE TAX RATES**

Per cent



1. Not applicable for small firms.
2. Including fishing and forestry.

Source: Chen and Mc Kenzie (1996).

Table 22. **Average effective tax rates on labour**[1]

Per cent

	1970	1980	1994
Canada	19	21	29
United States	18	21	23
United Kingdom	—	24	21
Germany	28	35	38
France	28	37	45
Italy	—	26	35
Japan	12	17	21

1. Calculated as income taxes on wages plus payroll taxes, divided by wages and salaries plus employers' contributions to social security and private pension schemes.

Source: OECD, *Revenue Statistics* and *National Accounts*.

– Finally, property and other local taxes are relatively high in Canada compared with competing jurisdictions, a situation that is unlikely to be mitigated in the near future as provincial governments pass on part of their deficits to municipalities by reducing transfers.

The design of tax policy to enhance competitiveness, innovation and growth must be tempered by other objectives of the system, such as the need to achieve desired norms of redistribution, discussed below. In this regard, the openness of the economy implies the following constraints for tax policy:

– To the extent that immigration rules are less strict for certain types of labour, mobility is restricted more by attachment to country and the costs of moving. Some types of labour might be relatively mobile, such as entrepreneurs, executives, or professionals. If so, imposing high taxes on them for, say, redistributive purposes, can be partially frustrated, a consideration that is particularly relevant for Canada because of the proximity of the United States.
– Attempting to impose taxes on products and factors of production that are mobile across borders will be frustrated as much of the tax will be passed back to immobile factors and will end up distorting domestic production patterns in capricious ways. In Canada, capital is highly mobile, so taxing capital income at source can be largely thwarted, except to the extent that international crediting arrangements exist.

– The existence of a consolidated return of income for corporate groups in the United States has a bearing on the role and design of corporate taxes in Canada.[53] However, any advantage of conforming with the US corporate tax system to benefit from consolidation must be set against the distortion that a corporate tax system has on domestic investment. The best that can be hoped for is that the corporate tax at least does not differentiate among different sectors or types of investments.[54]
– Finally, the opportunities for some forms of avoidance are enhanced in an international environment. In particular, capital income earned abroad can be difficult to monitor, especially if it can be made to accrue in countries without exchange of information provisions on tax havens. As well, methods like transfer pricing and intra-group borrowing and lending can be used to boost business income reported in low-tax jurisdictions. As recent Canadian experience shows, some forms of commodity taxation can also be avoided by cross-border shopping or outright smuggling.

The tax mix

Model simulations can be used to gauge the relative economic costs of various tax sources. For Canada, they suggest that sales and payroll taxes are least distortionary (Table 23). Thus, shifting the tax mix towards these revenue sources could have a positive effect on economic performance. Moreover, as Canada relies more on income taxes and less on payroll and consumption taxes than most OECD countries, it might be possible to change the tax mix without running the risk of putting Canadian businesses at a competitive disadvantage. With this in mind, each of the main tax bases is examined in turn.

Table 23. **Illustrative real output loss from extra dollar of tax**[1]

	Cents
Corporate income tax	155
Personal income tax	56
Payroll tax	27
Sales tax	17

1. Preliminary estimates. The payroll tax estimates are for a generalised tax with a flat rate on all payroll in contrast to the current EI contribution with a maximum ceiling.

Source: Department of Finance.

Income taxes. The fact that personal and business income tax rates are high relative to the United States suggests that there may be scope for reducing reliance on these tax sources. This would involve, however, some co-ordination between the two levels of government because unilateral federal tax reductions could be undone by the provinces, with no effect on the overall tax mix. This has not been a problem in recent years as many provinces have reduced their income tax rates (see below), but it is difficult to predict what will happen in the future as the provinces adjust to lower federal transfers. Significant cuts in income tax rates would probably require increased reliance on other taxes.

Sales taxes. Despite the fact that indirect taxation is higher than in the United States, there would seem to be scope for raising the share of sales taxes as revenue sources, provided possible adverse consequences could be mitigated. This means, in particular, that such tax increases should take the VAT form. Thus, the prospects for extending the use of sales taxes are tied to the success in harmonising the system with participation of the larger provinces, which may imply a higher provincial share of indirect tax revenues (see below). Increased use of sales taxation might be objected to because of its regressivity relative to, say, income taxation. But the adverse consequences for lower income persons could be overcome by enhancing refundable tax credits.

Payroll taxes. Among the large OECD countries, Canada has the lowest level of payroll taxes despite their strong increase since the 1980s.[55] The federal government has used them mainly as a means of financing Employment Insurance (EI) and the Canada Pension Plan (CPP), while the provinces have relied on them for workers' compensation and some more generally for financing health and education. A case could be made for using payroll taxation also as a general revenue source.[56] It has a broad base, comparable to consumption, is relatively easy to administer, is an efficient source of revenues, and does not discourage saving or investment. Like consumption taxation, it is not progressive, but that too could be remedied by enhancing refundable credits. Moreover, it is a tax that is well suited to use by the provinces and so could contribute to the need to make tax room available to them. However, there are a number of concerns with payroll taxes: they may be onerous on small firms and companies that have cash flow problems, and there is evidence that they have adverse effects on employment in the short run, although this may well be dissipated in the longer run as the tax is absorbed into wages.[57]

Changes in the tax mix can also have intergenerational effects given that different taxes are collected at different times over the life cycle. Payroll taxes tend to be paid earlier in one's life than consumption and income taxes. This implies that substituting payroll taxes for consumption or income taxes is analogous to an intergenerational transfer from younger to older generations. Thus, the case for any change in the tax mix must weigh the intergenerational redistributive effects against the gains in efficiency and administrative convenience that might occur.

Taxation of capital income

Except for sheltered forms of savings, interest income is taxed at the same statutory rate as other forms of income under the personal income tax system, and can be subject to some additional taxation through the corporate tax. As noted, capital gains or dividend income receive preferential treatment. A very high proportion of assets are in fact treated on a consumption tax basis, either because their capital income is exempt from taxation or because they are implicitly taxed on a cash flow basis. The return on housing and other consumer durables is tax exempt. Much of personal investment in human capital accumulation is effectively taxed on a cash flow basis (given that forgone earnings are the most important category of costs of accumulating human capital). Moreover, the tax on savings for retirement through individual Registered Retirement Savings Plans (RRSPs) and company Registered Pension Plans (RPPs) is deferred until amounts are paid out of the plans. One of the implications of this is that the attempt to include the remaining forms of capital income in the tax system inevitably leads to a distortion in the allocation of assets among various types, sheltered versus unsheltered.

Though the move to a full-fledged personal consumption tax is highly unlikely in Canada, changes in the tax sheltering of capital income can still be important. From this perspective, the three areas of saving for retirement, housing and human capital investment remain on the reform agenda. Given the pressures on the public pension system (see the 1996 OECD Survey), the case for maintaining some form of tax assistance for retirement savings is strong. The current system of taxing pension accumulations on a cash-flow basis seems the fairest and most efficient. The option of adopting a crediting system for contributions to retirement saving plans gives rise to potentially enormous administrative com-

plexities when it comes to the treatment of accumulating funds and their taxation on withdrawal. By the same token, changing the tax treatment of housing or human capital accumulation would be wrought with difficulties. In this context, interest deductibility of the costs of borrowing to finance education, and the use of income-contingent loans to insure against the riskiness associated with human capital accumulation are measures that have been advocated recently in Canada.[58]

Recent federal budgets have modified somewhat the taxation of capital income. In the case of savings for retirement, the limits on contributions to Registered Retirement Savings Plans (RRSPs) and employer-sponsored Registered Pension Plans (RPPs) were reduced, and the maximum pension limit for defined-benefit RPPs affecting higher-income individuals was frozen. Though ostensibly introduced to enhance the fairness of the tax system, these measures detract from the long-run objective of affording equivalent tax treatment to the various forms of retirement savings. On the other hand, the tax treatment of human capital investment has been improved. Deductions for tuition and for students attending educational institutions have been enhanced, and contribution limits to Registered Education Savings Plans, devices for deferring tax on the return on amounts set aside for post-secondary education, have been increased.

The redistributive dimension

The tax system fulfils an important redistributive role, but it is only one of many policy instruments for achieving this goal.[59] A growing consensus is emerging that, given the adverse incentive effects of progressive taxation and its consequences for compliance, the income tax rate structure should not be too steep. The recent moves in many countries to broaden the base and flatten the rate structure accords well with this advice. At the same time, it has been increasingly recognised that a substantial amount of redistribution, perhaps the lion's share, takes place through the expenditure side of the budget.[60] As noted, although the income tax rate structure has been flattened in Canada, the broadening of the base,[61] the replacement of deductions by credits, and the use of surtaxes on higher income taxpayers has apparently increased, on balance, the progressivity of the tax system.

Two relevant aspects of redistributive policy concern the poorest and the richest. Even though the income tax rate structure may not be steep, it can still

have an important role to play in providing financial assistance to the poor and needy, including the working poor, the elderly, the disabled and children. Although it has long been recognised that the income tax system could be used as a vehicle for delivering transfers to persons in a non-taxpaying position as well as for collecting taxes, this potential has rarely been exploited. The enhancement of refundable tax credits to low-income families in Canada provides an opportunity to let the income tax system take effect for those in a negative tax position.

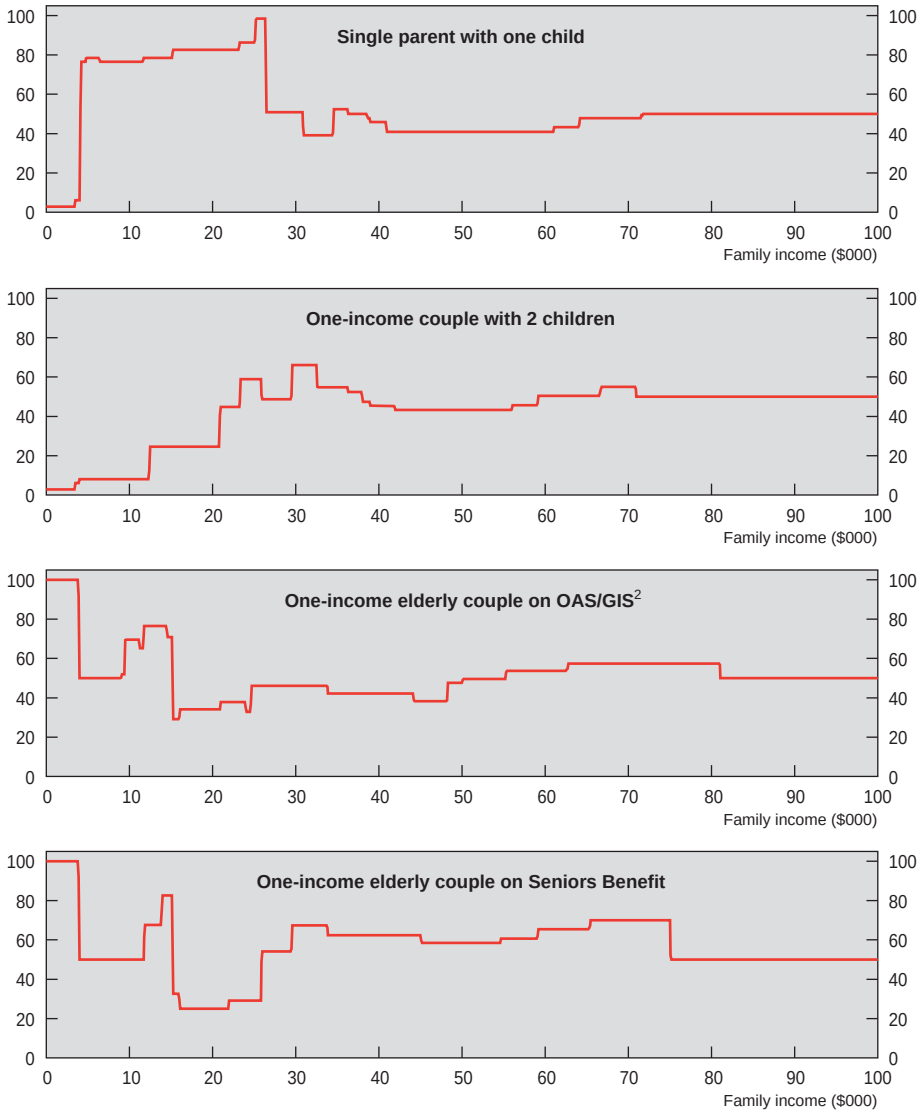
The federal government currently uses two refundable tax credits to assist low-income families. The GST credit provides for quarterly payments to low-income persons on the basis of their family income and marital status. The Child Tax Benefit is based on the number of children and also falls with family income. In addition, a Working Income Supplement is available to help defray the costs that low-income families incur when entering the labour force.

Revenue Canada has substantial experience in delivering income-tested benefits. Further use of refundable credits delivered through the tax system offers opportunities for improving the system of redistributing to the poor, now accomplished mainly through provincial welfare programmes. Integrating the tax and transfer systems allows transfers to be properly income-tested. In some instances, tax-back rates can be more easily tailored to trade-off the disincentive to work against the revenue costs and the adequacy of benefits provided. This may make it easier to avoid the very high marginal tax rates that still face some low-income persons (Figure 18). The issue becomes particularly relevant as the earnings distribution widens and the proportion of elderly in the population increases, while governments continue their recent trend of moving away from universal transfers to targeted ones.

Refundable tax credits would complement welfare programmes, not replace them. The latter have some advantages in terms of targeting transfers to those in need and responding quickly to changing circumstances of recipients. Given that the provinces and municipalities deliver welfare, the problem of work disincentives in transfer programmes cannot be addressed by the federal government in isolation. The high tax-back rates that can impose extraordinarily large effective marginal tax rates occur in provincial welfare systems: transfers to the poor can have tax-back rates of more than 100 per cent for persons beginning active employment.[62]

Figure 18. **EXAMPLES OF EFFECTIVE MARGINAL PERSONAL INCOME TAX RATE**[1]

Per cent



1. Combined federal-provincial rate. Ontario. 1998 tax year.
2. Old Age Security/Guaranteed Income Supplement.

Source: Department of Finance.

Some evidence exists that not only the income tax system but also the combined tax-transfer system has become somewhat more progressive in recent years. As Table 24 shows, higher-income families of all sizes had higher average tax rates in 1994 than in 1970, while almost all lower-income families with children had lower average tax rates. As noted, this reflects the combination of changes in the rate structure, the substitution of tax credits for many deductions, and especially the institution of refundable tax credits. Table 25, which lists average rates of income tax and transfer payments,[63] reveals that the transfer system has become much more progressive as well. Lower-income persons, especially those in the second and third lowest quintiles, have had a very large increase in the rate of transfers. When taxes and transfers are combined, the net effect is a definite increase in progressivity, at least for part of the income range. For the bottom three quintiles, transfers less taxes as a per cent of income have increased, while for the top group they have decreased by a modest amount.

Table 24. **Average income tax rates by family real income and size**

Per cent

Family income (C$ 1994)	Family size	1970	1994
18 200	2	9.4	3.4
	4	6.3	–10.2
	6	3.6	–24.5
30 400	2	16.1	15.1
	4	13.7	7.9
	6	11.4	–0.9
42 400	2	19.3	22.5
	4	17.7	19.2
	6	16.1	13.5
60 800	2	23.7	28.0
	4	22.1	27.1
	6	20.6	23.2
91 200	2	30.6	33.6
	4	29.3	33.6
	6	28.1	32.1
121 600	2	34.9	36.5
	4	33.8	36.5
	6	32.7	36.5

Source: Smith (1995).

Table 25. **Average income tax and transfer rates by quintile**

Per cent of income

	Quintile					Total
	Lowest	Second	Third	Fourth	Highest	
	Income tax					
1971	2.1	7.5	12.1	14.8	19.7	15.2
1986	2.5	8.7	14.4	17.6	21.8	17.2
1993	2.6	8.4	15.0	19.2	24.4	18.9
	Transfer payments					
1971	53.4	18.2	5.7	3.4	2.0	6.6
1986	59.5	31.2	13.0	6.9	3.3	11.3
1993	67.8	40.4	18.0	9.7	4.2	14.5
	Transfer payments minus income tax					
1971	51.3	10.7	–6.4	–11.4	–17.7	–8.6
1986	57.0	22.5	–1.4	–10.7	–18.5	–5.9
1993	65.2	32.0	3.0	–9.5	–20.2	–4.4

Source: Smith (1995); Statistics Canada, *Income After Tax, Distribution by Size in Canada* (1993).

While the income tax system can be fruitfully extended to address inequality at the lower end of the income distribution, it is not clear that it is sufficient for that at the upper end. Some of the inequality there is generated by differential lifetime earnings, and to deal with that the income tax system seems adequate. But some of it is generated by inherited wealth. Most OECD countries view inherited wealth as a legitimate base for redistributive taxation, either through transfer taxes on bequests or inheritances, or through wealth taxes. Canada stands out as an exception. In assessing the case for wealth transfer taxes, one must recognise that the tax system includes measures that have similar effects. Property tax rates on real property tend to be relatively high; accrued capital gains are generally taxed upon death; and capital income that is not sheltered is taxed, which is essentially equivalent to taxing the wealth itself. What might be accomplished by wealth transfer taxation is the taxation of large estates. Against this is the fact that wealth transfer taxes in other countries generate relatively little amounts of revenue, and they might have potentially adverse effects on wealth holdings within Canada.

Indexation

A final concern involves the indexation of the tax system, an issue that is mainly relevant for personal and corporate income taxes. Since the mid-1980s, partial indexation applies in the personal income tax system: tax brackets and the levels of exemptions and credits are indexed annually at the rate of CPI inflation in excess of 3 per cent. So too are the refundable credits and the threshold family income levels that apply to them. Because of low inflation rates, indexation has not been triggered since 1992. A consequence of partial indexation has been ''bracket creep'', whereby personal tax liabilities rise simply because an increase in nominal incomes causes a higher proportion of incomes to fall into higher tax brackets. It is estimated that, over the ten years to 1998, 18 per cent of tax filers will have either become taxable or been pushed into a higher tax bracket due to only partial indexation.[64] As a result, average tax rates have increased at all income levels, but proportionately more for low- and moderate-income individuals (Table 26). Overall, the absence of full indexation has pushed up the federal income tax rate by 1½ percentage points over the past decade.

Part of the motivation for the 3 per cent threshold is that the CPI likely overstates the actual rate of inflation, though the measurement error is probably less than 1 per cent (see Chapter II). More importantly, the revenue gain from

Table 26. **Impact of partial indexation on the basic federal income tax rate**
1988-1998

1998 individual income (C$)	Share of filers (per cent)	Basic federal tax rate (per cent)		
		Fully indexed system	Partially indexed system (CPI-3)	Difference
0 to 10 000	25.4	0.2	0.7	0.5
10 to 25 000	33.0	4.1	5.3	1.3
25 to 50 000	27.3	9.8	11.6	1.7
50 to 75 000	10.1	13.9	15.5	1.6
75 to 100 000	2.5	16.0	17.5	1.5
100 to 150 000	1.1	17.5	18.6	1.1
150 000 and over	0.8	19.1	19.5	0.4
All filers	100.0	10.8	12.2	1.4

Source: Department of Finance.

bracket creep serves the useful purpose of contributing to fiscal consolidation. However, with the budget approaching balance, the indexation of the tax system will need to be addressed in the context of other structural issues that have arisen in reducing the deficit.

There is also an absence of indexation of capital income taxation in both the personal and corporate tax systems. However, it should be noted that Canada has a 75 per cent inclusion rate for capital gains, with the 25 per cent exemption fully or partially offsetting the inflation portion of any such gains. Dividends are taxed on a received basis, and investors generally get a dividend tax credit. Thus, undistributed corporate profits can be reinvested in the business without generating tax or, if redistributed, generally benefit from the integration of the corporate and personal tax system. For individuals, both nominal interest and nominal capital gains are taxed, both of which include an element of loss of real capital value, all or a portion of which is offset by the 25 per cent exemption of capital gains. Inflation is also less of a problem in recent periods of low inflation and to the extent that capital income is sheltered. At the corporate level, absence of indexation has mixed effects, lowering the value of some deductions, such as depreciation and inventory usage, but increasing the value of others, such as interest payments. Studies have found that on average such effects roughly cancel out, but for some assets inflation remains a problem. The most notable is the case of inventories where the use of FIFO accounting methods for tax purposes understates the real cost of replacement. On the other hand, businesses that rely heavily on debt finance are helped by the absence of indexation because their interest deductions include both the inflation and real interest components.

Tax policy issues in a decentralised federation

The fact that independent revenue-raising authority rests with lower levels of government poses important issues for tax policy. Provincial taxes can cause, in addition to the usual distortionary effects of taxation, inefficiencies in the inter-provincial allocation of resources. These may simply be a result of uncoordinated provincial tax policies, but they may also reflect ‘‘beggar-thy-neighbour policies’’. Decentralised tax setting may be inefficient as well because of ‘‘vertical fiscal externalities’’: a tendency to underestimate the marginal cost of public funds induces provinces to set tax rates too high on bases that are relatively

elastic.[65] And, since the provinces co-occupy the main tax fields with the federal government, the possibility exists that collection and compliance costs will be much higher if taxpayers have to deal with more than one tax administration.

Despite the potential that the decentralisation of taxing authority has for causing inefficiency and administratively cumbersome tax systems, some mechanisms exist for mitigating such effects, notably the Equalization system[66] and the tax harmonisation arrangements. With respect to the former, the issue for the coming years is whether Equalization can continue to serve this purpose adequately. The decentralisation of revenue-raising responsibility has proceeded rapidly over the past few years as federal-provincial transfers have fallen as part of the federal government's deficit cutting strategy. All provinces now obtain a considerably higher proportion of their revenues from own sources (Table 27). This trend will continue as the phased-in curtailment of federal transfers to the provinces works its way through the economy in the coming years. Most of the reduction has been in the form of specific purpose transfers, especially those earmarked for health, post-secondary education and welfare. The share of provincial revenues accounted for by general purpose transfers, almost all of which comes from ''Equalization'', has changed relatively little until now. As provinces become more reliant on own source revenues, Equalization requirements could grow. Since entitlements increase with the average level of provincial tax rates

Table 27. **Federal transfers as a percentage of provincial revenues**

	General purpose transfers		Specific purpose transfers		Own-source revenues	
	1975/76	1996/97	1975/76	1996/97	1975/76	1996/97
Newfoundland	27.6	30.3	21.7	12.7	50.7	57.0
Prince Edward Island	27.1	23.9	17.7	12.5	45.2	63.6
Nova Scotia	28.3	27.5	19.4	12.8	52.3	59.8
New Brunswick	23.8	23.9	23.6	14.4	52.6	61.7
Quebec	11.8	9.9	10.8	7.7	77.4	82.4
Ontario	2.7	..	18.5	12.9	78.8	87.1
Manitoba	13.6	19.8	22.9	11.9	63.5	68.2
Saskatchewan	7.7	5.9	16.1	11.4	76.2	82.8
Alberta	1.7	..	11.8	10.1	86.5	89.9
British Columbia	1.4	..	17.2	9.2	81.4	90.8

Source: Department of Finance, provincial budgets.

(see Annex I), they could rise in the future, if provinces end up raising their tax rates to make up in part for their reduced transfer income.

The benefits of decentralised taxing authority can be achieved without sacrificing a harmonised and administratively simple tax system by federal-provincial tax collection agreements (TCAs). This has been a success story in the case of income taxes in Canada. The TCAs have preserved a single tax administration for participating provinces (that is, all provinces except Quebec), and have resulted in a fairly uniform base and rate structure for the personal and corporation income taxes. They have also enabled the provinces to exercise fiscal responsibility and accountability by selecting their own tax rates. In the case of sales taxation, harmonisation has begun with the implementation of the HST in three of the Atlantic provinces and the harmonisation of Quebec's sales tax with the GST. But at the same time the pressures on the TCAs are growing. Provinces are asking for, and being granted, an increasing number of special credits and other measures, some of which could reduce the efficiency of the internal common market.[67]

In this regard the federal government has a key role in co-ordinating the national tax system. Such a co-ordination function, which takes as a rule that the internal economic union must not be distorted by any measure, has benefits for both large and small provinces by helping to ensure the efficient allocation of resources. In carrying out this function, the federal government must of necessity limit the flexibility of a specific province, to the betterment of the union as a whole. For example, with the intent of increasing economic activity within their respective jurisdictions, several provinces have instituted tax credits that reduce the cost of equity financing for businesses which have a limited presence in their territories. In the case of venture capital and stock savings plan tax credits, investors earn a tax credit determined as a percentage of their investment, provided that the wages and salaries of the firm paid in the province concerned are above some predetermined minimum. These types of credits have the potential to segregate the Canadian capital market by encouraging people to invest only in firms located in their province. Where the pool of potential investors is large enough, such tax credit programmes can result in firms relocating to access these large capital pools to the detriment of economic activity in smaller provinces and in the country as whole. It will be important to ensure that the TCAs do not break down as a result of the provinces having to abide by the federal base and rate

structure. This would have a negative impact on the provinces, the federal government and the national economy.

The source of the tension in the TCAs is the gradual erosion of the share of the income tax bases occupied by the federal government as a consequence of decentralisation and related desire by the provinces for more say in setting income tax policy. Given that the fiscal responsibilities of the provinces are now comparable to those of the federal government, it is probably time for some fundamental rethinking both about the allotment of tax room between levels of government and about the institutional means by which tax harmonisation is achieved.[68]

The key issue here is the appropriate federal-provincial sharing of the main tax bases. Which tax fields, if any, ought the federal government to be dominant in? The main arguments for not letting the federal share of income tax deteriorate are twofold. *First*, the income tax is generally regarded as one whose assignment should be primarily to the federal government because of its importance as a redistributive device and because of the fact that it applies partly to capital and business income, which is mobile across provinces.[69] *Second*, the integrity of the TCAs as a device for harmonising income taxes depends upon the federal government maintaining some dominance in the personal and corporate income tax fields.

If the federal government is to maintain a majority share of the income tax fields, and if the federal government and the provinces raise comparable levels overall, then the federal government cannot have a majority share in other fields, such as sales taxation. This must be reconciled with the equally strong desire to harmonise federal and provincial sales taxes so as to capture all the advantages for business and competitiveness of the VAT form of tax. The solution may lie in the sort of harmonisation enacted with three of the Atlantic provinces through the CITCAs, that is a collaborative arrangement that need not rely on the dominance of the federal government. One scenario might be for this framework to be extended to all provinces, accompanied by an increased provincial share of sales tax revenues, with federal income tax revenues preserved or even enhanced. Such a scenario might be attractive to the provinces, despite the reduction in tax policy autonomy that it would entail, and would allow both for sales tax harmonisation and for a strengthening of the TCAs.

Reform initiatives and prospects

The discussion so far has identified a number of potential policy issues facing Canadian authorities as they strive to have a tax system in place that can cope with the demands of the rapidly evolving international economy. The Canadian tax system has moved to address some of them since the 1980s. As the major reforms initiated in both the income tax and the sales tax areas were reviewed in detail in the 1990 OECD Survey of Canada, the following discussion focuses on current policies and concludes with an assessment of outstanding issues.

Current policies

One of the federal government priorities in recent budgets has been to improve the fairness of the tax system (Department of Finance, 1997). On the business income tax side, the main objective has been to close loopholes and to ensure that corporations help deal with the fiscal problem. This has involved the reduction of possibilities to defer tax and increases to the corporate surtax and Large Corporation Tax. A number of base broadening measures have also been taken on the personal income tax side, including the elimination both of a lifetime exemption that applied to capital gains and of some tax advantages available through family trusts, and the expansion of the base for the alternative minimum tax. To ensure that Canadians who are better off pay their fair share of taxes, the treatment of savings for retirement[70] has been made less generous, with contribution limits to tax-assisted saving plans reduced in 1996 and frozen until at least 2002. At the same time, the potential for expanding the use of tax information to deliver assistance based on family income and circumstances is about to be realised with the implementation of the Seniors Benefit, a family-income tested transfer to the elderly that will replace the existing Old Age Security (OAS) and Guaranteed Income Supplement (GIS) programmes, completing the gradual transition from universal to targeted transfers to the elderly (see 1996 OECD Survey of Canada). There is also a joint federal-provincial initiative currently under way to replace in part the component of provincial welfare payments directed at families with children by an enlarged Canada Child Tax Benefit available for the working and non-working poor alike. One can imagine other ways of using the tax administration for delivering transfers in the future.

At the provincial level, there has been a recent movement to reduce personal income taxes. In their 1996 budgets, three provinces (British Columbia, Ontario and Nova Scotia) initiated reductions in their basic personal income rates and a fourth (Alberta) announced it would consider tax cuts as a way to respond to the stronger fiscal position of the province. Ontario's planned income tax reduction is particularly large, with the tax rate scheduled to fall by 30 per cent (from 58 per cent of basic federal tax in 1995 to 40.5 per cent in 1999). At the same time, however, a new ''health levy'' was imposed, replacing the existing surtax. The effect of this is largely to offset the tax rate reduction for high-income taxpayers. A similar change was enacted in British Columbia, with a modest income tax rate reduction accompanied by an increased surtax on higher incomes, leaving the effective tax rate on high-income taxpayers little changed. Most recently, Quebec has announced income tax cuts which will be financed largely by an increase in the VAT rate. As provincial budgets are brought into balance or even surplus positions, the impetus for income tax rate reductions is likely to remain high. This may also be true at the federal level, although the government so far has excluded broad-based tax cuts.

While most components of the tax system have been subject to considerable change, the taxation of foreign-source income has not been reviewed since the basic structure was put in place in the early 1970s. However, more recently, a number of concerns have been raised with respect to the existing rules. A study commissioned by the Government's Technical Committee on Business Taxation (Arnold *et al.*, 1996) found a number of flaws in the details of the system, concerning, among other things, the deductibility of interest in respect of borrowed funds used to earn dividends out of exempt surplus, the ability to earn low-taxed or exempt income in designated treaty countries, and the ability to deduct foreign branch losses. All of these issues are complex but important, given the significant growth in outbound investment over the years, all of which has been in equity. To the extent that these equity investments have been financed by debt located in Canada, the Canadian tax base has been eroded. For these reasons, a review of foreign source income taxation would seem desirable to ensure that favourable treatment of outbound (and inbound) investment does not discriminate against domestic investment.

In 1995, the Federal Government initiated a review of the resource allowance for oil, gas and mining industries, which concluded that most corporations

in the resource sector and provincial authorities prefer the current tax structure to other options. It was also recognised that the replacement of the resource allowance with royalty deductibility would have adversely affected frontier petroleum projects and many non-conventional oil developments, particularly during the early phases of their production. The 1996 Budget, nevertheless, proposed a number of changes to tighten the resource allowance. In addition, the Technical Committee on Business Taxation was asked to re-examine the resource taxation provisions as part of its review of the business tax system.

Despite the federal government's continued efforts to harmonise sales taxation, five provinces still maintain retail sales taxes that have relatively narrow bases and suffer from the problems associated with single-stage taxes. One way to address these deficiencies would be for them to join the new HST system.[71] However, there are problems with that option from some provinces' perspective, the main one being that they would be required to abide by both a common base and common rate level and structure. The non-participating provinces, especially the larger ones like Ontario and British Columbia, have found it difficult to accept that. The HST system has opted basically for a revenue-sharing approach, similar to that used in Germany, which buys a high degree of harmonisation and administrative simplification at the expense of provincial fiscal autonomy. Of course, the Canadian provinces have some input in sales tax policy through a consultative body known as the Tax Policy Review Committee. The purpose of this body is to review issues related to the tax and to advise the Ministers of Finance of the federal government and participating provinces. Changes in the tax will be referred to this committee for consideration. The provincial tax rate can be changed if all participating provinces agree. And the latter have some say in introducing rebates that do not impose significant administrative complexities. It remains to be seen, however, how effective such a procedure would be if most provinces participated.

The alternative of allowing provinces who join the HST to have some say over their own rate structures would be very difficult to administer, given the system of crediting that must apply. It could entail having separate tax administrations in each province as is done in Quebec. However, it is known from experience elsewhere that decentralised administration of a VAT in a federal setting is inefficient. Thus, while the provinces would retain the ability to set

their own sales tax rate, the full benefits of value-added taxation would not be captured.

Recent developments in payroll taxation have been characterised by diverging movements in its two major components, that is, contributions to the Canada Pension Plan (CPP) and to Employment Insurance (EI). With the reforms to the CPP, which are intended to move the system to fuller funding (see Chapter III), the rise in pension contributions has been accelerated. On the other hand, EI contributions rates have been lowered cautiously after their strong increase in the first half of the 1990s. The main issue is ensuring that the premium rates are chosen so that EI is fully funded over time, *i.e.* that surpluses in low unemployment years cover deficits in high unemployment years. In the past, there has been a tendency not to build up a large enough surplus to cover contingencies during recessions, with the consequence that payroll tax rates have had to be increased when they were the most burdensome. Of course, part of this is due to the persistence of high unemployment rates. The lesson is that the authorities should be prudent about the size of required surplus in the EI account, and should be cautious about lowering EI premium rates. Having said this, the surplus should not be excessive: EI tax rates are regressive, given the ceiling on taxable wages and salaries; they may distort hiring versus overtime choices; and they could have persistent disemployment effects in the presence of wage rigidities. The current situation is, however, not yet in balance. Furthermore, a number of structural issues, among which is the growing cumulative surplus in the EI programme, will need to be dealt with as part of an overall strategy, in order to maximise the gains in economic efficiency and to achieve income distribution objectives.

The macroeconomic dimension

As discussed in Chapter II, in the recent past, the over-riding imperative has been to halt the accumulation of public debt by eliminating budget deficits. The federal government chose to do so by cutting expenditures rather than by raising tax rates, notwithstanding some measures to enhance tax fairness. As an important source of the spending restraint was a reduction in transfers to the provinces, those were in turn forced to focus on balancing their own budgets. For the most part, they too chose to cut expenditures rather than to increase tax rates. Transfers to municipalities also took cuts, leaving some with the need to raise revenues: to

some extent, this has been done by finding new revenue sources in user fees of various sorts, but property taxes have not been left untouched.

The end result of this series of responses has been a reduction and rationalisation of expenditures at all levels of government. While federal and provincial governments in general have been able to hold the line on tax increases, the municipalities had initially some difficulty in this respect, exacerbating Canada's position as a relatively high property tax country. Although tax rates have held steady, income tax revenues have risen more rapidly than GDP largely because of the absence of full indexation. Bracket creep is one factor that has pushed more and more income into higher tax brackets, reduced the real value of tax credits and increased liability for surtaxes. As mentioned, this has contributed significantly to deficit reduction, while at the same time increasing the burden of taxes.

For the future, the only major tax rate increase in sight is the proposal to raise the contribution rate to the CPP. The purpose of this is to enhance the funding of the pension system, as recommended in the 1996 OECD Survey of Canada. At the same time, it is conceivable that EI premiums, another compulsory payroll levy, could fall significantly in the coming years if the unemployment rate moves to a lower plateau and the EI fund continues to be in a surplus position. Moreover, with budget balance already achieved in some provinces and within reach for the federal government, there should be some scope for lowering the tax burden and improving the tax system. Indeed, although the ''fiscal dividend'' available for tax measures is limited by the over-riding goal of keeping the public debt ratio on a clear downward trend, it provides some room for manoeuvre for adjustments to taxation, provided expenditure discipline is maintained.

Scope for further action

While there appear to be no major structural deficiencies in the overall Canadian tax system, the discussion above has, nonetheless, identified some outstanding issues. The scope for coping with them would be enhanced by shifts in the tax mix and further base-broadening. In addition, the improving fiscal situation should allow measures that are not entirely revenue neutral. The following aspects of the system, in particular, would deserve consideration:

- Personal income tax rates, both average and marginal, are quite high compared with the United States. The former are especially an issue with respect to higher-income persons, who are presumably also the most mobile across international borders. The latter have the standard adverse incentive effects, including those to avoid paying taxes. Of course, tax levels cannot be judged in isolation: if Canadian citizens want higher levels of public services than in the United States, tax rates have also to be higher. Nonetheless, now that the end of budget deficits is in sight, *the authorities should give some consideration both to the adverse locational effects that high average tax rates may have on the highly-skilled and entrepreneurial portion of the workforce, and to the negative consequences for economic performance of high and uneven marginal personal income tax rates.*
- Related to the above, and as noted, Canada relies more on income taxes and less on payroll and consumption taxes compared with other OECD countries, although the latter two taxes are less distorting. It is, however, not clear to what extent the tax mix should change. On the one hand, payroll taxes are more efficient, suggesting that they could be used not only to finance specific programmes but also as a general source of revenue. On the other hand, all broad-based taxes fall ultimately on personal incomes, and the income tax serves equity objectives relatively well, contrary to payroll taxes which also may have some adverse effects on employment in the short run. Moreover, the large surplus in the EI account rather suggests that premium rates should be lowered, and pension reform will tend to increase the payroll tax burden anyway. Regarding consumption taxes, increased reliance on them would have the desired effects only if it took the VAT form. In any case, *while a shift from income taxes to payroll and consumption taxes could be considered, any initiative to change the tax mix should involve a co-ordinated approach between the federal and provincial governments since attempts by the federal government to substitute one tax for another could be readily undone by the provinces.*
- Co-ordination with the provinces is also required to pursue further harmonisation of sales and income taxes. *Sales tax harmonisation is particularly important as a means of removing the distortions imposed by provincial sales taxes on business inputs and traded products.* But the

prospects for further progress in this area rely on convincing some of the larger provinces to participate, and that may not be easy, given the loss in autonomy that is involved. Meanwhile, income tax harmonisation is under continuing strain, and will remain so with the provinces occupying a significant proportion of the income tax base. Some of the demands of the provinces could be met by enabling them to have some independent choice of their own basic credits and exemptions. Thus, *the federal government, in conjunction with the provinces, should strive to protect the integrity of the Tax Collection Agreements by continuing to explore ways to meet the policy aspirations of the provinces in ways that do not damage the functioning of the internal economic union while maintaining a harmonised base and tax administration system.*

– The opportunities for exploiting the experience gained by authorities in administering refundable tax credits based on family incomes appear promising.[72] This form of delivering transfers to low-income families and children has great potential for improving the efficiency of the redistribution process by targeting money to those most in need, and doing so in a way that mitigates, to the extent desired, the adverse incentives on work effort. Indeed, as the tax system moves to one that responds to the pressures of competitiveness, refundable tax credits offer a cost-effective means of offsetting the detrimental consequences that low income persons would otherwise face. Co-operation with the provinces is important here as well, given the role they have assumed in delivering welfare to the needy. Thus, *the federal government should continue to explore, in co-operation with the provinces, ways to enhance the efficiency of the redistributive process by using the tax system to deliver refundable credits as a complement to provincial welfare programmes.*
– Apart from concerns over the rate structure, there remain preoccupations about the treatment of capital income under the income tax system. Full rationalisation of the system of contributions to tax-assisted retirement saving plans has been delayed to 2005. This is mainly for fiscal reasons, though some groups see these forms of sheltering of income as being excessively favourable to the better-off. The economic case for sheltering retirement savings, and for pursuing rationalisation across forms of such savings, is strong. Apart from its effect on national saving, it can be seen as a complement to pension reform: the more households provide for their

own retirement, the less reliant will they be on government in the future.[73] Thus, *the existing form of tax sheltering of savings for retirement should be preserved, and the process of making the tax treatment of all forms of such savings comparable should be continued.*

– There are various ways in which the taxation of business income is distortionary and may detract from competitiveness of Canadian industry and economic growth. Effective tax rates differ across sectors and types of investments. Hence, *there continues to be scope for improving the neutrality of the business tax system and, thereby, resource allocation.* This would include a review of various tax preferences in order to determine if their cost both in terms of foregone revenue and reduced efficiency outweigh their benefits. In the area of international taxation, it will be important to address the deficiencies in the current regime that lead to an erosion of the corporate income tax base. The issue of business taxation is currently under review by a Technical Committee which is due to report to the Department of Finance later this year.
– The absence of full indexation for inflation is also a matter of concern, despite the low rates of inflation Canada has experienced in recent years. The threshold of 3 per cent above which indexation takes effect by far overstates the extent to which the CPI misrepresents the true rate of inflation and has been partly responsible for the increasing tax burden. Once deficit reduction ceases to be an over-riding concern, the case for restricting indexing to general price increases above 3 per cent is considerably weakened. Therefore, *the government ought to move to remove the threshold on indexing the personal income tax system to arrest the resulting rise in the tax burden.* Indexation at the corporate level would be very complex and involve adjustments to depreciation, inventories and interest expense, thus making it much more difficult to accomplish, as evidenced by the fact that it is not commonly done elsewhere. Some of the more blatant problems of indexation, such as those affecting inventories, may be addressed by the above-mentioned Technical Committee on Business Taxation.
– Finally, apart from the concern with the existing provincial retail sales taxes, issues remain with three other forms of provincial and municipal taxation. *First,* property taxes on businesses are relatively high in Canada and their rates vary considerably across provinces, which constitutes a

potential drag on business investment. *Second,* resource taxes are also in many cases still inefficient, although provinces are increasingly moving towards net profit taxation. For example, many of them rely on some measure of production which clearly induce inefficiencies in the exploitation of resources. *Third,* the capital taxes used by many provinces can be detrimental to business. Their impact should be examined in conjunction with provincial corporate income tax rates, and it might seem odd that provinces choose to use such taxes, given the mobility of capital. Addressing the problems raised by these taxes is difficult, given that they are in the domain of provincial and municipal government, but *at the least, the provinces ought to review the structure and tax rates of existing property, resource and capital tax regimes.*

In summary, while the emerging ''fiscal dividend'' is unlikely to allow broad-based tax cuts in the near future, there is scope for arresting the rise in the tax burden and making adjustments to the tax system that are conducive to enhancing both economic efficiency and fairness. In particular, there could be a move to fuller indexing the income tax system; to a more even marginal tax rate structure; to more neutrality in business taxation across sectors and activities; and to greater harmonisation in indirect taxation. But most of these desirable initiatives require a co-ordinated approach between the various levels of government which co-occupy the main tax fields.

Notes

1. An additional factor may be the continued low level of profitability for the housing industry, likely in part the result of high fixed costs associated with government taxes and other levies.
2. That is, the non-accelerating wage rate of unemployment (NAWRU).
3. About one-quarter, according to OECD Secretariat calculations (see *Employment Outlook*, 1995).
4. Over the same period, the population of post-secondary graduates rose by 3½ per cent while that of persons with less than a high school diploma fell by 1½ per cent.
5. S. Gera *et al.*, ''Information Technology and Productivity Growth: An Empirical Analysis for Canada and the United States'', Industry Canada Working Paper (1997).
6. For instance, the percentage of firms using at least one technology rose from 58 per cent in 1989 to 73 per cent in 1993, while the corresponding proportion for the United States increased from 74 per cent to 81 per cent.
7. However, a difficulty in interpreting this measure is that it is also affected by factors such as liquidity and investor preferences as well as inflation uncertainty.
8. The Bank of Canada's assessment of the currently prevailing output gap is similar to that of the OECD Secretariat. The Department of Finance's estimates, however, suggest that the gap is smaller.
9. It should be noted, however, that Canada is one of the few countries where there appears to be evidence of such ''asymmetric'' inflation effects.
10. Crawford, A., J.-F. Fillion and T. Laflèche, 1997, ''L'IPC est-il une mesure adéquate pour la définition de la stabilité des prix?'', presented at the conference Price Stability, Inflation Targets and Monetary Policy, 3-4 May 1997, Bank of Canada, Ottawa.
11. Black, R., D. Coletti and S. Monnier, 1997, ''On the Costs and Benefits of Price Stability'', presented at the conference Price Stability, Inflation Targets and Monetary Policy, 3-4 May 1997, Bank of Canada, Ottawa.
12. Akerlof, George, William Dickens and George Perry, 1996, ''The macroeconomics of low inflation'', Brookings Papers in Economic Activity, 1, pp. 1-59. Fortin, Pierre, 1996, ''The Great Canadian Slump'', Canadian Journal of Economics, XXIX No. 4, November 1996, pp. 761-787.
13. Crawford, A. and A. Harrison, 1997, ''Testing for Downward Rigidity in Nominal Wage Rates'', presented at the conference Price Stability, Inflation Targets and Monetary Policy,

3-4 May 1997, Bank of Canada, Ottawa. Hogan, S., 1997, ''What Does Nominal Wage Rigidity Imply from Monetary Policy?'', Bank of Canada, Working Paper 97-13.

14. Fortin, Pierre, 1991, ''The Phillips Curve, Macroeconomic Policy and the Welfare of Canadians'', Canadian Journal of Economics, XXIV No. 4, November 1991, pp. 774-803. King, Phillip, ''A Reduced-Form Model of Canadian Inflation'', Department of Finance, Canada, mimeo, 1996. Dupasquier, C. and N. Ricketts, 1997, ''Non-Linearities in the Output-Inflation Relationship'', presented at the conference Price Stability, Inflation Targets and Monetary Policy, 3-4 May 1997, Bank of Canada, Ottawa.

15. The relevance of such results will depend, *inter alia*, on the extent that the new inflation targeting regime leads to behavioural changes, and the extent that the band width itself influences inflation volatility.

16. Black, R., T. Macklem and D. Rose, 1997, ''On Policy Rules for Price Stability'', presented at the conference Price Stability, Inflation Targets and Monetary Policy, 3-4 May 1997, Bank of Canada, Ottawa.

17. Industry Canada has recently launched a major research project directed at better understanding Canada's productivity puzzle.

18. See for example, OECD, *Implementing the OECD Jobs Strategy: Lessons from Member Countries' Experience*, Paris, 1997.

19. See for example, Lipsey, Richard, ''Economic Growth, Technological Change, and Canadian Economic Policy'', CD Howe Institute, *Benefactors Lecture, 1996*, November 1996, p. 28. See also Howitt, Peter (ed.), *The Implications of Knowledge-Based Growth for Micro-Economic Policies*, University of Calgary Press, 1996.

20. The temporary income support for claimants, while they are searching for work, is referred to as the income benefit of EI, as opposed to the employment benefit which includes the various programmes under ALMPs, such as: wage subsidies; earnings supplements; self-employment assistance; job creation partnerships; and skills loans and grants.

21. A requirement for monitoring and assessing impacts of EI implementation is included in the new legislation and the first report to Parliament is scheduled for January 1998.

22. The extent of EI benefits is determined by a minimum divisor rule, which is equivalent to the number of 35-hour work weeks required (between 14 and 22 weeks, depending on the region), plus 2 or the actual number of weeks at work in this period. When the benefit rate is calculated, total earnings are divided by the higher of the number of weeks of work or the divisor. As such, if a claimant has a combination of regular weeks and ''small weeks'' (with low earnings), the benefit rate could be lowered and this can create a disincentive to accept shorter work weeks.

23. By the year 2001/02 when EI is fully implemented, C$ 2.7 billion per annum will be available to support ALMPs.

24. British Columbia, Prince Edward Island, Newfoundland and Nova Scotia have opted for a co-management approach to ALMPs, while Alberta, Manitoba, New Brunswick and Quebec have chosen to accept full ALMP responsibility.

25. Between 1983 and 1992 the average duration of completed unemployment spells rose by 17 per cent in Ontario, while in the rest of Canada it fell by 8 per cent. As noted in last year's

Survey, this is probably due to the combined effects of a prolonged cyclical downturn in the labour market and the existence of an ''unemployment trap'', whereby the level of benefits available indefinitely under SA provided little incentive to take work once the recovery was under way.

26. See Ontario, Ministry of Community and Social Services, ''Final Report: A Survey of People Who Have Left Social Assistance''.

27. See Kapsalis, Constantine, ''Social assistance and the Employment Rate of Lone Mothers: An Analysis of Ontario's Live experiment'', Data Probe Economic Consulting Inc, April 1997.

28. The province is currently in the process of defining who is ''employable''.

29. Proposed changes to CPP benefits come into effect on 1 January 1998, while the changes to contribution rates will be retroactive to 1 January 1997.

30. There are actually two time frames for trade and investment liberalisation in APEC, one for industrial country members, 2010, and one for developing country members, 2020.

31. See Schwanen, Daniel, ''Trading Up: The impact of Increased Continental Integration on Trade, Investment, and Jobs in Canada'', CD Howe Institute, *Commentary*, No. 89, March 1997.

32. For example, in 1996 tariff rates for above-quota supplies were: 278.3 per cent for chicken; 265.5 per cent for yoghurt; 266.4 per cent for eggs; 309.7 per cent for ice cream; and 350.5 per cent for butter and other fats and oil derived from milk. See World Trade Organisation, *Trade Policy Review, Canada 1996*, Geneva, 1997.

33. Israel signed a free trade agreement with the United States in 1985 and with the European Union in December 1995.

34. Parallel agreements on environmental and labour co-operation were also included within the context of the bilateral trade arrangement.

35. Canada's position in this respect is being challenged in international fora. It recently lost a trade dispute with the United States at the WTO concerning the preferences given to Canadian magazine publishers, but not to US publishers, doing split-runs of their publications in Canada.

36. The Foundation will not fund the ''operating'' costs of research, such as salaries, regular maintenance or the ongoing operation of facilities. In some case, new construction may be financed where this contributes to cost-effective realisation of research objectives.

37. To receive funding, projects will be evaluated on a number of criteria, including: the planned partnership with other contributors; long-term sustainability; and cost-effectiveness and meaningful performance review mechanisms and benchmarks.

38. The issues involved with municipal revenues, mainly property taxes, licenses and user fees, are somewhat separate, though not unimportant. Their relevance will be referred to from time to time.

39. Moreover, these credits are enhanced if some income is earned from employment, to improve incentives to work despite the steep decline in transfer receipts as income increases.

40. The federal surtax rates are currently 3 per cent and 8 per cent.

41. On average, the provinces obtain about one-fifth of their total revenues as transfers from the federal government, with the proportion varying considerably across provinces because of the equalising nature of transfers (in Newfoundland, for instance, about half of revenues come from the federal government).

42. Equalisation entitlements are based on the extent to which per capita tax revenues that would be collected by applying common tax rates to a representative definition of tax bases used across all provinces fall short of the per capita revenue that would be collected by following the same procedure nationwide. The system relies on being able to define representative tax bases that are reasonably comparable across provinces.

43. In the case of the personal income tax, participating provinces (all but Quebec) choose a single rate of tax to be applied to basic federal tax liabilities for each taxpayer. Taxpayers file a single form to Revenue Canada. For the corporate tax, participating provinces (all except Alberta, Ontario and Quebec) set a provincial tax rate to be applied to the federal tax base. In both cases, the federal government also administers provincial tax credits, provided they are deemed not to be discriminatory or difficult to administer.

44. The HST follows similar principles as the GST, but uses a rate of 15 per cent applied to the GST base instead of 7 per cent applied in all other provinces. The additional revenue collected is paid to the provinces using a formula relying on provincial consumption data. The system is in effect similar to a revenue-sharing system where the sharing formula is based simply on the source of the tax revenues.

45. This implies that they should not either be obviously exaggerated (as when certain expenditures like exploration and development, advertising, and some depreciable capital expenditures are written off very rapidly) or understated (as when deductions are based on historic cost or when some types of taxes, such as taxes on resources, do not allow full capital deductions).

46. It is also true that middle-income workers pay higher marginal taxes in Canada (see Table 19). But these persons are much less mobile across international borders.

47. Although the combined federal-provincial statutory tax rate on manufacturing in Canada is about 4 percentage points lower than the in the United States, there are offsetting factors, such as a less generous depreciation regime and a large corporation tax (LCT).

48. In addition to the write-offs for exploration and development, this sector also receives a resource allowance in lieu of the deductibility of provincial oil and gas royalties and mining taxes. Flow-through shares are also a feature of the system for smaller resource companies enabling investors to utilise the tax deductions of the companies in which they have invested. While the resource allowance deduction has exceeded provincial charges in the mining sector, this is not the case for the oil and gas industry as a whole. The resource allowance is calculated as 25 per cent of net resource production revenue, which is determined before the write-off of exploration and development expenses, as well as interest charges, thus providing an added incentive for such expenses.

49. In Quebec, the corporate tax rate was deliberately reduced to below the levels prevailing in most other provinces while payroll and capital tax rates were increased, in an attempt both to attract capital and to take advantage of the fact that the latter two taxes are deductible against

federal corporate taxes. This illustrates the potential for decentralised taxation to induce inefficiencies in the economic union.

50. Tax assistance for retirement savings is estimated to have amounted to C$ 14.6 billion in 1996.

51. Interest income is taxable on an accrual basis. Dividends are taxed on a received basis, but with a dividend crediting system that generally ensures that the corporate business income is not taxed twice (*i.e.* partial integration exists). In the case of capital gains, the gain inclusion into income is limited to 75 per cent of an investor's capital gain. Also, capital gains are generally taxed on a realised basis rather than on an accrual basis. A Canadian investor may be entitled to a C$ 500 000 capital gain exemption in the case of gains on shares of a small business corporation or qualified farm property. Because of these differences, dividend and capital gain receipts are more favourably treated in Canada than is the case with interest income.

52. Of course, in an open economy, a reduction in savings does not necessarily translate into an equivalent reduction in investment because of the ability to obtain financing from external sources.

53. Taxes on corporations can fulfil a useful withholding function if *i)* their design conforms closely enough with those of creditor nations, so that full advantage can be taken of the implicit transfer of revenues from foreign treasuries to the domestic one, and *ii)* if the tax is integrated with personal income taxation so that the double taxation of savings is avoided and the incentive to retain income within the corporation is mitigated.

54. These issues are discussed in more detail in Boadway, Bruce and Mintz (1987). For a recent discussion of the withholding role of the corporate tax in an open economy, see Bird (1996).

55. According a recent study (Lin, Picot and Beach, 1996), the ''effective payroll tax rate'' rose gradually from 2 per cent in 1961 to 5½ per cent in 1980 and then jumped to 11½ per cent in 1993.

56. One problem that could arise, however, is that provinces as employers (including hospitals and municipalities) could refuse to pay the tax on constitutional grounds.

57. See Di Matteo and Shannon (1995) and Kesselman (1996).

58. See the recent report for the Ontario government of the Advisory Panel on Future Directions for Post-secondary Education (1996).

59. Others include transfers delivered to the least well-off through the provincial social assistance programmes, transfers to the elderly, social insurance programmes, like unemployment and health services, and transfers in kind like education, social services, nursing homes, hospitals, and so on.

60. See the recent World Bank volume devoted to studying this issue (van de Walle and Nead, 1995).

61. Most of the base broadening concerns capital income: a deduction for interest income has been eliminated, the proportion of capital gains subject to tax increased.

62. This has arisen because under the Canada Assistance Plan (CAP), the former federal-provincial cost-sharing programme for welfare, to be eligible for federal support, provincial welfare payments had to be based on the ''need'' of recipients. This was interpreted literally as

suggesting that each dollar of income earned reduced need by one dollar. The CAP transfer has now been abolished and partly replaced by a federal block grant called the Canada Health and Social Transfer (CHST). The CHST no longer restricts welfare payments to be based on need, and gradually the provinces are introducing into their welfare programmes incentives for recipients to seek employment by not reducing their welfare payments on a one-for-one basis with part-time employment earnings.

63. These comprise both federal and provincial taxes and transfers of all forms, including welfare assistance, unemployment insurance and public pensions.

64. Over that period, partial indexation of basic credits and tax bracket thresholds has made 1.4 million individuals taxable on the basis of nominal, not real, income growth. It has also pushed another 1.9 million individuals from the 17 per cent to the 26 per cent tax bracket, and a further 0.6 million from the 26 per cent to the 29 per cent bracket.

65. These fiscal externality effects are summarised in Dahlby (1996). Recent work by Bird and Smart (1996) show that the Equalization system further exaggerates the underestimation of the marginal cost of increasing provincial tax rates.

66. This programme is largely responsible for the fact that, despite widely differing tax capacities, average tax rates for most bases diverge remarkably little across provinces. A recent study has calculated that the Equalization system succeeds in raising provincial per capita tax capacities of each of the low-income provinces to 98 per cent of the national average (Boadway and Hobson, 1993). Tax capacity is defined as the revenues that would be raised by applying national average provincial tax rates to standardised provincial tax bases.

67. Examples include venture capital tax credits, investment tax credits, and tax holidays.

68. Regarding the latter, the federal and provincial governments are discussing the establishment of a new agency, the Canada Customs and Revenue Agency, to administer income, payroll and sales taxes.

69. See, for example, McLure (1983) and Boadway and Bruce (1992).

70. Some of the issues surrounding the tax treatment of retirement savings have been addressed in the 1996 OECD Survey of Canada.

71. As mentioned earlier and discussed in the Annex I, the HST is an augmented version of the GST that applies in New Brunswick, Newfoundland and Nova Scotia beginning in April 1997. Part of the revenue is turned over to the three provinces in proportion to the final consumption expenditures.

72. Revenue Canada's administration of these credits has not been without problems. The Auditor General's report of 1996 pointed out some of these, including problems with effectively auditing recipients. These lessons have presumably been learned.

73. The case for sheltering retirement savings is discussed in some detail in various papers in Banting and Boadway (1996), especially that by Mintz and Wilson.

List of acronyms

CAP	Canada Assistance Plan
CHST	Canadian Health and Social Transfer
CITCA	Comprehensive Integrated Tax Coordination Agreement
CPP	Canada Pension Plan
EI	Employment Insurance
GIS	Guaranteed Income Supplement
GST	Goods and Services Tax
HST	Harmonised Sales Tax
OAS	Old Age Security
QPP	Quebec Pension Plan
RPP	Registered Pension Plan
RRSP	Registered Retirement Savings Plan
TCA	Tax Collection Agreement

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Annex I

The Canadian tax system

A. A description of the Canadian tax system

Personal income tax

The personal income tax applies to all sources of income earned worldwide by individuals resident in Canada. Each taxpayer calculates their taxable income (total income less allowable deductions), applies a rate schedule to determine tax liabilities, and then subtracts various credits to yield ''basic federal tax''. Some refundable tax credits are subtracted for low-income taxpayers and surtaxes are added for high-income taxpayers to determine taxes owing to the federal government. In all provinces, except Quebec, taxpayers then compute provincial tax liabilities by applying a provincial tax rate to basic federal tax, and then deducting provincial-specific tax credits, and adding surtaxes and flat taxes. This procedure ensures that these provinces use the same base and rate schedule as the federal government, in return for which Revenue Canada acts as their tax collecting agency. Quebec operates its provincial tax base system, but its structure is similar to that used in other provinces. There are several features of the personal tax system that are worth highlighting:

- Not all sources of income are included in taxable income. Notable sources of sheltered income include: a maximum amount of saving for retirement income done through an employer, Registered Pension Plans (RPPs), or individually, Registered Retirement Savings Plans (RRSPs); imputed income and capital gains on owner-occupied housing; and inheritances, (except capital gains on assets not transferred to a spouse which are generally taxed on death). Only three-quarters of capital gains are taxable.
- Progressivity is accomplished through a combination of a three-tier rate schedule (17, 26 and 29 per cent), non-refundable credits for basic living expenses, family size, age and disability, and surtaxes that are higher for highly taxed persons. Progressivity is also enhanced by the use of credits rather than deductions for expenditures such as charitable donations, medical expenses, political contributions, tuition fees, and social security contributions.

- Refundable tax credits are used to deliver assistance to low-income taxpayers. A GST Credit is given on the basis of family income and family size. A Child Tax Benefit is based on the number of children and also on family income. A third refundable credit based on family income, the Seniors Benefit, will replace existing transfers to the elderly that are financed from general revenues. These credits are significant as they have the potential for delivering social policy through the tax system.
- The tax system is indexed annually for inflation using the change in the Consumer Price Index (CPI) less 3 per cent. The indexation factor is applied to the tax brackets and tax credits. As a result, income tax revenues have been enhanced by ''bracket creep''. Indexation has not been triggered since 1992, given the low rates of inflation in recent years.
- A dividend tax credit is available on dividends received from taxable Canadian corporations. The credit against basic federal tax is 13⅓ per cent of ''grossed-up'' dividends (125 per cent of actual dividends), and a corresponding amount against provincial tax liabilities. The dividend tax credit is a measure for integrating the personal and corporation income taxes, and is intended to compensate shareholders for corporate taxes withheld on their behalf.

Corporation income tax

Most corporations resident in Canada are taxable on income earned worldwide, and non-resident corporations are taxed on income earned in Canada. Taxable income reflects income accrued on behalf of shareholders (equity income) and is defined in the same way as business income earned by unincorporated taxpayers. The base is accrued revenues less current costs less accrued capital costs. The latter includes nominal interest payments, capital cost (depreciation) allowances, inventories used and many of the costs of acquiring resource properties. The federal corporation income tax rate of 28 per cent is applied to taxable income, though some forms of income are taxed preferentially: manufacturing and processing income earned in Canada is taxed at 21 per cent and the first C$ 200 000 of active business income of qualifying Canadian controlled private corporations is taxed at 12 per cent. As well, there is currently a surtax of 4 per cent applicable at the federal level. Various tax credits are available, including for certain new investments, expenditures on R&D, political contributions, and taxes paid to foreign governments.

All provinces also implement corporation income taxes. Alberta, Ontario and Quebec administer their own corporation taxes, but maintain structures that are quite similar to the federal one. The other seven have their taxes administered by Revenue Canada. They apply a discretionary provincial rate to federal taxable income earned within their jurisdictions, and are allowed to implement their own provincial tax credits, provided they are non-discriminatory and do not unduly distort the inter-provincial allocation of capital. As with the personal system, there are some features of the corporate tax system worth noting:

- Partly by intent and partly by consequence, the corporation tax system provides incentives for certain types of activities relative to others. For example, because of the way that accrued capital costs are defined, resource industries and industries using machinery and equipment intensively tend to be favoured relative to those that rely more on buildings and inventory. Firms investing in manufacturing and processing are favoured by lower income tax rates, while those investing in Atlantic Canada and in R&D can take advantage of preferential tax credits. Studies have calculated the extent to which the tax system provides differential incentives for various types of investments using the methodology of marginal effective tax rates (METRs). They typically find that manufacturing and resource industries are favoured relative to service and tertiary industries, and investments in machinery and equipment, intangible assets and resource properties tend to be favoured relative to buildings and inventories.
- As with corporation tax systems elsewhere, losses are not treated symmetrically with gains. Negative tax liabilities can be carried forward and backwards for limited periods, but without interest. A consequence is that firms that are more likely to be experiencing losses are treated relatively less favourably than those whose income streams are steady. Examples include small growing firms and firms in relatively risky businesses. Coincidentally, these types of firms are often relatively important for employment growth and innovation. Large firms are able to offset losses in some areas of operation against profits in others.
- There is no indexation of taxable income or deductions for inflation. This has important implications for various capital deductions in periods of high inflation. The ability to deduct nominal interest income provides an incentive for debt finance, while basing capital cost allowances and inventory deductions on historical values discourages investment.
- While the dividend tax credit provides a measure of integration of the personal and corporation tax systems, it is an imperfect one. The amount of the credit does not vary with the tax rate actually paid by the corporation. For example, it applies regardless of whether the corporation is in a taxpaying position or not, or whether it is engaging in a tax-favoured activity.
- Canadian taxpayers are subject to tax on their worldwide income. The treatment of foreign source income earned by Canadian resident corporations through foreign subsidiaries is a mix of credits and exemptions. Dividends paid out of active business income of foreign affiliates, that are residents of a country with which Canada has a tax treaty in force, are effectively exempt from tax in Canada. Dividends paid out of active business income of other foreign affiliates are taxable in Canada, subject to a credit for foreign withholding taxes as well as underlying foreign corporate income taxes. For this purpose a foreign affiliate is defined as a non-resident corporation in which the sum of the Canadian shareholder's direct and indirect shareholding is not less than 10 per cent. In addition, a taxpayer resident in Canada is required to include in his income for

any year his proportionate share of the ''foreign accrual property income'' (FAPI) of a controlled foreign affiliate whether or not any such income is actually distributed to him by the affiliate. These anti-avoidance provisions effectively bring back into the Canadian tax base passive income (income from property, income from other than active businesses, and the taxable portion of certain capital gains) earned by controlled foreign subsidiaries. For this purpose a controlled foreign affiliate is defined to mean one that is controlled directly or indirectly by the taxpayer, by the taxpayer and four or fewer persons resident in Canada or by a related group of persons, wherever resident, that includes the taxpayer. Finally, if the foreign operations of a Canadian resident are conducted through a branch, the branch's income is included in the resident's income subject to tax. A business foreign tax credit provides double taxation relief in respect of income taxes paid by the branch in the foreign jurisdiction. While Canadian taxation cannot be deferred when a branch structure is elected, foreign losses may be offset against Canadian source income.

– Non-resident businesses are liable for taxes on income earned in Canada. As well, withholding taxes are applicable on investment income at a rate of 25 per cent except where reduced by a treaty.
– Some corporations are liable for federal or provincial capital tax. The federal government imposes a tax on large corporations (LCT) with more than C$ 10 million of capital employed in Canada. The tax is reduced by the amount of the corporation's federal surtax. In addition to the LCT, large financial institutions are subject to a tax (Part VI tax) on taxable capital employed in Canada exceeding C$ 200 million, imposed at a rate of 1 per cent on taxable capital between C$ 200 million and C$ 300 million and 1.25 per cent above that. The Part VI tax is reduced by the corporation's income tax liability. A temporary surcharge of 12 per cent also applies on large deposit-taking institutions on their capital in excess of C$ 400 million. The surcharge is not reduced by the corporation's income tax liability. All provinces levy a capital tax on banks and trust companies at rates ranging from 1 per cent to 4 per cent. As well, five provinces impose smaller capital taxes on all corporations.

General sales taxation

General sales taxes are levied at both the federal and provincial levels of government. The federal GST is a form of value-added taxation similar to that used in most OECD countries, but with some distinct features. The base includes almost all consumption goods and services on a destination basis. The rate of tax is 7 per cent, with exports, food and purchases by provincial governments zero-rated. Sales by very small firms are exempt, as are residential rents, health services and educational services. Rebates are also given on inputs purchased by municipalities, universities, schools and hospitals. The tax

rate applicable on new housing sales under C$ 350 000 is 4.5 per cent. Resales of homes are exempt. The consequence is a fairly uniform tax applicable on a very broad base.

All provinces but Alberta also levy general sales taxes. The sales tax in Quebec is harmonised with that of the federal government, using almost the same final consumption base and charging a uniform rate of 6.5 per cent of the GST-inclusive price (which is to be raised to 7.5 per cent). Firms in Quebec pay both the federal and provincial sales tax to a single tax-collecting authority, Revenue Quebec. In the other provinces, retail sales taxes have been applied to a much narrower base than for the GST, exempting most services and various goods as well, and having a differentiated rate structure. As a result, services are treated preferentially, many business inputs are taxed and exports contain taxes paid at earlier stages of production. Because of these disadvantages, New Brunswick, Newfoundland and Nova Scotia have agreed to harmonise their taxes fully with the federal GST. Beginning in April 1997, a single multi-staged tax is applicable in these provinces, referred to as the Harmonised Sales Tax (HST), which is essentially equivalent to the GST except that the rate is 15 per cent. The HST effectively operates like a revenue-sharing system. Of the 15 percentage points of tax collected, 8 are paid to the provinces using a formula relying on final consumption data in each province. While the base and rate structure must ultimately be enacted in federal legislation, the provinces participate in the formulation of policy through an advisory Tax Review Committee. The federal government is encouraging other provinces to enter into similar arrangements for harmonising federal and provincial sales taxes.

Specific commodity taxes

Excise taxes or duties are applicable on a limited range of goods. The most important of these are excises on alcoholic beverages, tobacco products and fuels. Apart from serving a revenue-raising objective, such taxes are often justified as correcting for adverse externalities or as benefit taxes (user fees) for large expenditure items such as transportation facilities, hospital and police services. Both the federal government and the provinces levy substantial taxes on these products sold within their jurisdictions.

Payroll taxes

Limited use is made of payroll taxes by the federal and provincial governments, usually to finance social insurance programmes. The federal government finances both Employment Insurance (EI) and the Canada Pension Plan (CPP) by forms of payroll taxes, both based on earnings and both subject to upper limits. The taxes are paid jointly by employers and employees. Several provinces levy modest payroll taxes, usually earmarked for health financing, as well as for the Quebec Pension Plan (QPP) in that province. Provincial workers' compensation programmes are also financed by employer contributions.

Natural resource revenues

Virtually all revenues from renewable and non-renewable resources accrue to the provinces in which the resources are located. Given that provinces are endowed with very differing types and quantities of resources, this is a significant source of differences in tax potential across provinces. Revenue-raising practices differ considerably both across resource types and across provinces. Revenues from oil and gas production take the forms of sales of leases, production taxes and royalties on production. Mining operations are subject to mining taxes and leases, which typically apply on some measure of profits but also differ by type of mineral as well as province. Forestry revenues include leases, stumpage fees and royalties. Fishing generates revenues through licenses and permits. water power revenues come from profits of provincially owned hydro-electric corporations and from water power rentals. Very few natural resource revenues are based on the true rent of the resource, though provinces are increasingly moving towards net profit taxation regimes. The consequence is that the tax system can have considerable influence on decisions to exploit natural resources.

Property taxes

Property taxes are levied either by municipalities or on a province-wide basis. They are used to finance local services as well as varying portions of education and welfare services. A tax rate is applied to some measure of assessed value for each piece of real property (land and buildings). Effective tax rates vary considerably across provinces as do assessment practices. Some provinces use relatively recent market values and update them frequently. Others use outdated property values from as long ago as 1950s. Tax rates differ on residential, commercial and industrial properties, with the former typically bearing the lowest rate. Separate business taxes are often levied on users (rather than owners) of business properties. The consequence is that property taxes can represent an important element of costs of doing business that can vary across jurisdiction.

B. The Equalization system

Equalization payments are made to those provinces whose tax capacities fall below a national norm. The calculation of entitlements is as follows. There are 37 different provincial revenue sources defined for Equalization purposes. For each one, a standardised base is defined, meant to reflect a representative base used across provinces. The size of each base is measured for each province. A national average tax rate is calculated by dividing total provincial revenues for each source by the sum of all provincial tax bases. For each province and each revenue source, the amount of tax revenues that would be collected by applying the national average tax rate to the province's base is calculated and converted to per capita terms. Then, for each revenue source, the amount of revenue that would be obtained by applied the national average tax

rate to the per capita tax base averaged over five representative provinces (British Columbia, Manitoba, Ontario, Quebec and Saskatchewan) is calculated. Each province's per capita Equalization entitlement from a given revenue source (positive or negative) is the difference between that calculated for the representative five provinces and that calculated for the provinces. This procedure is repeated for all 37 revenue sources, and the per capita Equalization entitlements for each province are added up over all revenue sources. For those provinces with positive aggregate Equalization entitlements, a transfer is made from the federal government equal to the aggregate per capita entitlement times provincial population. The remaining provinces receive no transfer.

C. Tax harmonisation measures

Tax Collection Agreements: personal income tax

A Tax Collection Agreement (TCA) negotiated between each participating province (all except Quebec) and the federal government is a means of harmonising taxes to reduce collection and compliance costs while preserving provincial responsibility for choosing their own tax rates. The personal income TCAs involve the federal government administering provincial taxes jointly with the federal tax using a single tax form for taxpayers. Provinces must abide by the federally chosen tax base and rate structure, and by a common criterion for allocating taxable income among provinces (*viz.*, to the province of taxpayer residence at the end of the tax year). They are allowed to chose their own provincial tax rates to apply to basic federal taxes. Revenue Canada administers provincial tax credits as long as they are non-discriminatory, easy to administer, non-distortionary, and respect the common base. Three provinces also levy flat rate taxes on net income through the TCAs.

Tax Collection Agreements: corporation income tax

Corporate TCAs negotiated between the federal government and participating provinces also allow provinces to turn over tax administration to Revenue Canada. Provinces must abide by the federal tax base, and by a common allocation formula in the case of corporations which earn income in more than one jurisdiction. Taxable income of most corporations is allocated to a province using an average of the proportion of its sales and the proportion of it payroll in the province. Provinces are allowed to choose their own corporation income tax rates, and are allowed to implement tax credits that satisfy the same criteria as with the personal tax case. As mentioned earlier, even the non-participating provinces tend to adopt similar tax bases and to abide by the common allocation formula, partly a consequence of the fact that the current system evolved from one in which the federal government was the sole collector of corporation income taxes.

Comprehensive Integrated Tax Coordination Agreements

Comprehensive Integrated Tax Coordination Agreements (CITCAs) are used to achieve sales tax harmonisation via the HST. As mentioned, the HST is a revenue-sharing system whereby the federal government applies a 15 per cent tax rate rather than the 7 per cent GST rate applying in non-participating provinces and turns over to the participating provinces their shares based on aggregate consumption estimates. The same procedure of taxing sales and giving rebates on inputs purchased is used, the only exception being that account must be taken of the rate difference between participating and non-participating provinces. What results is a fully harmonised tax both in terms of the base and the rate structure. The CITCAs also include provisions governing adjustments to the HST rate and tax base by the federal government for the participating provinces.

Note

1. A marginal effective tax rate indicates the proportion of returns of the marginal investment of a given type that go to the government. A fully neutral tax system would have zero marginal effective tax rates. The methodology and its application to Canada may be found in Boadway, Bruce and Mintz (1987).

Annex II

Chronology of economic events

1996

January

The Ontario government passes the Savings and Restructuring Act which has potentially a large impact on health services, transport and municipal taxation.

Banks lower their prime lending rate to 7 per cent at month-end.

February

From late February the Bank rate (the rate used by the Bank of Canada for advances to financial institutions) is set at the upper limit of the central banks' operating target band for the overnight financing rate. (Before, the Bank rate was set at 25 basis points above the average interest rate resulting from the weekly auctions of three-month treasury bills).

Federal government tabled new legislation (Bill C-82) to amend certain laws relating to financial institutions.

March

The federal government presents its budget, projecting a deficit of C$ 24.3 billion (3 per cent of GDP) in 1996/97 and C$ 17 billion (2 per cent of GDP) in 1997/98. There are no tax increases. Spending restraint, largely outlined in past budgets, remains in force, with new measures leading to additional net expenditure reductions of C$ ½ billion in 1998/99. Planning assumptions remain prudent and the budget incorporates large contingency reserves.

Banks lower their prime lending rate to 6¾ per cent.

April

Ontario civil servants return to work after a five-week strike.

Banks lower their prime lending rate to 6½ per cent.

May

The Saskatchewan and Nova Scotia governments both envisage a budget surplus in the 1996/97 fiscal year.

The federal government announces a deal under which New Brunswick, Nova Scotia and Newfoundland will harmonise their provincial sales taxes with the federal Goods and Services Tax (GST) next year. The new federal-provincial GST rate will be 15 per cent from April 1997.

Keeping to its election promise, the Ontario government cuts provincial income taxes by 15 per cent in its 1996 budget, providing half of the pledged tax reduction. To meet its deficit targets of C$ 8.2 billion in 1996/97 and C$ 6.6 billion in 1997/98, the government will implement spending cuts of C$ 6 billion over the next two years.

The Quebec budget aims at lowering the deficit to C$ 3.3 billion in 1996/97 and C$ 2.2 billion in 1997/98 by a mix of revenue enhancement measures and spending cuts. Balanced budget legislation is to ensure that government finances remain in equilibrium following the elimination of the deficit by 1999/2000.

The federal government announces the replacement of the Unemployment Insurance programme by the Employment Insurance programme effective 1 July. Eligibility will be based on a minimum of 420 to 700 hours worked (depending on regional unemployment rates). Part-time workers will also be able to insure. The length of benefits is reduced to 45 weeks (from 50) and the maximum falls to C$ 413 per week (from C$ 448). Premiums are reduced to C$ 2.95 per C$ 100 of insurable earnings (from C$ 3) and employers pay to C$ 4.13 (down from C$ 4.20).

June

The Alberta government announces that all of its C$ 1.13 billion budget surplus will be applied to its debt.

The British Columbia government's new budget calls for a balanced budget, based on a six-month freeze on capital spending, spending growth of only 2 per cent, and a 1 per cent cut in income tax rates.

July

Banks lower their prime lending rate to 6¼ per cent.

August

Banks lower their prime lending rate in two steps to 5¾ per cent.

September

About 13 000 Canadian Auto Workers (CAW) ratify a three-year contract with Chrysler Canada which includes 2 per cent wage increases in each year of the contract and an increase in the cost-of-living allowance. Chrysler agreed not to sell or close any plant for three years and workers get ten more paid days vacation over the life of the agreement.

Canadian Broadcasting Corporation announced a C$ 127 million budget-cutting plan, including 2 500 job losses over the next eighteen months.

Quebec's minimum hourly wage rose 25 cents effective 1 October to C$ 6.70.

Mortgage rates were cut across the board by up to ¼ percentage point at month-end, putting some short-term rates near the 30-year lows reached in January 1994.

October

General Motors Canada and the Canadian Autoworkers Union reach agreement to end a strike by 26 300 employees on 22 October. The strike, which shut down all of GM Canada's operations as well as many parts suppliers and trucking companies both in Canada and the US lasted 20 days. The agreement is similar to the one reached between Chrysler and the CAW in September.

The Quebec government announced plans to regulate gasoline prices to prevent major oil companies from undercutting independently owned service stations at the pump.

Banks lower their prime lending rate to 5 per cent.

Annual Financial Report for 1995/96 released confirming the federal deficit of C$ 28.6 billion for that year.

Release of the Economic and Fiscal Update in which the government announces a new deficit target of C$ 17 billion for 1997/98.

November

Canada and Chile sign a free trade agreement which is to come into effect in June 1997.

Banks lower their prime lending rate to 4¾ per cent, the lowest rate since 1956.

December

New Brunswick unveils its 1997-98 provincial budget which includes a total tax cut of C$ 166 million, with reductions in both income and sales tax. The budget also calls for a 1.5 per cent cut in expenditures and the elimination of a further 370 civil service positions.

1997

January

The Ontario proposals for restructuring provincial/municipal responsibilities were amended on 1 May 1997. Funding for education will be shared, with 50 per cent coming from property taxes and 50 per cent from provincial grants. Funding for social assistance will still be shared on an 80/20 basis between the province and the municipalities. Administrative costs, however, will be shared on a 50/50 basis. Social housing will be a municipal responsibility, but homes for special care and long-term care will be a provincial responsibility.

Employment Insurance moves to an hours-based system. To qualify for EI benefits claimants must work between 420 and 700 hours, depending on the region and 910 hours for new and re-entrants to the labour market.

February

The 1997 budget released by the federal government contained no new programme cuts and no new taxes. Other highlights include: increased spending of C$ 137 million for higher education and skills training; C$ 300 million over three years for home care and drug coverage, better access to medical information and a community action programme; C$ 600 million a year for child-tax benefits for low-income families; the Canada Foundation for Innovation with funding of C$ 800 million over five years; and incentives in the tax system to encourage charitable donations.

Draft legislation to reform the Canada Pension Plan was tabled.

March

The legislation for the harmonised federal and provincial sales tax in three Atlantic provinces was amended to not include the blended tax-in prices.

Quebec tabled its budget which included a 15 per cent reduction in personal income taxes for lower and middle-income earners, and a 1 percentage point rise in the sales tax to 7.5 per cent.

British Columbia released its budget which contained a 2 percentage point cut in personal income taxes and continuing with the freeze on car insurance, hydro and tuition fees.

The Manitoba government announced its 1997/98 fiscal-year budget which contained no new taxes or tax increases. The government anticipates a C$ 27 million surplus – its third straight – achievable due to a C$ 100 million transfer from the province's fiscal Stabilisation Fund. (The Manitoba surplus includes a C$ 75 million instalment into the debt retirement fund.)

Saskatchewan tabled it fourth balanced budget in a row, with plans to reduce the public debt by almost C$ 4 billion over five years while maintaining key social programmes. Retail sales taxes were cut from 9 per cent to 7 per cent.

The Newfoundland government released its budget which contained a three-year programme to eliminate 1 100 civil servant positions and balance the books by the end of the century. The budget contained no new tax increase, but many user fees were increased.

The federal government introduces changes to the EI programme, allowing for ''small weeks'' to be used in the calculation of EI benefits in 29 high unemployment regions.

April

Alberta tabled its budget which contained an expected C$ 154 million surplus following a C$ 2.3 billion surplus last year. It calls for a modest increase in spending and no new tax increases.

Nova Scotia tabled its budget with a forecast surplus of C$ 4 million over the next two years.

Prince Edward Island released its budget with a projected C$ 17 million deficit. It contained a rise in the corporate income tax rate and spending cuts. Gasoline taxes will increase by one cent a litre, license and registration fees were raised, and the personal income tax base was expanded.

May

Ontario tabled its budget which included a deficit target of C$ 6.6 billion; personal income tax cuts; increased spending for health care and job creation; C$ 500 million for a ten-year research and development fund; and tax credits for small businesses.

June

The Liberal government was re-elected for a second mandate but with a reduced majority. The Liberals retained 155 of the 301 seats of the House of Commons, down from 174 in their first term in office. The Reform Party replaced the Bloc Quebecois as the official opposition party.

The Bank of Canada raised the Bank Rate by ¼ percentage point to 3½ per cent. The operating band for the overnight interest rate has consequently been raised by the same amount to 3-3½ per cent.

Changes to Financial Institutions Legislation comes into force, including measures to strengthen consumer protection, ease the regulatory burden on financial institutions and ensure the financial institutions statutes remain current with evolving trends.

STATISTICAL ANNEX AND STRUCTURAL INDICATORS

Table A. **Selected background statistics**

	Average 1987-96	1987	1988	1989	1990	1991	1992	1993	1994	1995	1996
A. Percentage changes											
Private consumption [1]	2.1	4.4	4.5	3.4	1.0	–1.6	1.3	1.6	2.9	1.4	2.4
Gross fixed capital formation [1]	3.1	10.8	10.3	6.1	–3.5	–2.9	–1.5	0.6	5.9	–0.1	6.4
Public investment [1]	3.3	2.2	3.6	8.8	6.9	4.9	–0.1	0.7	6.4	2.7	–2.8
Private investment [1]	3.0	12.0	11.1	5.7	–4.8	–4.0	–1.7	0.5	5.9	–0.6	7.8
Residential [1]	–0.1	16.4	2.8	4.8	–9.7	–12.5	7.9	–4.2	1.8	–15.1	11.5
Non-residential [1]	4.4	9.7	15.8	6.2	–2.4	–0.2	–5.5	2.7	7.6	5.2	6.6
GDP [1]	2.0	4.2	5.0	2.4	–0.2	–1.8	0.8	2.2	4.1	2.3	1.5
GDP price deflator	2.6	4.7	4.6	4.8	3.1	2.9	1.2	1.1	0.7	1.5	1.3
Industrial production	2.0	4.9	5.3	–0.1	–3.3	–4.2	1.1	4.5	7.0	3.4	1.7
Employment	1.2	2.7	3.2	2.1	0.6	–1.9	–0.6	1.4	2.1	1.6	1.3
Compensation of employees (current prices)	4.7	8.7	9.7	7.9	5.2	2.8	2.4	1.7	2.7	3.2	2.8
Productivity (real GDP/employment)	0.8	1.4	1.7	0.4	–0.8	0.1	1.3	0.8	1.9	0.7	0.2
Unit labour costs (compensation/real GDP)	2.6	4.4	4.5	5.3	5.5	4.6	1.6	–0.5	–1.3	0.8	1.3
B. Percentage ratios											
Gross fixed capital formation as per cent of GDP at constant prices	22.0	21.4	22.4	23.2	22.5	22.2	21.7	21.4	21.7	21.2	22.2
Stockbuilding as per cent of GDP at constant prices	0.2	0.6	0.5	0.7	–0.3	–0.4	–0.7	0.2	0.6	0.9	0.1
Foreign balance as per cent of GDP at constant prices	–1.3	0.1	–1.1	–2.6	–2.1	–2.7	–2.2	–1.9	–1.0	0.2	–0.1
Compensation of employees as per cent of GDP at current prices	55.1	54.2	54.1	54.3	55.6	56.5	56.7	55.9	54.7	54.4	54.4
Direct taxes as per cent of household income	16.4	15.4	15.8	15.6	17.3	17.0	16.4	16.1	16.3	16.8	17.6
Household saving as per cent of disposable income	8.8	9.2	9.7	10.4	9.7	9.9	10.3	9.6	7.6	7.0	4.6
Unemployment rate	9.5	8.8	7.8	7.5	8.2	10.4	11.3	11.2	10.4	9.5	9.7
C. Other indicator											
Current balance (billion dollars)	–15.1	–13.3	–14.7	–21.6	–19.7	–22.2	–20.8	–21.7	–14.7	–5.4	2.8

1. At constant 1986 prices.

Table B. **Supply and use of resources**

Million Canadian dollars, current prices

	1987	1988	1989	1990	1991	1992	1993	1994	1995	1996
Private consumption	322 769	349 937	378 933	399 319	411 960	422 515	436 542	452 444	465 970	482 951
Public consumption	105 836	114 472	124 108	135 157	144 885	150 390	152 158	150 593	150 834	148 548
Gross fixed investment	116 717	132 790	146 075	141 376	132 001	128 865	128 884	138 118	134 341	139 971
Final domestic demand	545 322	597 199	649 116	675 852	688 846	701 770	717 584	741 155	751 145	771 470
	(9.2)	(9.5)	(8.7)	(4.1)	(1.9)	(1.9)	(2.3)	(3.3)	(1.3)	(2.7)
Stockbuilding	3 071	3 795	3 607	–2 835	–3 235	–3 701	1 103	3 993	7 026	1 235
	(0.1)	(0.1)	(0)	(–1.0)	(–0.1)	(–0.1)	(0.7)	(0.4)	(0.4)	(–0.7)
Total domestic demand	548 393	600 994	652 723	673 017	685 611	698 069	718 687	745 148	758 171	772 705
	(9.3)	(9.6)	(8.6)	(3.1)	(1.9)	(1.8)	(3.0)	(3.7)	(1.7)	(1.9)
Exports	145 416	159 309	163 903	168 917	164 849	181 189	209 370	250 877	290 325	306 534
Imports	140 502	156 384	166 079	171 223	172 805	187 254	212 534	246 455	271 291	281 356
Foreign balance	4 914	2 925	–2 176	–2 306	–7 956	–6 065	–3 164	4 422	19 034	25 178
	(0)	(–0.4)	(–0.8)	(0)	(–0.8)	(0.3)	(0.4)	(1.1)	(2.0)	(0.8)
Statistical discrepancy	–1 710	1 987	201	–1 244	–1 178	–1 882	–2 668	–2 310	–906	–94
GDP (market prices)	551 597	605 906	650 748	669 467	676 477	690 122	712 855	747 260	776 299	797 789
	(9.1)	(9.8)	(7.4)	(2.9)	(1.0)	(2.0)	(3.3)	(4.8)	(3.9)	(2.8)

Note: Figures in parentheses are annual growth rates; for stockbuilding and the foreign balance they are contributions to GDP growth.
Source: CANSIM – Statistics Canada.

Table B. **Supply and use of resources** *(cont.)*

Million Canadian dollars, 1986 prices

	1987	1988	1989	1990	1991	1992	1993	1994	1995	1996
Private consumption	310 453	324 301	335 284	338 717	333 396	337 619	342 858	352 913	357 827	366 446
Public consumption	101 857	106 060	110 331	113 890	116 958	118 126	118 660	116 592	115 732	113 699
Gross fixed investment	112 542	124 105	131 630	126 962	123 236	121 419	122 095	129 345	129 157	137 365
Final domestic demand	524 852	554 466	577 245	579 569	573 590	577 164	583 613	598 850	602 716	617 510
	(5.1)	(5.6)	(4.1)	(0.4)	(–1.0)	(0.6)	(1.1)	(2.6)	(0.6)	(2.5)
Stockbuilding	3 222	2 515	3 778	–1 737	–2 427	–3 766	1 006	3 729	5 635	567
	(0.1)	(–0.1)	(0.2)	(–1.0)	(–0.1)	(–0.2)	(0.9)	(0.5)	(0.3)	(–0.8)
Total domestic demand	528 074	556 981	581 023	577 832	571 163	573 398	584 619	602 579	608 351	618 077
	(5.3)	(5.5)	(4.3)	(–0.5)	(–1.2)	(0.4)	(2.0)	(3.1)	(1.0)	(1.6)
Exports	142 942	156 528	157 799	164 312	166 687	179 426	198 093	227 120	254 269	265 726
Imports	142 678	162 385	172 584	175 960	181 831	192 000	208 856	232 871	253 082	266 043
Foreign balance	264	–5 857	–14 785	–11 648	–15 144	–12 574	–10 763	–5 751	1 187	–317
	(–0.9)	(–1.2)	(–1.6)	(0.6)	(–0.6)	(0.5)	(0.3)	(0.9)	(1.2)	(–0.2)
Statistical discrepancy	–1 608	1 834	248	–1 029	–967	–1 519	–2 134	–1 838	–703	35
GDP (market prices)	526 730	552 958	566 486	565 155	555 052	559 305	571 722	594 990	608 835	617 795
	(4.2)	(5.0)	(2.4)	(–0.2)	(–1.8)	(0.8)	(2.2)	(4.1)	(2.3)	(1.5)

Note: Figures in parentheses are annual growth rates; for stockbuilding and the foreign balance they are contributions to GDP growth.
Source: CANSIM – Statistics Canada.

Table C. **Industrial production, employment and other business indicators**

Seasonally adjusted

	1992	1993	1994	1995	1996	1996 II	1996 III	1996 IV	1997 I	1997 II
Indices of industrial production (1990 = 100)										
Total	96.9	101.2	108.3	112.0	113.9	112.6	115.0	115.6	116.5	118.0
Durable manufactures	92.9	99.4	110.0	117.3	119.3	118.2	120.6	120.5	122.8	125.4
Non-durable manufactures	95.6	97.8	101.7	102.0	102.8	101.8	103.7	105.3	106.3	106.7
New Residential construction (annual rates)										
Building permits (million C$)	17 161	16 433	17 590	13 242	15 718	5 016	5 248	5 566	6 348	6 038
Starts (thousand units)	167.7	155.4	153.9	112.6	123.7	124.0	128.8	129.3	155.1	145.3
Employment and unemployment (thousands, monthly averages)										
Civilian labour force	14 483	14 665	14 827	14 926	15 149	15 116	15 164	15 240	15 226	15 331
Non-agricultural employment	12 404	12 565	12 866	13 076	13 224	13 217	13 227	13 258	13 325	13 470
Employment										
Mining	133	125	137	134	131	128	138	133	134	144
Manufacturing	1 620	1 617	1 655	1 697	1 738	1 754	1 774	1 748	1 727	1 812
Durables	864	850	893	931	963	969	978	973	962	1 009
Non-durables	756	767	762	766	775	785	796	775	765	803
Transportation, communication and other utilities[1]	860	851	858	857	839	846	833	840	836	857
Unemployment (thousands)	1 642	1 649	1 536	1 419	1 472	1 456	1 484	1 512	1 459	1 438
Unemployment (percentage of civilian labour force)	11.3	11.2	10.4	9.5	9.7	9.6	9.8	9.9	9.6	9.4
Average weekly hours worked in manufacturing	38.3	38.6	38.9	38.5	38.4	38.3	37.9	38.8	39.2	39.6
Retail sales ($ million, monthly averages)	15 421	16 151	17 236	17 633	18 086	17 909	18 122	18 581	18 966	19 267
Orders and inventories in manufacturing ($ million)										
New orders (monthly averages)[2]	23 924	26 107	29 868	32 428	33 424	33 148	34 040	34 115	35 631	35 925
Unfilled orders (end of period)	25 959	29 172	34 761	33 866	33 788	34 072	33 705	33 788	35 472	37 398
Total inventories (end of period)	42 181	44 069	48 580	53 407	53 888	52 831	53 221	53 888	53 952	54 761

1. Includes storage, electric power gas and water utilities.
2. 3-month averages for quarters.

Source: OECD, *Main Economic Indicators;* CANSIM-Statistics Canada.

Table D. **Prices, wages and finance**

	1992	1993	1994	1995	1996	1996 II	1996 III	1996 IV	1997 I	1997 II
Prices (1990 = 100)										
Consumer prices, all items	107.2	109.2	109.4	111.8	113.5	113.4	113.6	114.3	115.0	115.3
of which:										
Food	104.3	106.1	106.6	109.1	110.6	110.8	110.5	111.3	112.1	112.2
Non-food	107.8	109.8	110.0	112.3	114.1	114.1	114.3	115.0	115.6	116.0
Producer prices, manufactured goods	99.5	102.8	108.6	117.4	118.0	117.7	118.0	118.1	118.3	119.1
Wages and profits										
Hourly earnings in manufacturing (1990 = 100)	108.4	110.7	112.5	114.1	117.7	116.9	118.4	120.0	119.2	119.0
Corporate profits before tax ($ million, annual rates)	35 060	42 135	56 611	64 015	64 121	60 200	66 688	69 696	73 824	75 320
Banking ($ million, end of period)										
Chartered banks:										
Canadian dollar deposits	339 085	377 231	406 349	433 558	449 541	436 182	441 928	449 541	455 983	457 881
of which:										
Personal savings deposits	228 732	263 766	280 291	297 559	292 444	298 640	295 608	292 444	288 895	286 399
Liquid assets	57 865	82 181	87 238	91 788	87 099	87 362	88 657	87 099	84 443	82 885
Holdings of Government of Canada direct and guaranteed securities	17 388	32 829	42 291	49 284	55 361	49 765	50 758	55 361	54 169	53 918
Total loans	323 996	357 878	393 063	417 947	468 017	439 761	451 462	468 017	482 018	505 828
Currency outside banks	22 046	23 678	25 437	26 130	27 147	26 624	26 871	27 147	27 688	27 853
Interest rates (per cent, end of period)										
Prime corporate paper (3 month)	7.3	3.9	7.2	5.7	3.1	4.8	4.0	3.1	3.3	3.3
Yield of long-term Govt. bonds	8.6	7.3	9.2	7.5	6.8	8.0	7.6	6.8	6.9	6.6
Miscellaneous										
Share prices Toronto stock exchange (1990 = 100)	99.5	114.1	125.2	129.6	154.0	150.4	149.7	170.9	176.5	183.1

Source: OECD, *Main Economic Indicators;* CANSIM – Statistics Canada.

Table E. **Balance of payments**

Million US dollars

	1987	1988	1989	1990	1991	1992	1993	1994	1995	1996
Merchandise exports	99 141	116 582	124 114	130 289	128 839	135 219	147 561	166 843	193 035	205 727
Merchandise imports	89 972	107 794	117 572	120 816	122 722	127 746	137 648	152 712	168 458	175 670
Trade balance	9 169	8 787	6 543	9 473	6 118	7 473	9 914	14 131	24 576	30 055
Services, credits	13 074	15 648	17 547	19 178	20 349	20 781	22 091	23 857	26 163	28 513
Services, debits	17 642	21 007	24 428	28 292	30 313	30 809	32 450	32 510	33 751	35 767
Services, net	–4 569	–5 358	–6 880	–9 114	–9 962	–10 028	–10 359	–8 653	–7 588	–7 253
Investment income, credits	8 824	13 208	13 431	15 064	12 930	11 391	10 690	14 981	18 894	19 098
Investment income, debits	25 964	30 674	33 918	34 451	30 329	28 873	31 504	34 903	41 279	39 412
Investment income, net	–17 140	–17 464	–20 488	–19 385	–17 399	–17 482	–20 814	–19 923	–22 384	–20 315
Official transfers, net	–707	–569	–620	–594	–939	–851	–571	–429	–329	67
Private transfers, net	–53	–126	–204	–41	–6	55	115	186	312	249
Current account	–13 300	–14 730	–21 650	–19 663	–22 189	–20 833	–21 715	–14 686	–5 411	2 805
Capital account	2 803	3 914	4 628	5 315	5 593	7 092	8 296	7 498	4 946	5 921
Direct investment, assets	–7 119	–6 222	–5 266	–5 235	–5 833	–3 589	–5 867	–9 118	–11 145	–8 501
Portfolio investment, assets	–2 216	–3 640	–4 620	–2 224	–10 178	–9 719	–13 859	–6 470	–5 241	–13 529
Other investment, assets	–121	3 830	–6 100	–8 351	976	–3 369	–478	–20 577	–8 211	–22 582
Total assets, net flows	–9 456	–6 032	–15 986	–15 810	–15 035	–16 677	–20 204	–36 165	–24 597	–44 612
Direct investment, liabilities	8 113	6 123	6 010	7 580	2 879	4 722	4 747	8 457	10 761	6 398
Portfolio investment, liabilities	13 004	18 728	19 603	15 924	27 483	20 433	41 124	17 052	17 170	14 689
Other investment, liabilities	5 174	2	9 758	9 624	–255	–2 218	–6 101	16 421	–3 101	18 023
Total liabilities, net flows	26 291	24 853	35 371	33 128	30 107	22 937	39 770	41 930	24 830	39 110
Change in reserves	–3 901	–8 264	–690	–1 068	1 834	4 756	–935	358	–2 752	–5 498
Financial account	12 933	10 556	18 695	16 249	16 911	11 015	18 631	6 124	–2 521	–11 001
Errors and omissions	–2 435	258	–1 674	–1 903	–313	2 726	–5 212	1 062	2 985	2 275

Source: Statistics Canada; OECD Secretariat.

Table F. **Public sector**

	A. Budget indicators: general government accounts (Per cent of GDP)[1]				
	1960	1970	1980	1995	1996
Current receipts	26.0	34.3	36.1	42.4	42.9
Non-interest expenditure	26.1	31.2	34.9	38.1	36.7
Primary budget balance	–0.6	1.9	–0.9	1.7	3.5
Net interest payments	1.1	1.1	1.9	5.8	5.3
General government budget balance	–1.7	0.8	–2.8	–4.1	–1.8
of which:					
Federal	–0.6	0.3	–3.4	–3.4	–2.0
Provincial, local, hospitals	–1.1	–0.8	–0.3	–0.5	0.5
Pension plans[2]	0.0	1.3	1.0	–0.1	–0.3
General government debt					
Gross debt	67.8*	52.8	40.0	100.5	100.3
Net debt	26.7*	11.6	13.3	69.6	70.3
of which:					
Federal	18.9	7.3	16.0	54.9	55.1
	B. The structure of expenditure *(per cent of GDP)*				
Total general government expenditure	28.8	34.9	40.3	47.9	46.1
Current consumption	13.4	18.5	19.1	19.4	18.6
Transfers to persons	7.8	7.8	9.8	14.6	14.2
Subsidies (+ capital assistance)	0.8	1.1	3.0	1.5	1.5
Capital formation	3.8	3.5	2.7	2.3	2.1
Other programme expenditure	0.2	0.3	0.3	0.3	0.3
Total programme expenditure	26.1	31.2	34.9	38.1	36.7
Gross interest payments	2.8	3.6	5.4	9.8	9.4
	C. General government expenditure by function, financial management system basis *(per cent of GDP, Fiscal year)*				
	1965-66	1970-71	1980-81	1990-91	1995-96
Social services	5.4	6.7	8.6	10.2	11.0
Education	5.2	6.8	5.8	5.8	5.6
Health	2.9	4.8	5.2	6.2	6.0
Transport and communication	3.7	3.1	3.0	2.3	2.0
National defence	2.7	2.0	1.6	1.7	1.4
General services	1.7	2.6	2.8	2.7	2.3

* 1961.
1. National accounts basis.
2. Canada and Quebec Pension Plans.
Source: Department of Finance.

Table G. **Financial markets**

	1970	1980	1992	1996
Size of the financial sector (percentages)[1]				
Sector employment/total employment	4.8	5.7	6.2	5.8
Net financial assets/GDP	6.3	4.3	7.5	n.a.
Structure of financial assets and liabilities				
Financial institutions' share in domestic financial assets (per cent)	33.1	38.6	41.5	43.0
Government securities in NFB[2] total financial assets (per cent)	0.3	0.1	2.2	1.7
Structure of structure of NFB liabilities				
Debt to equity ratio[3]	1.17	1.42	1.48	1.41
Short-term:				
Securities and mortgages ($ billion)	8.7	33.4	118.9	122.9
Trade payables ($ billion)	12.4	54.9	101.6	116.9
Long-term:				
Bonds ($ billion)	13.7	28.7	97.5	124.5
Loans and corporate claims ($ billion)	27.0	105.5	252.5	297.5
Internationalisation of markets				
Share of foreign currency assets and liabilities in the banking sector[4]				
Assets	28.9	39.1	30.8	38.2
Liabilities	28.6	40.2	33.3	39.6
Foreign purchases of Canadian securities[5]	12.4	18.2	75.4	49.9
Canadian purchases of foreign securities[5]	–1.2	0.6	20.8	48.5
Debt (per cent of GDP)				
Private non-financial sector				
NFB[2,6]	77.3	81.6	87.8	93.3
Households[7]	45.3	52.5	63.2	n.a.

1. Public and private financial institutions and insurance.
2. NFB = non-financial corporate business, excluding farms.
3. (Liabilities-shares)/shares.
4. Per cent of consolidated balance sheet of chartered banks; excludes other deposit-taking institutions.
5. Per cent of net issues on domestic securities markets. Data include new issues as well as secondary market transactions.
6. Liabilities less shares.
7. Persons and unincorporated business liabilities less trade payables, other loans and other Canadian bonds.

Source: *Bank of Canada Review*; Labour Force Survey; National Authorities (National Balance Sheet; Security transactions with non-residents).

Table H. **Labour market indicators**

	A. Labour market performance			
	1976	1980	1990	1996
Standardised unemployment rate	7.2	7.5	8.1	9.7
Unemployment rate: Total	7.2	7.5	8.1	9.7
Male	6.4	6.9	8.1	9.9
Women	8.4	8.4	8.1	9.4
Youth[1]	12.6	13.1	12.7	16.1
Share of long-term unemployment in total unemployment[2]	2.5	3.7	5.5	13.6
Dispersion of regional unemployment rates[3]	3.0	3.0	3.5	3.8
	B. Structural or institutional characteristics			
	1976	1980	1990	1996
Participation rate:[4] Total	67.9	71.8	76.8	74.8
Male	84.5	85.9	84.9	81.5
Women	51.2	57.8	68.6	68.0
Employment/population (15-64 years)	63.0	66.4	70.5	67.5
Average hours worked	35.4	35.2	35.2	34.9
Part-time work (as per cent of dependent of total employment)	12.5	14.4	17.0	18.9
Non-wage labour costs[5] (as percentage of total compensation)	7.9	8.5	10.1	13.0
Government unemployment insurance replacement ratio[6]	44.9	41.5	45.4	42.8
Unionisation rate	29.2	28.6	27.2	..

	Average percentage changes (annual rates)		
	1980/1976	1990/1980	1996/1990
Labour force	3.3	1.8	0.9
Employment: Total	3.2	1.7	0.6
Goods-producing	2.0	0.1	–0.6
Services	3.8	2.5	1.1

1. People between 15 and 24 years as a percentage of the labour force of the same age group.
2. People looking for a job since one year or more.
3. Measured by standard deviation for 10 provinces
4. Labour force as a percentage of relevant population group, aged between 15 and 64 years.
5. Employers' contributions to social security, pension funds, medical and dental plans and other forms of supplementary labour income. Total compensation excludes military pay.
6. Average weekly benefits (regular)/average weekly earnings (including overtime) from SEPH.

Source: CANSIM – Statistics Canada; Canadian Authorities; OECD Secretariat.

Table I. **Production structure and performance indicators**

	A. Production structure					
	Per cent share of GDP at factor cost (constant prices)			Per cent share of total employment		
	1976	1985	1996	1976	1985	1996
Agriculture	2.9	2.1	2.2	4.9	4.1	3.3
Mining and quarrying	5.1	4.3	4.4	1.5	1.7	1.2
Manufacturing	20.6	19.7	18.8	20.3	17.6	15.2
of which:						
Food and beverages	3.1	2.7	2.4	2.6	2.2	1.8
Paper and paper products	2.1	1.6	1.4	1.5	1.1	0.9
Primary metal industries	1.6	1.4	1.5	1.3	1.1	0.8
Fabricated metal products, machinery and equipment	4.7	4.2	5.0	4.6	3.9	3.6
Chemicals and chemical products	1.2	1.4	1.5	1.0	0.8	0.7
Construction	6.8	6.1	4.8	6.7	5.2	5.3
Market services	57.1	59.7	63.4	64.5	69.5	73.1
of which:						
Transport, storage and communication	7.1	7.9	9.1	8.7	7.8	7.5
Wholesale and retail trade	11.3	11.2	12.2	17.3	17.8	17.3
Finance, insurance and real estate	13.5	15.1	16.3	5.3	5.6	5.8
Community, business social and personal services	22.0	22.0	22.5	27.1	32.3	27.1
Government services	7.9	7.1	5.7	7.2	7.1	7.2

	B. Manufacturing sector performance *(constant prices)*		
	Productivity growth by sector (GDP/employment, annual rate)		
	1980/1976	1990/1980	1996/1990
Manufacturing	–0.7	2.5	2.3
of which:			
Food and beverages	0.1	1.3	1.3
Paper and paper products	1.3	0.3	1.2
Primary metal industries	–3.2	2.6	4.1
Chemicals and chemical products	1.5	4.4	3.8

Source: Canadian authorities; OECD, *National Accounts.*

BASIC STATISTICS:

INTERNATIONAL COMPARISONS

BASIC STATISTICS: INTERNATIONAL COMPARISONS

	Units	Reference period[1]	Australia	Austria	Belgium	Canada	Denmark	Finland	France	Germany	Greece
Population											
Total	Thousands	1994	17 840	8 031	10 124	29 251	5 206	5 088	57 960	81 407	10 430
Inhabitants per sq. km	Number	1994	2	96	332	3	121	15	106	228	79
Net average annual increase over previous 10 years	%	1994	1.4	0.6	0.3	1.6	0.2	0.4	0.5	2.9	0.5
Employment											
Total civilian employment (TCE)[2]	Thousands	1994	7 680 (93)	3 737	3 724 (92)	13 292	2 508	2 015	21 781 (93)	35 894	3 790
of which: Agriculture	% of TCE	1994	5.3 (93)	7.2	2.6 (92)	4.1	5.1	8.3	5.1 (93)	3.3	20.8
Industry	% of TCE	1994	23.7 (93)	33.2	27.7 (92)	22.6	26.8	26.8	27.7 (93)	37.6	23.6
Services	% of TCE	1994	71 (93)	59.6	69.7 (92)	73.3	68.1	64.9	67.2 (93)	59.1	55.5
Gross domestic product (GDP)											
At current prices and current exchange rates	Bill. US$	1994	331.6	198.1	227.9	544	146.7	97.2	1 328.5	1 832.3	73.1 (93)
Per capita	US$	1994	18 588	24 670	22 515	18 598	28 181	19 106	22 944	27 826	7 051 (93)
At current prices using current PPP's[3]	Bill. US$	1994	327.9	162.3	204.2	596.7	107	82.5	1 111.8	1 601.7	118
Per capita	US$	1994	18 382	20 210	20 166	20 401	20 546	16 208	19 201	24 325	1 450
Average annual volume growth over previous 5 years	%	1994	2.2	2.5	1.6	1.1	1.9	–1.6	1.1	2.6	1.4 (93)
Gross fixed capital formation (GFCF)	% of GDP	1994	21.4	24.8	17.4	18.7	14.8	14.3	18.1	18.5	17.4 (93)
of which: Machinery and equipment	% of GDP	1993	9.8	8.7	7.8	6.2	7.2	5.90	8.10	7.5	7.8
Residential construction	% of GDP	1993	5.2	6.3	4.6	6.1	2.9	3.7	4.9	6.5	3.6
Average annual volume growth over previous 5 years	%	1994	0.8	3.7	0.4	–0.1	–2.8	–12.9	–1	0.8	2.7 (93)
Gross saving ratio[4]	% of GDP	1994	16.8	25.3	22	16	17	16.6	19	21	15.5 (93)
General government											
Current expenditure on goods and services	% of GDP	1994	17.5	18.8	15	20.2	25.3	22.4	19.6	17.7	19.1 (93)
Current disbursements[5]	% of GDP	1993	36.9	48.4	55.3	49	61.1	58.9	51.5	45.6	51.2
Current receipts	% of GDP	1993	33.5	48.6	50.1	43	58.3	52.5	46.8	45.7	40.2
Net official development assistance	% of GNP	1993	0.34	0.31	0.41	0.46	1.03	0.76	0.66	0.44	..
Indicators of living standards											
Private consumption per capita using current PPP's[3]	US$	1993	10 803	10 546	12 090	11 863	10 042	8 814	11 395	10 733	6 367
Passenger cars, per 1 000 inhabitants	Number	1990	430	382	387	469	311	386	413	480[8]	169
Telephones, per 1 000 inhabitants	Number	1991	464	432	410	586	577	544	511	420[8]	413
Television sets, per 1 000 inhabitants	Number	1991	480	478	451	639	536	501	407	556[8]	197
Doctors, per 1 000 inhabitants	Number	1993	2.2 (91)	2.3	3.7	2.2	2.8 (92)	2.6 (92)	2.8	3.2 (92)	3.8 (92)
Infant mortality per 1 000 live births	Number		6.1	6.5	8	6.8	5.4	4.4	6.5	5.8	8.5
Wages and prices (average annual increase over previous 5 years)											
Wages (earnings or rates according to availability)	%	1994	3	5.5	3.7	3.3	3.5	4.8	3.5	5.2	14.6
Consumer prices	%	1994	3	3.4	2.8	2.8	2.1	3.3	2.5	3.3	16.2
Foreign trade											
Exports of goods, fob*	Mill. US$	1994	47 363	44 881	137 259[7]	165 358	41 850	29 514	235 337	422 243	8 958
As % of GDP	%	1994	14.3	22.7	60.2	30.4	28.5	30.4	17.7	23	11.5 (93)
Average annual increase over previous 5 years	%	1994	5	6.7	6.5	7.1	8.3	4.9	5.6	4.4	3.4
Imports of goods, cif*	Mill. US$	1994	49 731	55 071	126 006[7]	148 297	35 932	23 091	220 508	376 566	21 111
As % of GDP	%	1994	15	27.8	55.3	27.3	24.5	23.8	16.6	20.6	30.1 (93)
Average annual increase over previous 5 years	%	1994	4	7.2	5	5.4	6.1	–1.3	3.5	6.9	5.4
Total official reserves[6]	Mill. SDRs	1994	7 730	11 523	9 505[7]	8 416	6 203	7 304	17 986	52 994	9 924
As ratio of average monthly imports of goods	Ratio	1994	1.9	2.5	0.9	0.7	2.1	3.8	1	1.7	5.6

* At current prices and exchange rates.
1. Unless otherwise stated.
2. According to the definitions used in OECD *Labour Force Statistics*.
3. PPPs = Purchasing Power Parities.
4. Gross saving = Gross national disposable income minus private and government consumption.
5. Current disbursements = Current expenditure on goods and services plus current transfers and payments of property income.
6. Gold included in reserves is valued at 35 SDRs per ounce. End of year.
7. Data refer to the Belgo-Luxembourg Economic Union.
8. Data refer to western Germany.
9. Refers to the public sector including public enterprises.
10. Including non-residential construction.

Sources: Population and Employment: OECD, *Labour Force Statistics*. GDP, GFCF, and General Government: OECD, *National Accounts*, Vol. I and *OECD Economic Outlook*, Historical Statistics. Indicators of living standards: Miscellaneous national publications. Wages and Prices: OECD, *Main Economic Indicators*. Foreign trade: OECD, *Monthly Foreign Trade Statistics*, series A. Total official reserves: IMF, *International Financial Statistics*.

BASIC STATISTICS: INTERNATIONAL COMPARISONS *(cont'd)*

	Units	Reference period[1]	Iceland	Ireland	Italy	Japan	Luxembourg	Mexico	Netherlands	New Zealand
Population										
Total	Thousands	1994	267	3 571	57 190	124 960	398	93 010	15 382	3 526
Inhabitants per sq. km	Number	1994	3	51	190	331	153	47	377	13
Net average annual increase over previous 10 years	%	1994	1.1	0.1	0	0.4	0.8	2	0.6	0.8
Employment										
Total civilian employment (TCE)[2]	Thousands	1994	138	1 168 (93)	20 152 (93)	64 530	162 (91)	32 439	6 631	1 560
of which: Agriculture	% of TCE	1994	9.4	12.7 (93)	7.5 (93)	5.8	3.7 (91)	25.8	4	10.4
Industry	% of TCE	1994	26.1	27.7 (93)	33 (93)	34	31.5 (91)	22.2	23	25
Services	% of TCE	1994	65.2	59.7 (93)	59.6 (93)	60.2	64.8 (91)	52.1	73	64.6
Gross domestic product (GDP)										
At current prices and current exchange rates	Bill. US$	1994	6.2	52	1 017.8	4 590	10.6 (92)	371.2	334.3	51.2
Per capita	US$	1994	23 199	14 550	17 796	36 732	27 073 (92)	3 991	21 733	14 513
At current prices using current PPP's[3]	Bill. US$	1994	5.1	54.3	1 068.4	2 593.7	11.7	673.3	285.9	57.3
Per capita	US$	1994	19 271	15 212	18 681	20 756	29 454	7 239	18 589	16 248
Average annual volume growth over previous 5 years	%	1994	0.6	4.7	1	2.1	4.1 (92)	3	2.3	2.5
Gross fixed capital formation (GFCF)	% of GDP	1994	15.2	15.1	16.4	28.6	20.4 (93)	20.7	19.3	20
of which: Machinery and equipment	% of GDP	1993	3.9	6.3	7.4	11.5	..	9.4	8.6	9.3
Residential construction	% of GDP	1993	4.4	4.1	5.3	5.4	..	4.9	5.1	4.9
Average annual volume growth over previous 5 years	%	1994	–4	1	–2.3	1.4	6.5 (92)	7.7	0.4	5.8
Gross saving ratio[4]	% of GDP	1994	16.9	19.5	18.8	31.2	60.2 (92)	15.1	24.4	20.7
General government										
Current expenditure on goods and services	% of GDP	1994	20.6	16	17.1	9.8	17.1 (92)	11.8[9]	14.2	14.7
Current disbursements[5]	% of GDP	1993	34.9	..	53.2	26.9	..	..	55.4	..
Current receipts	% of GDP	1993	35.9	..	47.1	32.9	..	..	54.5	..
Net official development assistance	% of GNP	1993	..	0.15	0.42	0.27	0.34 (92)	..	0.88	0.22
Indicators of living standards										
Private consumption per capita using current PPP's[3]	US$	1993	11 546	7 750	11 029	11 791	15 545	4 853	10 726	9 266
Passenger cars, per 1 000 inhabitants	Number	1990	464	228	478	282	470	85	356	440
Telephones, per 1 000 inhabitants	Number	1991	527	300	400	454	511	70	477	436
Television sets, per 1 000 inhabitants	Number	1991	319	276	421	613	267	148	485	443
Doctors, per 1 000 inhabitants	Number	1993	3	1.7 (92)	1.7 (91)	1.7 (92)	2.1 (92)	1	2.5 (90)	2
Infant mortality per 1 000 live births	Number		4.8	5.9	7.3	4.5 (92)	8.5 (92)	18	6.3	7.3
Wages and prices (average annual increase over previous 5 years)										
Wages (earnings or rates according to availability)	%	1994	..	4.6	5.9	2.4	..	5.3	3.2	2.1
Consumer prices	%	1994	6.3	2.7	5.2	2	3.1	16.1	2.8	2.5
Foreign trade										
Exports of goods, fob*	Mill. US$	1994	1 628	34 125	189 802	396 149	..	60 882	155 084	12 169
As % of GDP	%	1994	26.3	65.7	18.6	8.6	..	16.4	46.4	23.8
Average annual increase over previous 5 years	%	1994	2.7	10.5	6.2	7.6	..	21.7	7.6	6.5
Imports of goods, cif*	Mill. US$	1994	1 464	25 812	167 690	274 916	..	79 346	139 800	11 859
As % of GDP	%	1994	23.6	49.7	16.5	6	..	21.4	41.8	23.2
Average annual increase over previous 5 years	%	1994	0.7	8.1	1.9	5.5	..	25.5	6.1	6.1
Total official reserves[6]	Mill. SDRs	1994	201	4 189	22 102	86 214	..	4 301	23 655	2 540
As ratio of average monthly imports of goods	Ratio	1994	1.6	1.9	1.6	3.8	..	0.7	2	2.6

* At current prices and exchange rates.
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2. According to the definitions used in OECD *Labour Force Statistics*.
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4. Gross saving = Gross national disposable income minus private and government consumption.
5. Current disbursements = Current expenditure on goods and services plus current transfers and payments of property income.
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BASIC STATISTICS: INTERNATIONAL COMPARISONS *(cont'd)*

	Units	Reference period[1]	Norway	Portugal	Spain	Sweden	Switzerland	Turkey	United Kingdom	United States
Population										
Total	Thousands	1994	4 337	9 900	39 150	8 781	6 994	60 573	58 375	260 651
Inhabitants per sq. km	Number	1994	13	107	78	20	169	78	238	28
Net average annual increase over previous 10 years	%	1994	0.5	–0.1	0.2	0.5	0.7	2.1	0.3	1
Employment										
Total civilian employment (TCE)[2]	Thousands	1994	1970 (93)	4 372	11 760	3 926	3 772	19 664	25 044 (93)	123 060
of which: Agriculture	% of TCE	1994	5.6 (93)	11.5	9.8	3.4	4	44.8	2.2 (93)	2.9
Industry	% of TCE	1994	23.1 (93)	32.8	30.1	25	28.8	22.2	26.2 (93)	24
Services	% of TCE	1994	71.3 (93)	55.7	60.2	71.6	67.2	33	71.6 (93)	73.1
Gross domestic product (GDP)										
At current prices and current exchange rates	Bill. US$	1994	103.4 (93)	87	482.4	196.6	257.3	130.7	1 019.5	6 649.8
Per capita	US$	1994	23 984 (93)	8 792	12 321	22 389	36 790	2 157	17 468	25 512
At current prices using current PPP's[3]	Bill. US$	1994	95.3	122	531.7	153	167.4	319.3	1 030.2	6 649.8
Per capita	US$	1994	21 968	12 335	13 581	17 422	23 942	5 271	17 650	25 512
Average annual volume growth over previous 5 years	%	1994	2.1 (93)	1.4	1.5	–0.3	0.5	3.6	0.8	2.1
Gross fixed capital formation (GFCF)	% of GDP	1994	22 (93)	25.7	19.8	13.7	22.8	24.5	15	17.2
of which: Machinery and equipment	% of GDP	1993	..	..	5.7	5.7	7.5	10.3	7.3	7.7
Residential construction	% of GDP	1993	..	..	4.1	4	14.9[10]	9.1	3.1	4
Average annual volume growth over previous 5 years	%	1994	–3.93	2.7	–1.2	–7.6	–0.4	5.1	–2.1	4.6
Gross saving ratio[4]	% of GDP	1994	21.9 (93)	24.2	18.8	13.7	29.3	22.5	13.5	16.2
General government										
Current expenditure on goods and services	% of GDP	1994	22.1 (93)	17.2	16.9	27.3	14.1	11.7	21.6	16.4
Current disbursements[5]	% of GDP	1993	..	..	43.7	67.3	36.7	..	42.7	35.8
Current disbursements[5]	% of GDP	1993	..	..	40.1	59	36	..	36.8	31.7
Current receipts	% of GDP	1993	1.23	0.36	0.32	1.33	0.49	..	0.34	0.19
Net official development assistance	% of GNP	1993	9 826	7 780	8 412	9 240	13 730	3 617	10 942	16 444
Indicators of living standards										
Private consumption per capita using current PPP's[3]	US$	1993	378	260	307	418	441	29	361	568
Passenger cars, per 1 000 inhabitants	Number	1990	515	273	340	687	603	143	445	553
Telephones, per 1 000 inhabitants	Number	1991	423	187	400	468	406	175	434	814
Television sets, per 1 000 inhabitants	Number	1991	3.2 (92)	2.9	4.1	3	3	0.9	1.5 (92)	2.3 (92)
Doctors, per 1 000 inhabitants	Number	1993	5	8.7	7.6	4.8	5.6	52.6	6.6	8.5 (92)
Infant mortality per 1 000 live births	Number		4	..	7.2	5.4	..	..	6.7	2.8
Wages and prices (average annual increase over previous 5 years)										
Wages (earnings or rates according to availability)	%	1994	2.7	9	5.6	5.7	3.9	73	4.6	3.6
Consumer prices	%	1994	34 645	17 072	73 129	61 122	70 467	18 456	205 170	512 627
Foreign trade										
Exports of goods, fob*	Mill. US$	1994	30.9 (93)	19.6	15.2	31.1	27.4	14.1	20.1	7.7
As % of GDP	%	1994	5	6.1	10.5	3.4	6.4	9.5	6.1	7.1
Average annual increase over previous 5 years	%	1994	27 345	25 967	92 182	51 730	68 126	22 976	227 026	663 256
Imports of goods, cif*	Mill. US$	1994	23.3 (93)	29.9	19.1	26.3	26.5	17.6	22.3	10
As % of GDP	%	1994	2.9	6.6	5.2	1	3.2	37.9	2.8	7
Average annual increase over previous 5 years	%	1994	13 033	10 627	28 475	15 929	23 790	4 911	28 094	43 350
Total official reserves[6]	Mill. SDRs	1994	5.7	4.9	3.7	3.7	4.2	2.6	1.5	0.8

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